Algebraic Number Theory

Some Basics

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Part I

Dedekind Domains

1 Dedekind Domains

This chapter follows a local-to-global path. A DVR is the one-prime local model; a Dedekind domain is the global ring obtained by requiring this model at every nonzero prime. Fractional ideals then assemble the local valuations into global prime factorizations, divisors, and finally the class group.

1.1 Discrete Valuation Rings

The word *discrete* means that divisibility is measured by integers. A discrete valuation turns multiplication in a field into addition in $\mathbb{Z}$, so the order of a zero or a pole becomes an ordinary integer.

> **Definition 1.1.1 (Discrete Valuation).** Let K be a field. A *discrete valuation* on K is a surjective group homomorphism
>
> $$v : K^\times \to \mathbb{Z}$$
>
> such that
>
> $$v(x + y) \geq \min(v(x), v(y))$$
>
> whenever $x, y \in K^\times$ and $x + y \neq 0$. We extend v to all of K by setting $v(0) = +\infty$.

> **Definition 1.1.2 (Discrete Valuation Ring).** Let v be a discrete valuation on K. Its *valuation ring* is
>
> $$\mathcal{O}_v \coloneqq \{x \in K \mid v(x) \geq 0\}$$
>
> A domain A is a *discrete valuation ring*, abbreviated *DVR*, if its fraction field is K and $A = \mathcal{O}_v$ for some discrete valuation v on K.

The ring $\mathcal{O}_v$ is local. Its unique maximal ideal, its group of units, and its residue field are respectively

$$\mathfrak{m}_v = \{x \in K \mid v(x) > 0\}, \quad \mathcal{O}_v^\times = \{x \in K \mid v(x) = 0\}, \quad \kappa(v) = \mathcal{O}_v/\mathfrak{m}_v$$

Since v is surjective, there is an element $\pi \in K$ with $v(\pi) = 1$. Such an element automatically lies in $\mathcal{O}_v$ and generates $\mathfrak{m}_v$; it is called a *uniformizer* (or a *prime element*). A uniformizer is not unique, but any two uniformizers differ by a unit.

> **Remark (What a DVR remembers).** Fix a uniformizer π. Every $x \in K^\times$ has a unique expression
>
> $$x = u\pi^n, \quad u \in A^\times, \quad n \in \mathbb{Z}$$
>
> and $n = v(x)$. Thus $v(x) = 0$ means that x is a unit, $v(x) > 0$ records the order to which x vanishes, and $v(x) < 0$ records the order of its pole. Moreover, for nonzero $x, y \in A$,

$$x \mid y \Leftrightarrow v(x) \leq v(y)$$

In this sense a DVR is a one-dimensional local ring with only one direction of vanishing. Geometrically, the local ring of an integral curve at a regular closed point is a DVR, and its valuation is the order of vanishing at that point.

Theorem 1.1.1 (Equivalent characterizations of a DVR). Let A be a domain that is not a field, and let K be its fraction field. The following conditions are equivalent.

1. A is a DVR.
2. A is a Noetherian valuation ring of K; equivalently, A is Noetherian and for every $x \in K^\times$, either $x \in A$ or $x^{-1} \in A$.
3. A is a principal ideal domain with exactly one nonzero prime ideal.
4. A is a Noetherian local domain whose maximal ideal $\mathfrak{m}$ is generated by one nonzero element.
5. A is a regular local ring of Krull dimension one. Equivalently, A is Noetherian and local, $\dim(A) = 1$, and
$$\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$$
6. A is a Noetherian normal local domain of Krull dimension one.
7. There is a nonzero element $\pi \in A$ such that every $x \in K^\times$ can be written uniquely as
$$x = u\pi^n, \quad u \in A^\times, \quad n \in \mathbb{Z}$$
8. A is local with maximal ideal $\mathfrak{m}$, and its nonzero ideals are precisely
$$A = \mathfrak{m}^0, \mathfrak{m}, \mathfrak{m}^2, \mathfrak{m}^3, \ldots$$
with no repetitions.

Proof. Suppose first that $A = \mathcal{O}_v$ and choose π with $v(\pi) = 1$. For each $x \in K^\times$, the element $u = x\pi^{-v(x)}$ has valuation zero, hence is a unit of A. This gives the unique factorization $x = u\pi^{v(x)}$.

Now let I be a nonzero ideal of A. The set of valuations of its nonzero elements is a nonempty subset of $\mathbb{Z}_{\geq 0}$, so it has a least element n. An element of valuation n generates I, and therefore
$$I = (\pi^n) = \mathfrak{m}^n$$
Consequently all nonzero ideals are powers of $\mathfrak{m} = (\pi)$, A is a PID, and $\mathfrak{m}$ is its only nonzero prime ideal.

Conversely, if A is local and Noetherian with $\mathfrak{m} = (\pi)$, the Krull intersection theorem gives $\bigcap_{n\geq 0} \mathfrak{m}^n = (0)$. Repeatedly dividing a nonzero element by π must therefore stop, and every nonzero element of A has the form $u\pi^n$ with u a unit. Extending the exponent to quotients defines a discrete valuation on K, with valuation ring exactly A.

The factorization description makes the principal ideals of A linearly ordered, so A is a valuation ring. Conversely, every finitely generated ideal in a valuation ring is generated by an element of least valuation. Thus a Noetherian valuation ring that is not a field is a DVR.

A DVR has only the prime ideals (0) and $\mathfrak{m}$, so it has dimension one. It is also normal: if $x \in K$ had $v(x) < 0$, then in any monic equation for x over A, the leading term would have strictly smaller valuation than all the other terms, which is impossible.

Conversely, suppose that A is Noetherian, normal, local, and one-dimensional. Choose $0 \neq a \in \mathfrak{m}$. Since $\mathfrak{m}$ is the only prime ideal containing (a), some power of $\mathfrak{m}$ lies in (a). Choose n minimal with $\mathfrak{m}^n \subseteq (a)$, take $y \in \mathfrak{m}^{n-1} \setminus (a)$, and put $z = \frac{y}{a} \in K$. Then $z\mathfrak{m} \subseteq A$. If $z\mathfrak{m} \subseteq \mathfrak{m}$, the determinant trick would make z integral over A, contradicting normality and $z \notin A$. Hence zb is a unit for some $b \in \mathfrak{m}$. For every $c \in \mathfrak{m}$,

$$\frac{c}{b} = \frac{zc}{zb} \in A$$

so $\mathfrak{m} = (b)$. This recovers the principal maximal ideal criterion. Finally, Nakayama's lemma says that the minimal number of generators of $\mathfrak{m}$ is $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2)$, which gives the regular local characterization. □

Example (The basic examples). Let k be a field. The formal power series ring $k[[t]]$ is a DVR with uniformizer t. For a nonzero series $f = \sum_{i \geq 0} a_i t^i$, its valuation is the least index i for which $a_i \neq 0$.

The localization $k[t]_{(t)}$ is also a DVR with uniformizer t. It consists of rational functions that are regular at $t = 0$, and its valuation measures the order of vanishing at the origin.

Similarly, for a prime number p, the local ring $\mathbb{Z}_{(p)}$ is a DVR with fraction field $\mathbb{Q}$, uniformizer p, and residue field $\mathbb{F}_p$.

Example (A nearby non-example). The local ring $k[x, y]_{(x,y)}$ is not a DVR. Its maximal ideal (x, y) needs two generators, and the ring has Krull dimension two. There are two independent directions of vanishing rather than one.

1.2 Dedekind Domains: Definitions

A Dedekind domain can be recognized either from its global ring-theoretic properties or from its local rings at nonzero prime ideals.

Definition 1.2.1 (Dedekind Domain). Let A be a domain that is not a field, and let K be its fraction field. The following conditions are equivalent.

1. The ring A is Noetherian, normal, and of Krull dimension one. Here *normal* means that A is integrally closed in K.
2. The ring A is Noetherian, and for every nonzero prime ideal $\mathfrak{p}$ of A, the localization $A_{\mathfrak{p}}$ is a DVR.

A domain satisfying either condition is called a *Dedekind domain*.

Proof. Suppose first that A is Noetherian, normal, and one-dimensional. Every nonzero prime ideal $\mathfrak{p}$ is maximal. Localization preserves both the Noetherian and normal properties, and

$$\dim\left(A_{\mathfrak{p}}\right) = \operatorname{ht}(\mathfrak{p}) = 1$$

Thus $A_{\mathfrak{p}}$ is a Noetherian normal local domain of dimension one, so it is a DVR by the equivalent characterizations in the previous section.

Conversely, suppose that A is Noetherian and that $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$. We first show that A is one-dimensional. If

$$(0) \subset \mathfrak{p} \subseteq \mathfrak{q}$$

is a chain of prime ideals, then after localization at $\mathfrak{q}$, the ideal $\mathfrak{p}A_{\mathfrak{q}}$ is a nonzero prime ideal of the DVR $A_{\mathfrak{q}}$. A DVR has only one nonzero prime ideal, so $\mathfrak{p}A_{\mathfrak{q}} = \mathfrak{q}A_{\mathfrak{q}}$. Contracting back to A gives $\mathfrak{p} = \mathfrak{q}$. Hence every nonzero prime ideal is maximal and $\dim(A) = 1$.

It remains to prove normality. Let $z \in K$ be integral over A. Integrality is preserved by localization, so z is integral over $A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. Since A is not a field, each maximal ideal is nonzero; hence every $A_{\mathfrak{m}}$ is a DVR and is integrally closed. It follows that $z \in A_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$.

A domain is the intersection of its localizations at all maximal ideals inside its fraction field. Indeed, if $z \notin A$, then

$$I = \{a \in A \mid az \in A\}$$

is a proper ideal. Choose a maximal ideal $\mathfrak{m}$ containing I. If $z \in A_{\mathfrak{m}}$, then $sz \in A$ for some $s \notin \mathfrak{m}$, which would give $s \in I \subseteq \mathfrak{m}$, a contradiction. Therefore $z \in A$, and A is normal. □

> **Remark.** Some authors allow a field to be a Dedekind domain and replace $\dim(A) = 1$ by $\dim(A) \leq 1$. In these notes a Dedekind domain is not a field, so the dimension is exactly one.

1.3 Fractional Ideals and Prime Factorization

Let A be a Dedekind domain with fraction field K. Ordinary ideals live inside A, but taking an inverse usually introduces denominators. Fractional ideals enlarge the collection of ideals just enough to make ideal multiplication into a group operation.

> **Definition 1.3.1 (Fractional Ideal).** A *fractional ideal* of A is a nonzero A-submodule $I \subseteq K$ for which there exists $0 \neq d \in A$ such that
>
> $$dI \subseteq A$$
>
> It is *integral* if $I \subseteq A$, and *principal* if $I = xA$ for some $x \in K^{\times}$. We also write $(x) = xA$.

> **Proposition 1.3.1 (Clearing Denominators).** Every fractional ideal L can be written as
>
> $$L = \frac{1}{a} I$$
>
> for some $0 \neq a \in A$ and some nonzero integral ideal $I \subseteq A$. Conversely, every module of this form is a fractional ideal.

Proof. Choose $0 \neq a \in A$ such that $aL \subseteq A$, and set $I = aL$. Then I is a nonzero integral ideal and $L = \frac{1}{a}I$. Conversely, if $I \subseteq A$ is a nonzero ideal, then $a\left(\frac{1}{a}I\right) = I \subseteq A$, so $\frac{1}{a}I$ satisfies the definition of a fractional ideal. □

Thus a fractional ideal is simply an ordinary ideal with one common denominator. Since A is Noetherian, it is also a finitely generated A-module.

> **Remark (Why introduce denominators?).** An integral ideal can have an inverse only when it is the unit ideal: if $I \subseteq \mathfrak{m}$ is proper and $J \subseteq A$, then $IJ \subseteq \mathfrak{m}$, so $IJ \neq A$. Allowing submodules of K makes room for negative orders of vanishing. Fractional ideals therefore play the same role for ideals that negative exponents play for numbers.

1.3.1 Ideal Operations

Definition 1.3.2 (Sum, Product, and Inverse). For nonzero fractional ideals $I, J \subseteq K$, define their sum and product by

$$I + J = \{x + y \mid x \in I, y \in J\}$$

$$IJ = \left\{\sum_{i=1}^{n} x_i y_i \mid n \geq 1, x_i \in I, y_i \in J\right\}$$

The inverse candidate of I is the ideal quotient

$$I^{-1} = (A : I) := \{x \in K \mid xI \subseteq A\}$$

Their intersection $I \cap J$ is again a nonzero fractional ideal.

The finite sums in the definition of IJ are essential: the set of single products xy need not be closed under addition. With this definition, multiplication is associative and commutative, distributes over sums, and has identity A. Principal fractional ideals behave exactly as expected:

$$(x)(y) = (xy), \quad (x)^{-1} = \left(x^{-1}\right)$$

Theorem 1.3.1 (Invertibility and Unique Prime-Ideal Factorization). Every nonzero fractional ideal I of A is invertible:

$$II^{-1} = A$$

For each nonzero prime ideal $\mathfrak{p}$, there is a unique integer $v_{\mathfrak{p}}(I)$ such that

$$IA_{\mathfrak{p}} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)}$$

Only finitely many of these integers are nonzero, and

$$I = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(I)}$$

Consequently, the nonzero fractional ideals form an abelian group under ideal multiplication. They are also finite projective A-modules of rank one.

Proof. The localization $A_{\mathfrak{p}}$ is a DVR. A finitely generated fractional ideal over a DVR is generated by an element of least valuation, so it is a unique integer power of the maximal ideal. This defines $v_{\mathfrak{p}}(I)$.

The inverse candidate is itself fractional. Indeed, the denominator-clearing element d belongs to I^{-1}, while any $0 \neq x \in I$ gives $I^{-1} \subseteq x^{-1}A$.

The equality $(I^{-1})A_{\mathfrak{p}} = \left(IA_{\mathfrak{p}}\right)^{-1}$ holds because I is finitely generated. Hence

$$(II^{-1})A_{\mathfrak{p}} = A_{\mathfrak{p}}$$

for every maximal ideal $\mathfrak{p}$. Equality of fractional ideals can be checked after localization at all maximal ideals, so $II^{-1} = A$.

Choose $0 \neq d \in A$ with $dI \subseteq A$. The exponent $v_{\mathfrak{p}}(I)$ can be nonzero only if $\mathfrak{p}$ contains dI or (d). In a one-dimensional Noetherian domain, only finitely many prime ideals contain a fixed nonzero ideal. Thus the displayed product is finite and defines a fractional ideal J. At every $\mathfrak{p}$, the ideals I and J have the same localization, so $I = J$. Uniqueness follows by localizing a proposed factorization at each $\mathfrak{p}$.

Finally, an invertible ideal is locally free of rank one, hence finite projective of rank one. □

> **Remark (Why the Dedekind Hypotheses Matter).** The argument is local-to-global. The DVR condition produces one integer exponent at each nonzero prime, Noetherianity makes the support finite, and invertibility permits negative exponents and cancellation. Consequently, prime-ideal factorization survives even when factorization of elements does not.

Corollary 1.3.1 (The Exponent Dictionary). For nonzero fractional ideals I and J and every nonzero prime $\mathfrak{p}$,

Operation	Rule at $\mathfrak{p}$
IJ	$v_{\mathfrak{p}}(IJ) = v_{\mathfrak{p}}(I) + v_{\mathfrak{p}}(J)$
I^{-1}	$v_{\mathfrak{p}}\left(I^{-1}\right) = -v_{\mathfrak{p}}(I)$
$I + J$	$v_{\mathfrak{p}}(I + J) = \min\left(v_{\mathfrak{p}}(I), v_{\mathfrak{p}}(J)\right)$
$I \cap J$	$v_{\mathfrak{p}}(I \cap J) = \max\left(v_{\mathfrak{p}}(I), v_{\mathfrak{p}}(J)\right)$

Moreover,

$$I \subseteq J \Leftrightarrow v_{\mathfrak{p}}(I) \geq v_{\mathfrak{p}}(J) \text{ for every } \mathfrak{p}$$

The ideal I is integral if and only if every $v_{\mathfrak{p}}(I)$ is nonnegative. For integral ideals, divisibility reverses containment:

$$I \text{ divides } J \Leftrightarrow J \subseteq I$$

Proof. All statements may be checked after localization at $\mathfrak{p}$. In a DVR, powers of the maximal ideal satisfy

$$\mathfrak{m}^a\mathfrak{m}^b = \mathfrak{m}^{a+b}, \quad \mathfrak{m}^a + \mathfrak{m}^b = \mathfrak{m}^{\min(a,b)}, \quad \mathfrak{m}^a \cap \mathfrak{m}^b = \mathfrak{m}^{\max(a,b)}$$

The formulas and the reversed inequality for containment follow immediately. If $J = IL$ with L integral, then the exponent of every prime in J is at least its exponent in I, and conversely $L = JI^{-1}$ is integral whenever $J \subseteq I$. □

> **Remark.** The sum $I + J$ is the ideal-theoretic analogue of a greatest common divisor, while $I \cap J$ behaves like a least common multiple. The reversal comes from $(a) \subseteq (b)$ precisely when b divides a.

1.4 Arithmetic of Integral Ideals

The prime exponents introduced for fractional ideals turn containment, divisibility, coprimality, and congruences into elementary integer arithmetic. We now collect the computational consequences used later in algebraic number theory.

1.4.1 Exponents and Divisibility

Proposition 1.4.1 (Computing the Prime Exponents). If I is a nonzero integral ideal, then the exponent of a nonzero prime $\mathfrak{p}$ in its factorization is

$$v_{\mathfrak{p}}(I) = \max\{n \geq 0 \mid I \subseteq \mathfrak{p}^n\}$$

It can also be read locally as

$$v_{\mathfrak{p}}(I) = \operatorname{length}_{A_{\mathfrak{p}}}\left(A_{\mathfrak{p}}/IA_{\mathfrak{p}}\right)$$

In particular,

$$v_{\mathfrak{p}}(I) > 0 \Leftrightarrow I \subseteq \mathfrak{p}$$

Thus the prime ideals occurring in the factorization of I are exactly the prime ideals containing I.

Proof. Localizing the factorization of I at $\mathfrak{p}$ turns every prime factor other than $\mathfrak{p}$ into the unit ideal, so

$$IA_{\mathfrak{p}} = \left(\mathfrak{p}A_{\mathfrak{p}}\right)^{v_{\mathfrak{p}}(I)} = \left(\pi^{v_{\mathfrak{p}}(I)}\right)$$

The containment criterion in a DVR gives the maximum formula. The quotient of a DVR by (π^n) has length n, which gives the second formula. The last equivalence is the case $n = 1$. □

> **Remark (The gcd and lcm Viewpoint).** These names have the same divisibility meaning as they do for integers. Recall that D divides I precisely when $I \subseteq D$. Therefore:
>
> 1. an ideal D divides both I and J if and only if it divides $I + J = \gcd(I, J)$;
> 2. both I and J divide an ideal M if and only if $I \cap J = \operatorname{lcm}(I, J)$ divides M.
>
> The apparently reversed operations come from the reversed divisibility order on ideals. For example, in $\mathbb{Z}$,
>
> $$(a) + (b) = (\gcd(a, b)), \quad (a) \cap (b) = (\operatorname{lcm}(a, b))$$

> In a general Dedekind domain the ideals $I + J$ and $I \cap J$ need not be principal, so their gcd and lcm are naturally ideals rather than elements.

1.4.2 Coprime Factors and Remainders

Definition 1.4.1 (Coprime Ideals). Two ideals I and J are *coprime* if

$$I + J = A$$

Equivalently, no nonzero prime ideal occurs in both prime factorizations.

Proposition 1.4.2 (Intersection, Product, and the Chinese Remainder Theorem). If I and J are coprime, then

$$I \cap J = IJ$$

More generally, if

$$I = \prod_{i=1}^{r} \mathfrak{p}_i^{n_i}$$

with distinct prime ideals $\mathfrak{p}_i$, then

$$I = \bigcap_{i=1}^{r} \mathfrak{p}_i^{n_i}$$

and the natural residue map induces an isomorphism

$$A/I \to \prod_{i=1}^{r} A/\mathfrak{p}_i^{n_i}$$

Proof. Coprimality means that at every prime, at least one of the two exponents is zero. Their sum therefore equals their maximum, so the exponent dictionary gives $IJ = I \cap J$. Distinct nonzero prime ideals are maximal and hence pairwise coprime; the intersection formula follows by induction. The final isomorphism is the Chinese remainder theorem. □

Corollary 1.4.1 (The Radical of an Ideal). If

$$I = \prod_{i=1}^{r} \mathfrak{p}_i^{n_i}, \quad n_i \geq 1$$

then

$$\sqrt{I} = \prod_{i=1}^{r} \mathfrak{p}_i = \bigcap_{i=1}^{r} \mathfrak{p}_i$$

Thus I is radical if and only if every exponent in its prime factorization is 1.

1.4.3 Examples

Example (Ideals of the Integers). The nonzero prime ideals of $\mathbb{Z}$ are (p) for rational primes p. If

$$n = \pm \prod_p p^{e_p}$$

is the ordinary prime factorization of a nonzero integer, then

$$(n) = \prod_p (p)^{e_p}$$

Thus unique factorization of ideals in $\mathbb{Z}$ is exactly ordinary unique factorization, written in ideal language.

Example (Prime-Ideal Factorization in $\mathbb{Z}\left[\sqrt{-5}\right]$). In $A = \mathbb{Z}\left[\sqrt{-5}\right]$, put

$$\mathfrak{p}_2 = \left(2, 1 + \sqrt{-5}\right)$$

$$\mathfrak{p}_3 = \left(3, 1 + \sqrt{-5}\right), \quad \mathfrak{q}_3 = \left(3, 1 - \sqrt{-5}\right)$$

Direct multiplication gives

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3\mathfrak{q}_3$$

and

$$\left(1 + \sqrt{-5}\right) = \mathfrak{p}_2\mathfrak{p}_3$$

$$\left(1 - \sqrt{-5}\right) = \mathfrak{p}_2\mathfrak{q}_3$$

Consequently, both element factorizations of 6 induce the same unique factorization of its principal ideal:

$$(6) = \mathfrak{p}_2^2\mathfrak{p}_3\mathfrak{q}_3$$

The ambiguity among elements disappears after passing to ideals.

1.5 Divisors and the Class Group

1.5.1 The Correspondence with Divisors

Definition 1.5.1 (Divisors). The *divisor group* of A, denoted $\mathrm{Div}(A)$, is the free abelian group on the nonzero prime ideals of A. Its elements are finite formal sums

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}], \quad n_{\mathfrak{p}} \in \mathbb{Z}$$

A divisor is *effective* if every coefficient is nonnegative. For $x \in K^\times$, its *principal divisor* is

$$\operatorname{div}(x) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(x)[\mathfrak{p}]$$

where $v_{\mathfrak{p}}(x)$ is the normalized valuation of x in the DVR $A_{\mathfrak{p}}$.

Because A is one-dimensional, its nonzero prime ideals are exactly the codimension-one points of $\operatorname{Spec} A$. Thus this is also the group of Weil divisors on $\operatorname{Spec} A$.

Theorem 1.5.1 (Fractional Ideals as Divisors). Let $\operatorname{FracId}(A)$ denote the group of nonzero fractional ideals. The map

$$\Phi(I) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(I)[\mathfrak{p}]$$

is an isomorphism of abelian groups

$$\Phi : \operatorname{FracId}(A) \to \operatorname{Div}(A)$$

Under this correspondence:

1. ideal multiplication corresponds to addition of divisors;
2. integral ideals correspond to effective divisors;
3. the principal fractional ideal (x) corresponds to the principal divisor $\operatorname{div}(x)$.

Proof. Multiplicativity follows from $v_{\mathfrak{p}}(IJ) = v_{\mathfrak{p}}(I) + v_{\mathfrak{p}}(J)$. The prime-ideal factorization above shows that Φ is injective. Conversely, for a finite divisor $D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$, the fractional ideal

$$I_D = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

satisfies $\Phi(I_D) = D$, proving surjectivity. The remaining claims follow from the exponent dictionary. □

1.5.2 Two Viewpoints on the Class Group

Definition 1.5.2 (Ideal Class Group: The Ideal-Theoretic Viewpoint). Let $\operatorname{PrinId}(A)$ be the subgroup of $\operatorname{FracId}(A)$ consisting of principal fractional ideals $(x) = xA$ with $x \in K^\times$. The *ideal class group* of A is

$$\operatorname{Cl}_{\mathrm{id}}(A) := \operatorname{FracId}(A)/\operatorname{PrinId}(A)$$

Thus two fractional ideals I and J determine the same ideal class if and only if $I = xJ$ for some $x \in K^\times$. We write the quotient law additively, so $[IJ] = [I] + [J]$.

Multiplying by x merely rescales an A-lattice inside the one-dimensional K-vector space K. The quotient forgets this choice of scale and remembers only the intrinsic isomorphism class of the ideal. Equivalently, the ideal class group completes the exact sequence

$$0 \to A^\times \to K^\times \to \operatorname{FracId}(A) \to \operatorname{Cl}_{\mathrm{id}}(A) \to 0$$

where $x \in K^\times$ maps to (x).

Definition 1.5.3 (Divisor Class Group: The Geometric Viewpoint). Put $X = \operatorname{Spec} A$. The subgroup of *principal divisors* is

$$\operatorname{Prin}(X) = \{\operatorname{div}(x) \mid x \in K^\times\} \subseteq \operatorname{Div}(X)$$

Two divisors D and E are *linearly equivalent* if

$$D - E = \operatorname{div}(x)$$

for some $x \in K^\times$. The *divisor class group* is

$$\operatorname{Cl}_{\mathrm{div}}(X) := \operatorname{Div}(X)/\operatorname{Prin}(X)$$

Its elements are therefore linear-equivalence classes of divisors.

This second definition has a geometric interpretation. A nonzero prime ideal $\mathfrak{p}$ is a closed codimension-one point of the one-dimensional scheme X. A divisor

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$$

is a finite collection of closed points with integer multiplicities. Positive coefficients describe zeros, while negative coefficients describe poles. For a rational function $x \in K^\times$, the coefficient of $[\mathfrak{p}]$ in $\operatorname{div}(x)$ is its order of vanishing at $\mathfrak{p}$. Passing to divisor classes means that divisors differing only by the zeros and poles of a global rational function are regarded as geometrically equivalent.

Proposition 1.5.1 (Effective Divisors and Their Ideal Sheaves). Let

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}[\mathfrak{p}]$$

be an effective divisor, and set

$$I_D = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

Then I_D is the integral ideal cutting out the corresponding zero-dimensional closed subscheme of X. At $\mathfrak{p}$ its multiplicity is

$$\operatorname{length}_{A_{\mathfrak{p}}}\left(A_{\mathfrak{p}}/I_D A_{\mathfrak{p}}\right) = n_{\mathfrak{p}}$$

The associated ideal sheaf is

$$\widetilde{I_D} = \mathcal{O}_X(-D)$$

More generally, for an arbitrary divisor D, the same product defines a fractional ideal I_D, and the global sections of its associated line bundle are

$$\Gamma(X, \mathcal{O}_X(D)) = \left\{x \in K \mid v_{\mathfrak{p}}(x) + n_{\mathfrak{p}} \geq 0 \;\text{ for every }\; \mathfrak{p}\right\} = I_D^{-1}$$

Proof. Localizing at $\mathfrak{p}$ gives $I_D A_{\mathfrak{p}} = \mathfrak{p}^{n_{\mathfrak{p}}} A_{\mathfrak{p}}$. If π is a uniformizer of the DVR $A_{\mathfrak{p}}$, then the quotient is $A_{\mathfrak{p}}/\left(\pi^{n_{\mathfrak{p}}}\right)$, which has length $n_{\mathfrak{p}}$. The descriptions of $\mathcal{O}_X(-D)$ and $\mathcal{O}_X(D)$ follow from the same local valuation inequalities. □

Theorem 1.5.2 (Comparison of the Two Viewpoints). The fractional-ideal–divisor isomorphism Φ sends principal fractional ideals to principal divisors and therefore induces an isomorphism

$$\Phi_{\mathrm{cl}} : \mathrm{Cl}_{\mathrm{id}}(A) \to \mathrm{Cl}_{\mathrm{div}}(\operatorname{Spec} A)$$

Hence both groups are denoted simply by $\mathrm{Cl}(A)$. Moreover, there is a canonical isomorphism between $\mathrm{Cl}(A)$ and the Picard group of $X = \operatorname{Spec} A$, the group of isomorphism classes of line bundles on X.

Proof. We have already seen that $\Phi((x)) = \operatorname{div}(x)$. Since Φ is an isomorphism before taking quotients, it remains an isomorphism after quotienting by the corresponding principal subgroups.

Every fractional ideal is a projective A-module of rank one, so its associated sheaf is a line bundle on X. Conversely, every line bundle on an affine scheme arises from a projective module of rank one; after choosing a trivialization over the generic point, that module becomes a fractional ideal in K. Changing the trivialization multiplies the ideal by an element of $K^\times$, so its ideal class is well-defined. □

Remark (The Sign Convention). If $\Phi(I) = D$, then

$$\tilde{I} = \mathcal{O}_X(-D)$$

Thus the direct identification of ideal classes with divisor classes uses $I \mapsto D$, while the line bundle naturally attached to I is $\mathcal{O}_X(-D)$. Equivalently, $\mathcal{O}_X(D)$ corresponds to I^{-1}. This minus sign is the only convention one must watch when moving among ideals, divisors, and line bundles.

Remark (What Does the Class Group Measure?). The class group packages several related obstructions.

1. *Failure of ideals to be principal.* The class $[I]$ is zero exactly when I is principal. Hence
$$\mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID}$$
2. *Failure of unique factorization of elements.* Ideals in a Dedekind domain always factor uniquely into prime ideals, but the prime ideals need not be generated by prime elements. The class group records precisely this obstruction. In fact,
$$\mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID} \Leftrightarrow A \text{ is a UFD}$$
Relations among the classes of prime ideals explain how distinct factorizations of elements can produce the same principal ideal.
3. *Nontrivial line bundles.* The group $\mathrm{Cl}(A)$ classifies rank-one projective A-modules, or equivalently line bundles on $\operatorname{Spec} A$. The zero class is the trivial line bundle $\mathcal{O}_X$.
4. *Arithmetic complexity.* When A is the ring of integers of a number field, $\mathrm{Cl}(A)$ is finite. Its order is the *class number*. It gives a numerical measure of how far the ring is from being principal.

Example (The Integers). Every nonzero fractional ideal of $\mathbb{Z}$ is $q\mathbb{Z}$ for some $q \in \mathbb{Q}^\times$. If $q = \pm \prod_p p^{n_p}$, then $\Phi(q\mathbb{Z}) = \sum_p n_p[p]$. Every divisor is principal, so $\mathrm{Cl}(\mathbb{Z}) = 0$.

Example (The Class Group of $\mathbb{Z}\left[\sqrt{-5}\right]$). Consider the Dedekind domain $A = \mathbb{Z}\left[\sqrt{-5}\right]$, the ring of integers of $\mathbb{Q}\left(\sqrt{-5}\right)$. The ideal

$$\mathfrak{p} = \left(2, 1 + \sqrt{-5}\right)$$

satisfies $\mathfrak{p}^2 = (2)$, but $\mathfrak{p}$ is not principal: a generator would have norm 2, whereas the equation $a^2 + 5b^2 = 2$ has no integer solutions. Therefore $[\mathfrak{p}]$ is a nonzero class with

$$2[\mathfrak{p}] = 0$$

We now show that there are no other ideal classes. For a nonzero integral ideal J, write $\mathrm{N}(J) = |A/J|$. The field $\mathbb{Q}\left(\sqrt{-5}\right)$ has discriminant $D_K = -20$. The Minkowski bound for an imaginary quadratic field says that every ideal class contains a nonzero integral ideal J such that

$$\mathrm{N}(J) \le \left(\frac{2}{\pi}\right)\sqrt{|D_K|} = \left(\frac{2}{\pi}\right)\sqrt{20} < 3$$

Since $\mathrm{N}(J)$ is a positive integer, it is either 1 or 2. If $\mathrm{N}(J) = 1$, then $J = A$ and its class is zero. If $\mathrm{N}(J) = 2$, then A/J is the field $\mathbb{F}_2$. The image of $\sqrt{-5}$ in this field must satisfy $t^2 + 1 = 0$, whose only root is $t = 1$. Hence the quotient map has kernel

$$J = \left(2, 1 + \sqrt{-5}\right) = \mathfrak{p}$$

Thus every ideal class is either $[A]$ or $[\mathfrak{p}]$. These classes are distinct because $\mathfrak{p}$ is not principal, and $2[\mathfrak{p}] = 0$. Therefore

$$\mathrm{Cl}\left(\mathbb{Z}\left[\sqrt{-5}\right]\right) = \{[A], [\mathfrak{p}]\}$$

As a group, it is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, and the class number is 2.

The prime-ideal factorizations computed above show how this nonzero two-torsion class accounts for the two element factorizations $6 = 2 \cdot 3 = \left(1 + \sqrt{-5}\right)\left(1 - \sqrt{-5}\right)$: uniqueness remains true for ideals, while the nonprincipal prime ideals obstruct uniqueness for elements.

1.5.3 Class Number and an Ideal-Theoretic Characterization

Definition 1.5.4 (Class Number). The *class number* of a Dedekind domain A is the cardinality of its class group:

$$h(A) := |\mathrm{Cl}(A)|$$

For a general Dedekind domain this cardinality may be infinite. When it is finite, $h(A)$ is a positive integer. In particular,

$$h(A) = 1 \Leftrightarrow \mathrm{Cl}(A) = 0 \Leftrightarrow A \text{ is a PID} \Leftrightarrow A \text{ is a UFD}$$

If A is the ring of integers of a number field, then $h(A)$ is always finite.

Theorem 1.5.3 (Dedekind Domains via Invertible Ideals). Let A be a domain that is not a field, with fraction field K. The following conditions are equivalent.

1. The ring A is a Dedekind domain.
2. Every nonzero integral ideal $I \subseteq A$ is invertible as a fractional ideal; equivalently,
$$I(A : I) = A$$
where $(A : I) = \{x \in K \mid xI \subseteq A\}$.
3. Every nonzero fractional ideal of A is invertible.

Thus invertibility of all nonzero ideals may be taken as another equivalent definition of a Dedekind domain.

Proof. If A is Dedekind, every nonzero fractional ideal is invertible by the invertibility theorem above, so the first condition implies the third, and the third plainly implies the second.

Now suppose that every nonzero integral ideal is invertible. Such an ideal is automatically finitely generated. Indeed, write
$$1 = \sum_{i=1}^{n} x_i y_i, \quad x_i \in I, y_i \in (A : I)$$
For any $x \in I$,
$$x = \sum_{i=1}^{n} (y_i x) x_i$$
and every coefficient $y_i x$ lies in A. Hence I is generated by $x_1, \ldots, x_n$. Since every ideal of A is finitely generated, A is Noetherian.

Let $\mathfrak{p}$ be a nonzero prime ideal and put $R = A_{\mathfrak{p}}$. Every nonzero ideal of R has the form IR for a nonzero ideal I of A. Localization preserves invertibility, so IR is an invertible ideal of the local ring R. Every invertible ideal of a local ring is principal: if
$$1 = \sum_{i=1}^{n} u_i v_i, \quad u_i \in IR, v_i \in (IR)^{-1}$$
then some product $u_i v_i$ is a unit, and the corresponding u_i generates IR. Thus every nonzero ideal of R is principal. Since R is local and not a field, it is a DVR. Therefore $A_{\mathfrak{p}}$ is a DVR for every nonzero prime $\mathfrak{p}$, and the local characterization proved earlier shows that A is Dedekind. □

2 Étale Algebras

The chapter begins with separability for field extensions, packages that notion geometrically as étaleness, and then develops trace and norm as the additive and multiplicative operations along finite maps. Each layer motivates the next while keeping the field-theoretic, geometric, and arithmetic viewpoints visible.

2.1 Separability

Separability says that the algebraic conjugates of an element remain distinct. It is the condition that allows a finite field extension to behave, after passing to an algebraic closure, like a finite collection of distinct points. We begin with a review of the field-theoretic definitions and their most useful consequences.

2.1.1 Basic Definitions

Definition 2.1.1 (Separable Polynomials, Elements, and Extensions). Let K be a field and fix an algebraic closure $\overline{K}$.

1. A nonconstant polynomial $f \in K[X]$ is *separable over* K if its roots in $\overline{K}$ are distinct.
2. An algebraic element $\alpha \in \overline{K}$ is *separable over* K if its minimal polynomial $m_{\alpha,K} \in K[X]$ is separable.
3. An algebraic extension L/K is *separable* if every element of L is separable over K.

For a general, not necessarily algebraic extension, separability means that every element of L that is algebraic over K is separable over K.

Remark (Separable Does Not Mean Split). A polynomial may be separable without having any root in its ground field. For example, $X^2 - 2$ is separable over $\mathbb{Q}$ but does not split over $\mathbb{Q}$. Separability asks whether the roots are *distinct* in an algebraic closure, whereas splitting asks whether they already lie in the chosen field.

Proposition 2.1.1 (The Derivative Criterion). Let $f \in K[X]$ be nonconstant and let f' be its formal derivative. The following conditions are equivalent.

1. The polynomial f is separable.
2. The polynomials f and f' have no common root in $\overline{K}$.
3. We have

$$\gcd(f, f') = 1$$

If f is irreducible, these are also equivalent to $f' \neq 0$.

Proof. Over $\overline{K}$, write

$$f(X) = c \prod_{i=1}^{r} (X - \alpha_i)^{e_i}$$

with distinct α_i. The factor $X - \alpha_i$ divides both f and f' if and only if $e_i \geq 2$. Thus f has no repeated root precisely when $\gcd(f, f') = 1$. If f is irreducible, its greatest common divisor with f' is either 1 or f; the second possibility occurs exactly when $f' = 0$. □

Corollary 2.1.1 (The Role of the Characteristic). If $\mathrm{char}(K) = 0$, every irreducible polynomial over K is separable. If $\mathrm{char}(K) = p > 0$, then

$$f' = 0 \Leftrightarrow f(X) = g(X^p)$$

for some $g \in K[X]$. Hence inseparability can occur only in positive characteristic.

Proof. In characteristic zero, a nonconstant polynomial has nonzero derivative. In characteristic p, the derivative of X^n vanishes exactly when p divides n. Therefore $f' = 0$ precisely when every exponent occurring in f is divisible by p. □

2.1.2 Perfect Fields

Definition 2.1.2 (Perfect Field). A field K is *perfect* if every algebraic extension of K is separable. Equivalently, every irreducible polynomial in $K[X]$ is separable.

Theorem 2.1.1 (Characterization of Perfect Fields). A field K is perfect if and only if either

1. $\mathrm{char}(K) = 0$; or
2. $\mathrm{char}(K) = p > 0$ and the Frobenius map

$$\mathrm{Frob}_K : K \to K, \quad a \mapsto a^p$$

   is surjective.

In particular, every field of characteristic zero and every finite field is perfect.

Proof. The characteristic-zero statement follows from the derivative criterion. Suppose $\mathrm{char}(K) = p$ and Frobenius is surjective. If an irreducible polynomial f had $f' = 0$, then

$$f(X) = \sum_i a_i X^{pi} = \left(\sum_i b_i X^i \right)^p$$

after choosing $b_i \in K$ with $b_i^p = a_i$. This contradicts irreducibility unless f is linear. Conversely, if Frobenius is not surjective, choose $a \in K$ that is not a p-th power. Then $X^p - a$ is irreducible and has derivative zero, so K is not perfect. Finally, Frobenius is injective on every field and is therefore bijective when the field is finite. □

Example (A Basic Inseparable Extension). Let $K = \mathbb{F}_p(t)$ and let

$$L = K(u), \quad u^p = t$$

The element t is not a p-th power in K, so $X^p - t$ is irreducible. However,

$$X^p - t = (X - u)^p$$

over $\overline{K}$. Thus L/K has degree p but is not separable. It is a *purely inseparable* extension: every element of L has some p-power lying in K.

2.1.3 Separable Degree

Definition 2.1.3 (Separable and Inseparable Degrees). Let L/K be a finite extension and fix an algebraic closure $\overline{K}$. Its *separable degree* is

$$[L : K]_{\mathrm{s}} \coloneqq \left|\mathrm{Hom}_K\left(L, \overline{K}\right)\right|$$

and its *inseparable degree* is

$$[L : K]_{\mathrm{i}} \coloneqq [L : K]/[L : K]_{\mathrm{s}}$$

Thus

$$[L : K] = [L : K]_{\mathrm{s}} \cdot [L : K]_{\mathrm{i}}$$

Proposition 2.1.2 (Basic Properties of the Separable Degree). Let L_{s} be the set of elements of L that are separable over K. Then L_{s} is the largest intermediate field separable over K, the extension L/L_{s} is purely inseparable, and

$$[L : K]_{\mathrm{s}} = [L_{\mathrm{s}} : K], \quad [L : K]_{\mathrm{i}} = [L : L_{\mathrm{s}}]$$

Consequently,

1. L/K is separable exactly when $[L : K]_{\mathrm{s}} = [L : K]$;
2. L/K is purely inseparable exactly when $[L : K]_{\mathrm{s}} = 1$;
3. in characteristic $p > 0$, the inseparable degree is a power of p;
4. for every finite tower $K \subseteq E \subseteq L$, both degrees are multiplicative:

$$[L : K]_{\mathrm{s}} = [L : E]_{\mathrm{s}} \cdot [E : K]_{\mathrm{s}}, \quad [L : K]_{\mathrm{i}} = [L : E]_{\mathrm{i}} \cdot [E : K]_{\mathrm{i}}$$

Remark (Distinct Geometric Points versus Their Lengths). The algebra $L \otimes_K \overline{K}$ has exactly $[L : K]_{\mathrm{s}}$ geometric points, and every point has scheme-theoretic length $[L : K]_{\mathrm{i}}$. Thus the ordinary degree counts points with multiplicity:

$$[L : K] = \underbrace{[L : K]_{\mathrm{s}}}_{\text{distinct points}} \cdot \underbrace{[L : K]_{\mathrm{i}}}_{\text{length of each point}}$$

2.1.4 Finite Separable Extensions

Theorem 2.1.2 (Primitive Element Theorem). Every finite separable extension is simple: if L/K is finite separable, then $L = K(\alpha)$ for some $\alpha \in L$.

Theorem 2.1.3 (Equivalent Forms of Separability). Let L/K be a finite extension of degree n, and let $\overline{K}$ be an algebraic closure of K. The following conditions are equivalent.

1. The extension L/K is separable.
2. There are exactly n distinct K-embeddings
$$\sigma : L \to \overline{K}$$
3. After extending scalars to $\overline{K}$, there is an isomorphism of $\overline{K}$-algebras
$$L \otimes_K \overline{K} \simeq \prod_{\sigma : L \to \overline{K}} \overline{K} \simeq \overline{K}^n$$

Proof. Suppose first that L/K is separable. By the primitive element theorem, $L = K(\alpha)$ for some α, and the minimal polynomial $m_{\alpha,K}$ has n distinct roots $\alpha_1, \ldots, \alpha_n$ in $\overline{K}$. Each embedding is determined by the image of α, which may be any one of these roots. Hence there are exactly n embeddings.

Moreover,
$$L \otimes_K \overline{K} \simeq \overline{K}[X]/\left(m_{\alpha,K}\right)$$
Since the linear factors $X - \alpha_i$ are pairwise coprime, the Chinese remainder theorem gives
$$\overline{K}[X]/\left(m_{\alpha,K}\right) \simeq \prod_{i=1}^{n} \overline{K}[X]/(X - \alpha_i) \simeq \overline{K}^n$$
Conversely, the number of K-embeddings of any finite extension into $\overline{K}$ is at most its degree, with equality exactly when all minimal polynomials involved have distinct roots. If L/K is not separable, this number is strictly less than n, and the scalar extension acquires nilpotents. This proves all equivalences. □

Remark (The Geometric Base-Change Criterion). Put $X = \operatorname{Spec} K$ and $Y = \operatorname{Spec} L$. For every field extension K'/K, base change gives

$$Y_{K'} = Y \times_X \operatorname{Spec}(K') = \operatorname{Spec}(L \otimes_K K')$$

> "Separability is reducedness that survives arbitrary base change."

The extension L/K is separable if and only if Y is *geometrically reduced* over K. Concretely, the following conditions are equivalent:

1. L/K is separable;
2. $L \otimes_K K'$ is reduced for every field extension K'/K;
3. $L \otimes_K \overline{K}$ is reduced;
4. after base change to $\overline{K}$, the scheme Y becomes a disjoint union of $n = [L : K]$ geometric points:

$$Y_{\overline{K}} \simeq \sqcup_{\sigma: L \to \overline{K}} \operatorname{Spec}\left(\overline{K}\right)$$

Indeed, a separable minimal polynomial becomes a product of distinct linear factors over $\overline{K}$, so its coordinate algebra becomes $\overline{K}^n$. A repeated root instead produces a nonzero nilpotent, and the geometric fiber is nonreduced. Thus separability is exactly the assertion that distinct points remain distinct after arbitrary base change.

If K^{sep} denotes a separable closure, there is also the splitting criterion

$$L/K \text{ is separable} \Leftrightarrow L \otimes_K K^{\mathrm{sep}} \simeq \left(K^{\mathrm{sep}}\right)^n$$

One should not weaken this to saying merely that $L \otimes_K K^{\mathrm{sep}}$ is reduced. For example, a purely inseparable extension can remain a field after base change to K^{sep}. Passing to an algebraic closure, or requiring reducedness after *every* field extension, is what detects inseparability.

2.1.5 Stability Properties

Proposition 2.1.3 (Separability in Towers and under Base Change). Let $K \subseteq E \subseteq L$ be algebraic field extensions.

1. If L/K is separable, then both E/K and L/E are separable.
2. If E/K and L/E are separable, then L/K is separable.
3. The compositum of two separable extensions of K is separable over K.
4. If L/K is finite separable and K'/K is any field extension, then

$$L \otimes_K K' \simeq \prod_{i=1}^{r} L_i$$

   for finite separable field extensions L_i/K'. Thus separability is preserved by arbitrary extension of the ground field.

Proof. For the first assertion, elements of E are already separable over K, and the minimal polynomial over E of an element of L divides its separable minimal polynomial over K. Transitivity follows by counting embeddings in the two stages of the tower. If L_1/K and L_2/K are separable, then L_1L_2/L_1 is separable because minimal polynomials over L_1 divide the corresponding separable polynomials over K. Transitivity now shows that L_1L_2/K is separable.

For base change, write a finite separable extension as $L = K(\alpha)$. The polynomial $m_{\alpha,K}$ remains square-free over K' and factors there as a product of distinct irreducible polynomials $f_1 \dots f_r$. Consequently,

$$L \otimes_K K' \simeq K'[X]/\big(m_{\alpha,K}\big) \simeq \prod_{i=1}^{r} K'[X]/(f_i)$$

and each factor is a finite separable extension of K'. □

Remark (Why Separability Matters). Separability connects field theory, arithmetic, and geometry.

1. *Distinct conjugates.* A degree-n separable extension has the expected n embeddings into an algebraic closure. These embeddings make trace and norm computable as sums and products of conjugates.
2. *Nondegenerate arithmetic.* The trace pairing and discriminant detect linear independence and control integral bases, ramification, and the geometry of numbers. Number fields are automatically separable because they have characteristic zero.
3. *Reduced geometric fibers.* A finite separable field extension becomes a product of copies of an algebraic closure after base change. There are no nilpotent directions, so the corresponding geometric points are distinct.
4. *Galois theory.* A finite extension is Galois precisely when it is both normal and separable. Separability supplies the full set of distinct embeddings; normality ensures that their images remain inside the field.
5. *Étale algebras.* Finite étale algebras over a field are exactly finite products of finite separable field extensions. Thus separability is the field-theoretic model for the word *étale*.

Example (Number Fields and Finite Fields). If K is a number field of degree n over $\mathbb{Q}$, then $K/\mathbb{Q}$ is separable and has exactly n embeddings into $\mathbb{C}$. If r_1 of them are real and $2r_2$ are nonreal, paired by complex conjugation, then

$$n = r_1 + 2r_2$$

This is the starting point of the Minkowski embedding.

Likewise, every extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is separable. Its embeddings into an algebraic closure are generated by Frobenius:

$$x \mapsto x^q$$

Example (What Inseparability Looks Like Geometrically). Return to $K = \mathbb{F}_p(t)$ and $L = K(u)$ with $u^p = t$. There is only one K-embedding $L \to \overline{K}$, although $[L : K] = p$. After base change,

$$L \otimes_K \overline{K} \simeq \overline{K}[X]/\big((X-u)^p\big) \simeq \overline{K}[\varepsilon]/(\varepsilon^p)$$

The class of ε is a nonzero nilpotent. Thus the single geometric point occurs with multiplicity p instead of splitting into p distinct points. This is exactly the behavior that separability, and later étaleness, rules out.

2.2 Étale Algebras and Étale Morphisms

The word *étale* means that a map has no ramification and no infinitesimal thickening. It is the algebro-geometric analogue of a local diffeomorphism, or of a covering map when the morphism is finite. In short, étale morphisms are flat families of discrete reduced points.

2.2.1 Étale Algebras over a Field

Definition 2.2.1 (Finite Étale Algebra over a Field). Let K be a field. A *finite étale K-algebra* is a finite-dimensional commutative K-algebra of the form

$$B = L_1 \times \ldots \times L_r$$

where each L_i/K is a finite separable field extension. Its degree is $\deg_K(B) = \dim_K(B)$. In particular, a field L is a finite étale K-algebra precisely when L/K is finite separable.

Theorem 2.2.1 (Characterizations over a Field). Let B be a finite-dimensional commutative K-algebra, and put $n = \dim_K(B)$. The following conditions are equivalent.

1. The algebra B is finite étale over K.
2. The algebra B is *geometrically reduced*: for every field extension K'/K, the algebra $B \otimes_K K'$ is reduced.
3. The algebra obtained over an algebraic closure splits as
$$B \otimes_K \overline{K} \simeq \overline{K}^n$$
4. The module of Kähler differentials vanishes:
$$\Omega_{B/K} = 0$$

Proof. A finite-dimensional commutative K-algebra is Artinian. If it is reduced, it is a finite product of fields. Geometric reducedness therefore gives

$$B = L_1 \times \ldots \times L_r$$

and each L_i/K is separable by the geometric base-change criterion. Conversely, a product of finite separable extensions becomes $\overline{K}^n$ over $\overline{K}$ and remains reduced after every base change.

For a finite field extension L/K, the equality $\Omega_{L/K} = 0$ is equivalent to separability; this can be checked from a presentation $L = K[\alpha]$, since

$$\Omega_{L/K} \simeq L \,\mathrm{d}\alpha / \left(m_{(\alpha,K)}{}'(\alpha) \,\mathrm{d}\alpha \right)$$

Products give the assertion for B. □

Example (A Polynomial Algebra). For a monic polynomial $f \in K[X]$, the algebra

$$B = K[X]/(f)$$

is finite étale over K if and only if f is separable, equivalently

$$\gcd(f, f') = 1$$

In contrast, the dual numbers $K[\varepsilon]/\left(\varepsilon^2\right)$ are not étale: they contain a nonzero nilpotent and have $\Omega_{B/K} \neq 0$.

Proposition 2.2.1 (Finite Étale Algebras as Galois Sets). Let

$$G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$$

be the absolute Galois group. The contravariant functor

$$B \mapsto \mathrm{Hom}_{\mathbf{Alg}_K}(B, K^{\mathrm{sep}})$$

gives an anti-equivalence between finite étale K-algebras and finite sets with a continuous action of G_K. Under this correspondence, field extensions correspond to transitive G_K-sets.

Remark. This is the first appearance of a central principle: finite étale geometry over a field is another language for finite Galois actions. Passing from an algebra to its spectrum reverses arrows, which explains the anti-equivalence.

2.2.2 Étale Morphisms of Schemes

Definition 2.2.2 (Étale Morphism). A morphism of schemes $f : X \to Y$ is *étale* if it is locally of finite presentation, flat, and unramified. For a morphism locally of finite presentation, the unramified condition is equivalent to

$$\Omega_{X/Y} = 0$$

where $\Omega_{X/Y}$ is the sheaf of relative Kähler differentials.

Flatness says that the fibers vary without algebraic jumps, while $\Omega_{X/Y} = 0$ says that there are no nonzero tangent directions along the fibers. Together they describe a family of discrete reduced points.

Theorem 2.2.2 (Equivalent Characterizations of Étale Morphisms). For a morphism $f : X \to Y$, the following conditions are equivalent.

1. The morphism f is étale.
2. The morphism f is smooth of relative dimension zero.

3. The morphism f is locally of finite presentation and formally étale: maps into X lift uniquely across nilpotent thickenings over Y.
4. Locally on X and Y, the morphism has a standard étale presentation

$$B = (A[T_1, ..., T_r]/(F_1, ..., F_r))_g$$

for which the Jacobian determinant

$$\det\left(\left(\frac{\partial F_i}{\partial T_j}\right)_{1 \leq i,j \leq r}\right)$$

is a unit in B.
5. The morphism is flat and locally of finite presentation, and every geometric fiber is a zero-dimensional reduced scheme.

Proposition 2.2.2 (Stability and Locality). Étale morphisms satisfy the following fundamental properties.

1. A composition of étale morphisms is étale.
2. Étaleness is preserved by arbitrary base change. If $X \to Y$ is étale and $Y' \to Y$ is any morphism, then

$$X \times_Y Y' \to Y'$$

is étale.
3. Open immersions are étale, and every étale morphism is an open map.
4. Finite étale morphisms remain finite étale after arbitrary base change.

Proof sketch. Local finite presentation and flatness are preserved by composition and base change. Relative differentials satisfy the corresponding base-change and transitivity formulas, so their vanishing is preserved as well. The openness statement follows from the general theorem that a flat morphism locally of finite presentation is open. □

Remark (The Infinitesimal Lifting Property). On affine schemes, formal étaleness says that for every A-algebra C and every nilpotent ideal $N \subseteq C$, reduction induces a bijection

$$\mathrm{Hom}_{\mathbf{Alg}_A}(B, C) \to \mathrm{Hom}_{\mathbf{Alg}_A}(B, C/N)$$

Existence means that infinitesimal deformations lift; uniqueness means that there are no relative tangent directions. An open immersion is étale, which also shows that an étale morphism need not be finite.

Remark (Why Flatness Is Necessary). Vanishing differentials alone do not imply étaleness. For example,

$$\Omega_{\mathbb{F}_p/\mathbb{Z}} = 0$$

but $\mathbb{Z} \to \mathbb{F}_p$ is not flat. Geometrically, the closed point $\mathrm{Spec}(\mathbb{F}_p) \to \mathrm{Spec}(\mathbb{Z})$ does not form a locally constant family over the base.

2.2.3 Finite Étale Algebras over a Ring

Definition 2.2.3 (Finite Étale Algebra). Let A be a commutative ring. A commutative A-algebra B is *finite étale* if the associated morphism

$$\mathrm{Spec}\ B \to \mathrm{Spec}\ A$$

is finite and étale.

Theorem 2.2.3 (Algebraic and Fiberwise Characterizations). Let B be a finite A-algebra. The following conditions are equivalent.

1. The algebra B is finite étale over A.
2. The A-module B is finite projective and
$$\Omega_{B/A} = 0$$
3. The A-module B is finite projective and, for every prime ideal $\mathfrak{p} \subseteq A$, the fiber
$$B \otimes_A \kappa(\mathfrak{p})$$
is a finite product of finite separable extensions of $\kappa(\mathfrak{p})$.

Proof sketch. Finite projectivity is the algebraic form of finite flatness. The vanishing of $\Omega_{B/A}$ is the unramified condition. After base change to a residue field, the field-theoretic theorem identifies the fibers with finite products of separable extensions. Conversely, the fiberwise condition and finite projectivity imply that the morphism is étale. □

Example (Ramification Appears Where Étaleness Fails). The algebra $\mathbb{Z}[i] = \mathbb{Z}[T]/(T^2 + 1)$ is not étale over $\mathbb{Z}$ at the prime 2, because the derivative $2T$ vanishes in the fiber over $\mathbb{F}_2$. After inverting 2, however,

$$\mathbb{Z}\left[\frac{1}{2}, i\right] \text{ is finite étale over } \mathbb{Z}\left[\frac{1}{2}\right]$$

2.2.4 Arithmetic Outlook

Remark (Étale Geometry in Number Theory). Étale morphisms organize several central ideas in arithmetic.

1. *Unramified primes.* Let L/K be a finite extension of number fields. The finite morphism
$$\mathrm{Spec}(\mathcal{O}_L) \to \mathrm{Spec}(\mathcal{O}_K)$$
is étale away from the relative discriminant ideal $\mathfrak{d}_{L/K}$. At a prime $\mathfrak{p}$ outside the discriminant, the fiber is a product of finite separable residue-field extensions:

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \kappa(\mathfrak{p}) \simeq \prod_{\mathfrak{q}|\mathfrak{p}} \kappa(\mathfrak{q})$$

Thus étaleness is the scheme-theoretic form of being unramified.

2. *Frobenius.* In a finite Galois extension, the action of Frobenius on the geometric fiber above an unramified prime records the residue degrees and produces the Frobenius conjugacy class used in decomposition laws, Chebotarev's theorem, and Euler factors of L-functions.
3. *The étale fundamental group.* For a geometric point $\overline{\eta}$ of Spec K,

$$\pi_1^{\text{étale}}(\operatorname{Spec} K, \overline{\eta}) \simeq \operatorname{Gal}(K^{\text{sep}}/K)$$

More generally, finite connected étale covers of a normal connected arithmetic scheme correspond to finite extensions unramified along that scheme. If $\mathcal{O}_{K,S}$ denotes the ring of S-integers, passing to $\operatorname{Spec}(\mathcal{O}_{K,S})$ permits ramification precisely at the primes in S.

4. *Étale cohomology.* The groups $H^i_{\text{étale}}(X_{\overline{K}}, \mathbb{Q}_\ell)$ carry continuous Galois actions. They connect geometry with Galois representations, Frobenius traces, point counts, zeta functions, and arithmetic L-functions.
5. *Local-to-global arithmetic.* Étale localization is fine enough to split finite separable phenomena but coarse enough to retain arithmetic information. It is therefore the natural topology for descent, torsors, and cohomological formulations of class field theory. In arithmetic geometry, étale means locally unramified, infinitesimally rigid, and stable under base change.

2.3 Trace and Norm

Trace and norm turn an element upstairs into an element downstairs. They are defined by linear algebra, computed from conjugates, and interpreted geometrically by summing or multiplying values along a finite fiber.

> “Trace is additive integration along a finite fiber; norm is multiplicative integration along that fiber.”

2.3.1 Linear-Algebraic Definitions

Definition 2.3.1 (Field Trace, Norm, and Characteristic Polynomial). Let L/K be a finite extension of degree n, not necessarily separable. For $x \in L$, multiplication by x is the K-linear map

$$m_x : L \to L, \quad y \mapsto xy$$

The *trace* and *norm* of x are

$$\operatorname{Tr}_{L/K}(x) \coloneqq \operatorname{tr}(m_x)$$

$$\operatorname{N}_{L/K}(x) \coloneqq \det(m_x)$$

Its characteristic polynomial over K is

$$\chi_{x,L/K}(T) \coloneqq \det(T\mathrm{id}_L - m_x)$$

Thus

$$\chi_{x,L/K}(T) = T^n - \mathrm{Tr}_{L/K}(x)T^{n-1} + \ldots + (-1)^n \mathrm{N}_{L/K}(x)$$

These definitions do not depend on a choice of basis. In practice, choose a K-basis $e_1, \ldots, e_n$ of L, form the multiplication matrix M_x from

$$xe_j = \sum_{i=1}^{n} (M_x)_{ij} e_i$$

and compute its trace and determinant.

Proposition 2.3.1 (Basic Identities). For $x, y \in L$ and $a \in K$,

1. trace is K-linear:

$$\mathrm{Tr}_{L/K}(x+y) = \mathrm{Tr}_{L/K}(x) + \mathrm{Tr}_{L/K}(y)$$

$$\mathrm{Tr}_{L/K}(ax) = a\mathrm{Tr}_{L/K}(x)$$

2. norm is multiplicative:

$$\mathrm{N}_{L/K}(xy) = \mathrm{N}_{L/K}(x)\mathrm{N}_{L/K}(y)$$

3. scalars satisfy

$$\mathrm{Tr}_{L/K}(a) = na, \quad \mathrm{N}_{L/K}(a) = a^n$$

4. for $x \neq 0$,

$$\mathrm{N}_{L/K}\left(x^{-1}\right) = \mathrm{N}_{L/K}(x)^{-1}$$

5. every K-automorphism τ of L preserves trace and norm.

Proof. The identities

$$m_{x+y} = m_x + m_y, \quad m_{ax} = am_x, \quad m_{xy} = m_x \circ m_y$$

reduce the first three assertions to the linearity of matrix trace and the multiplicativity of determinant. For $a \in K$, multiplication by a is the scalar matrix aI_n. Finally,

$$m_{\tau(x)} = \tau \circ m_x \circ \tau^{-1}$$

so the two multiplication maps have the same characteristic polynomial. □

Remark (Trace Is Additive, Norm Is Multiplicative). Norm is generally not additive, and trace is generally not multiplicative. The two constructions are parallel only after replacing addition on one side by multiplication on the other. Moreover,

$$\mathrm{N}_{L/K}(x) = 0 \Leftrightarrow x = 0$$

because L is a field and m_x is invertible exactly when $x \neq 0$.

Theorem 2.3.1 (Transitivity in a Tower). For a tower of finite extensions $K \subseteq E \subseteq L$,

$$\mathrm{Tr}_{L/K} = \mathrm{Tr}_{E/K} \circ \mathrm{Tr}_{L/E}$$

$$\mathrm{N}_{L/K} = \mathrm{N}_{E/K} \circ \mathrm{N}_{L/E}$$

Proof sketch. Choose an E-basis of L and a K-basis of E. Multiplication by an element of L first gives a matrix over E; replacing each entry by its multiplication matrix over K produces the full matrix over K. Taking traces and determinants gives the two formulas. □

2.3.2 Computation from Minimal Polynomials and Conjugates

Proposition 2.3.2 (The Minimal-Polynomial Formula). Let $x \in L$, put $E = K(x)$, and write

$$m_{x,K}(T) = T^d + c_{d-1}T^{d-1} + \ldots + c_0$$

for the minimal polynomial of x over K. If $r = [L : E]$, then

$$\chi_{x,L/K}(T) = m_{x,K}(T)^r$$

Consequently,

$$\mathrm{Tr}_{L/K}(x) = -rc_{d-1}$$

$$\mathrm{N}_{L/K}(x) = \left((-1)^d c_0\right)^r$$

In particular, when $L = K(x)$,

$$\mathrm{Tr}_{L/K}(x) = -c_{n-1}, \quad \mathrm{N}_{L/K}(x) = (-1)^n c_0$$

Proof. As an E-vector space, L has dimension r, and multiplication by x acts as the scalar x on each copy of E. Therefore its characteristic polynomial over K is the r-th power of the characteristic polynomial of multiplication by x on E. Since $E = K(x)$, the latter is precisely $m_{x,K}$. Comparing the coefficient of T^{n-1} and the constant term gives the formulas. □

Theorem 2.3.2 (The Conjugate Formula). Suppose that L/K is separable of degree n. If $\sigma_1, \ldots, \sigma_n : L \to \overline{K}$ are its K-embeddings, then

$$\chi_{x,L/K}(T) = \prod_{i=1}^{n}(T - \sigma_i(x))$$

and hence

$$\mathrm{Tr}_{L/K}(x) = \sum_{i=1}^{n} \sigma_i(x)$$

$$\mathrm{N}_{L/K}(x) = \prod_{i=1}^{n} \sigma_i(x)$$

Proof. After base change to $\overline{K}$, the étale algebra $L \otimes_K \overline{K}$ becomes $\overline{K}^n$. Multiplication by x becomes the diagonal operator with entries $\sigma_1(x), \ldots, \sigma_n(x)$. The formulas are its characteristic polynomial, trace, and determinant. □

Corollary 2.3.1 (Trace Pairing and Discriminant). For a finite extension L/K, the trace pairing

$$L \times L \to K, \quad (x, y) \mapsto \mathrm{Tr}_{L/K}(xy)$$

is nondegenerate if and only if L/K is separable. Hence, for any K-basis $b_1, \ldots, b_n$ of L, the discriminant

$$\mathrm{disc}(b_1, \ldots, b_n) := \det\left(\mathrm{Tr}_{L/K}(b_i b_j)\right)_{1 \leq i,j \leq n}$$

is nonzero exactly when L/K is separable.

Proof. In the separable case, base change to $\overline{K}$ identifies the pairing with the ordinary dot product on $\overline{K}^n$. In the inseparable case, $L \otimes_K \overline{K}$ has a nonzero nilpotent z. For every b, the operator m_{zb} is nilpotent, so its trace is zero. Thus z lies in the radical of the base-changed trace pairing, which must be degenerate. □

Example (Quadratic Extensions). Assume $\mathrm{char}(K) \neq 2$ and let $L = K\left(\sqrt{d}\right)$, where d is not a square. The two embeddings send $\sqrt{d}$ to $\pm\sqrt{d}$. Hence

$$\mathrm{Tr}_{L/K}\left(a + b\sqrt{d}\right) = 2a$$

$$\mathrm{N}_{L/K}\left(a + b\sqrt{d}\right) = a^2 - db^2$$

In particular, for $L = \mathbb{Q}\left(\sqrt{-5}\right)$,

$$\mathrm{N}_{L/\mathbb{Q}}\left(a + b\sqrt{-5}\right) = a^2 + 5b^2$$

This is the norm calculation used earlier to show that $\left(2, 1 + \sqrt{-5}\right)$ is not principal.

Example (Finite Fields). The embeddings of $\mathbb{F}_{q^n}$ over $\mathbb{F}_q$ are the powers of Frobenius. Thus

$$\mathrm{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x) = x + x^q + \ldots + x^{q^{n-1}}$$

$$\mathrm{N}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(x) = x^{1+q+\ldots+q^{n-1}} = x^{\frac{q^n-1}{q-1}}$$

for $x \neq 0$. The norm is a surjective homomorphism $\mathbb{F}_{q^n}^\times \to \mathbb{F}_q^\times$.

Example (The Purely Inseparable Contrast). Let $K = \mathbb{F}_p(t)$ and $L = K(u)$ with $u^p = t$. For every $x \in L$,

$$\chi_{x,L/K}(T) = (T - x)^p = T^p - x^p$$

Thus

$$\mathrm{Tr}_{L/K}(x) = 0, \quad \mathrm{N}_{L/K}(x) = x^p$$

The trace pairing is identically zero. This illustrates why separability is exactly the condition needed for a nondegenerate trace pairing.

2.3.3 Finite Locally Free Algebras

Definition 2.3.2 (Trace and Norm over a Ring). Let B be a finite locally free A-algebra of constant rank n. For $b \in B$, multiplication defines an A-linear endomorphism

$$m_b : B \to B$$

Its trace and determinant define

$$\mathrm{Tr}_{B/A}(b) := \mathrm{tr}(m_b) \in A$$

$$\mathrm{N}_{B/A}(b) := \det(m_b) \in A$$

These definitions make sense for finite projective modules because trace and determinant may be checked after localization, where B is free.

Proposition 2.3.3 (Formulas over a Ring). Let B be finite locally free of rank n over A.

1. The trace map $B \to A$ is A-linear, while norm gives a multiplicative map $B \to A$.
2. For $a \in A$,

$$\mathrm{Tr}_{B/A}(a) = na, \quad \mathrm{N}_{B/A}(a) = a^n$$

3. An element $b \in B$ is a unit if and only if $\mathrm{N}_{B/A}(b)$ is a unit in A.
4. For finite locally free algebras B_1 and B_2,

$$\mathrm{Tr}_{(B_1 \times B_2)/A}(b_1, b_2) = \mathrm{Tr}_{B_1/A}(b_1) + \mathrm{Tr}_{B_2/A}(b_2)$$

$$\mathrm{N}_{(B_1 \times B_2)/A}(b_1, b_2) = \mathrm{N}_{B_1/A}(b_1)\mathrm{N}_{B_2/A}(b_2)$$

5. Trace, norm, and characteristic polynomial commute with arbitrary base change. For an A-algebra A',

$$\mathrm{Tr}_{(B \otimes_A A')/A'}(b \otimes 1) = \mathrm{Tr}_{B/A}(b) \otimes 1$$

$$\mathrm{N}_{(B \otimes_A A')/A'}(b \otimes 1) = \mathrm{N}_{B/A}(b) \otimes 1$$

Proof sketch. All assertions are local on Spec A, so one may choose a basis of B. They then follow from the corresponding matrix identities. For the unit criterion, m_b is invertible exactly when its determinant is a unit; if m_b is invertible, applying its inverse to 1 produces an inverse of b. □

Theorem 2.3.3 (The Trace Criterion for Finite Étale Algebras). Let B be finite locally free over A. Then B is finite étale over A if and only if the trace pairing is perfect; equivalently, the map

$$B \to \mathrm{Hom}_A(B, A), \quad b \mapsto \big(c \mapsto \mathrm{Tr}_{B/A}(bc)\big)$$

is an isomorphism.

Proof sketch. The displayed map is a map between finite locally free modules of the same rank, so it is an isomorphism exactly when it is an isomorphism on every residue-field fiber. Over a field, a finite commutative algebra becomes a product of local Artin algebras after passing to an algebraic closure. Its trace pairing is perfect exactly when every local factor has length one, that is, exactly when the algebra is a product of finite separable fields. The fiberwise characterization of finite étale algebras now gives the result. □

2.3.4 Geometric Meaning along Finite Fibers

Let $f : X \to S$ be a finite locally free morphism of constant degree n. The sheaf $f_*\mathcal{O}_X$ is a rank-n vector bundle on S. Multiplication by a regular function b on X is an endomorphism of this vector bundle, and its trace and determinant are regular functions on S:

$$\mathrm{Tr}_f(b), \mathrm{N}_f(b) \in \Gamma(S, \mathcal{O}_S)$$

Theorem 2.3.4 (Fiberwise Sum and Product). Let $\bar{s} \to S$ be a geometric point. Write the finite geometric fiber as a collection of points $x \in X_{\bar{s}}$, and let ℓ_x be the length of the local Artin ring at x. Then

$$\mathrm{Tr}_f(b)(\bar{s}) = \sum_{x \in X_{\bar{s}}} \ell_x b(x)$$

$$\mathrm{N}_f(b)(\bar{s}) = \prod_{x \in X_{\bar{s}}} b(x)^{\ell_x}$$

If f is finite étale, every $\ell_x = 1$, so its geometric fibers consist of n distinct points and

$$\mathrm{Tr}_f(b)(\bar{s}) = \sum_{x \in X_{\bar{s}}} b(x), \quad \mathrm{N}_f(b)(\bar{s}) = \prod_{x \in X_{\bar{s}}} b(x)$$

Proof. Trace and norm commute with base change, so it is enough to work over the algebraically closed residue field of $\bar{s}$. The coordinate algebra of the fiber is a product of local Artin algebras. On the factor supported at x, write $b = b(x) + v$, where v is nilpotent. Multiplication by b is the scalar operator $b(x)$ plus a nilpotent operator. Its trace is $\ell_x b(x)$ and its determinant is $b(x)^{\ell_x}$. Summing traces and multiplying determinants over the factors gives the formulas. □

Remark (What the Multiplicities Remember). For a finite étale map, trace and norm literally sum and multiply over the distinct points of a fiber. For a ramified or nonreduced fiber, the same point is counted with its scheme-theoretic length. Thus the linear-algebraic multiplicity in the characteristic polynomial is exactly the geometric multiplicity of the fiber.

Proposition 2.3.4 (Norm and Pushforward of Divisors). Let $f : X \to Y$ be a finite dominant morphism of normal integral curves, with function-field extension $K(X)/K(Y)$. For $g \in K(X)^\times$,

$$\mathrm{div}_Y\big(\mathrm{N}_{K(X)/K(Y)}(g)\big) = f_*\mathrm{div}_X(g)$$

Here the pushforward of a closed point is

$$f_*[x] = [\kappa(x) : \kappa(f(x))][f(x)]$$

Equivalently, for a closed point $y \in Y$,

$$v_y\big(\mathrm{N}_{K(X)/K(Y)}(g)\big) = \sum_{x \in X, f(x)=y} [\kappa(x) : \kappa(y)] v_x(g)$$

Remark. This identity says that the norm pushes zeros and poles down the map. It also extends from rational functions to a norm homomorphism on line bundles,

$$\mathrm{Nm}_f : \mathrm{Pic}(X) \to \mathrm{Pic}(Y)$$

whose effect on divisor classes is induced by f_*. Trace is the additive pushforward on functions; norm is its multiplicative companion.

2.3.5 Arithmetic Consequences

Proposition 2.3.5 (Integrality and Principal Ideals). Let L/K be a finite extension of number fields. If $\alpha \in \mathcal{O}_L$, then

$$\mathrm{Tr}_{L/K}(\alpha), \mathrm{N}_{L/K}(\alpha) \in \mathcal{O}_K$$

For $K = \mathbb{Q}$ and $0 \neq \alpha \in \mathcal{O}_L$,

$$|\mathcal{O}_L/\alpha\mathcal{O}_L| = |\mathrm{N}_{L/\mathbb{Q}}(\alpha)|$$

Thus the absolute norm of the principal ideal (α) is the absolute value of the field norm of α.

Proof. The conjugates of an algebraic integer are algebraic integers, so their sum and product are integral over $\mathcal{O}_K$. They also lie in K; since $\mathcal{O}_K$ is integrally closed, they lie in $\mathcal{O}_K$. For the second formula, multiplication by α is an injective endomorphism of the lattice $\mathcal{O}_L$. The index of its image is the absolute value of its determinant, which is $|\mathrm{N}_{L/\mathbb{Q}}(\alpha)|$. □

3 Dedekind Extensions

The local structure of a Dedekind domain is stable under finite separable extensions, but proving this requires a global finiteness argument. Dual modules and lattices provide exactly that argument: the trace pairing places the integral closure between two finite lattices. Once the new ring is known to be Dedekind, unique factorization of ideals describes how every prime splits upstairs.

3.1 Dual Modules and Pairings

Duality reverses arrows and exchanges a rank-one object with its inverse. This is already familiar for fractional ideals, and it is the same operation that appears geometrically in the Picard group.

Definition 3.1.1 (Dual Module and Dual Homomorphism). Let A be a commutative ring and let M be an A-module. The *dual module* of M is

$$M^\vee := \mathrm{Hom}_A(M, A)$$

For an A-linear map $\varphi : M \to N$, its *dual map* is

$$\varphi^\vee : N^\vee \to M^\vee, \quad g \mapsto g \circ \varphi$$

Thus duality is a contravariant functor

$$(-)^\vee : \mathrm{Mod}_A^{\mathrm{op}} \to \mathrm{Mod}_A$$

The reversal of arrows follows from

$$(\psi \circ \varphi)^\vee = \varphi^\vee \circ \psi^\vee$$

There is also a canonical evaluation map

$$\mathrm{ev}_M : M \to \left(M^\vee\right)^\vee, \quad m \mapsto (f \mapsto f(m))$$

An A-module is *reflexive* if this map is an isomorphism.

Proposition 3.1.1 (Basic Properties of Dual Modules). Let M and N be A-modules.

1. Finite direct sums commute with duality:

$$(M \oplus N)^\vee \simeq M^\vee \oplus N^\vee$$

2. If M is finite projective, then $M^\vee$ is finite projective and the evaluation map is an isomorphism:

$$M \simeq \left(M^\vee\right)^\vee$$

3. If M is free with basis $e_1, \dots, e_n$, then $M^\vee$ has the unique dual basis $e_1^\vee, \dots, e_n^\vee$ satisfying

$$e_i^\vee\left(e_j\right) = \delta_{ij}$$

Proof. A map $M \oplus N \to A$ is uniquely determined by its restrictions to the two summands, which proves the first assertion. The remaining assertions are immediate for a finite free module. A finite projective module is locally finite free, and a map between finite projective modules is an isomorphism exactly when it is so after localization at every prime. The free case therefore proves the projective case. □

Example (Why Finiteness Matters). As a $\mathbb{Z}$-module, $\mathbb{Q}^\vee = 0$: every homomorphism $\mathbb{Q} \to \mathbb{Z}$ is zero because $\mathbb{Q}$ is divisible and $\mathbb{Z}$ has no nonzero divisible subgroup. Hence $(\mathbb{Q}^\vee)^\vee = 0$, so the evaluation map is very far from an isomorphism. Finite projectivity, rather than duality alone, is what makes double dualization recover the original module.

Proposition 3.1.2 (The Dual of a Fractional Ideal). Let A be a domain with fraction field K, and let $I \subseteq K$ be a nonzero A-submodule. Multiplication induces a canonical isomorphism

$$(A : I) := \{x \in K \mid xI \subseteq A\} \simeq I^\vee$$

If I is an invertible fractional ideal, then

$$I^\vee \simeq I^{-1}, \quad (I^\vee)^\vee \simeq I$$

Proof. Every $x \in (A : I)$ defines the functional $m \mapsto xm$. Conversely, choose $0 \neq a \in I$. If $f : I \to A$ is A-linear, tensoring with K gives a K-linear endomorphism of $K \otimes_A I \simeq K$. It is multiplication by $\frac{f(a)}{a}$, and therefore

$$f(m) = \frac{f(a)}{a} \cdot m$$

for every $m \in I$. Hence f is multiplication by $\frac{f(a)}{a} \in (A : I)$. The two constructions are inverse. For an invertible ideal, $(A : I) = I^{-1}$, and dualizing once more gives I. □

Remark (The Picard-Group Analogy). Let $X = \operatorname{Spec} A$. A finite projective module P of rank one determines a line bundle $\tilde{P}$ on X, and its module dual determines the dual line bundle

$$\widetilde{P^\vee} \simeq \operatorname{Hom}_{\mathcal{O}_X}(\tilde{P}, \mathcal{O}_X)$$

Evaluation gives

$$\tilde{P} \otimes_{\mathcal{O}_X} \widetilde{P^\vee} \simeq \mathcal{O}_X$$

so dualization is inversion in the Picard group:

$$\left[\widetilde{P^\vee}\right] = -[\tilde{P}] \text{ in } \operatorname{Pic}(X)$$

For a Dedekind domain, a fractional ideal I is a rank-one projective module and $I^\vee \simeq I^{-1}$. Under $\operatorname{Cl}(A) \simeq \operatorname{Pic}(X)$, the passage $I \mapsto I^\vee$ is therefore exactly the passage from an ideal class to its inverse.

Definition 3.1.2 (Pairings: Nondegenerate versus Perfect). An A-bilinear pairing on M is a map

$$\beta : M \times M \to A$$

It induces an A-linear map

$$\lambda_\beta : M \to M^\vee, \quad x \mapsto (y \mapsto \beta(x, y))$$

The pairing is *nondegenerate* if λ_β is injective and *perfect* if λ_β is an isomorphism.

Remark (Nondegenerate Need Not Mean Perfect). On a finite-dimensional vector space over a field, injectivity and bijectivity are equivalent, so a nondegenerate pairing is perfect. Over a general ring they differ. The pairing

$$\beta : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}, \quad (x, y) \mapsto 2xy$$

is nondegenerate, but the induced map $\mathbb{Z} \to \mathbb{Z}^\vee \simeq \mathbb{Z}$ has image $2\mathbb{Z}$ and is not surjective.

3.2 Lattices and Dual Lattices

Let A be a domain with fraction field K, and let V be a finite-dimensional K-vector space. A lattice is an integral structure inside V: it remembers which vectors should count as having integral coordinates without requiring a preferred basis.

Definition 3.2.1 (Full Lattice). An A-*lattice* in V is a finitely generated A-submodule $M \subseteq V$ such that

$$KM = V$$

Equivalently, the natural map

$$K \otimes_A M \to V$$

is an isomorphism. We always use lattice to mean a full lattice.

Remark (A Lattice Need Not Be Free). If A is a PID, every lattice is finite free because it is finitely generated and torsion-free. Over a nonprincipal Dedekind domain, lattices need not be free. A nonprincipal fractional ideal is already a rank-one lattice in K: it is projective and locally free, but it has no global basis. This is precisely the phenomenon measured by $\mathrm{Pic}(\mathrm{Spec}\, A)$.

Proposition 3.2.1 (Commensurability of Lattices). If M and N are two A-lattices in V, then there exist nonzero $a, b \in A$ such that

$$aM \subseteq N, \quad bN \subseteq M$$

Thus any two lattices differ only by bounded denominators.

Proof. Choose generators of M and express them in terms of elements of N that span V over K. Clearing the finitely many denominators gives $0 \neq a \in A$ with $aM \subseteq N$. Interchanging M and N gives b. □

Definition 3.2.2 (Dual Lattice). Assume now that A is Noetherian and that V carries a perfect K-bilinear pairing

$$\beta : V \times V \to K$$

For an A-lattice $M \subseteq V$, its *dual lattice* with respect to β is

$$M^* := \{x \in V \mid \beta(x, M) \subseteq A\}$$

Theorem 3.2.1 (The Dual Lattice Is the Dual Module). The module M^* is an A-lattice in V, and the map

$$M^* \to M^\vee, \quad x \mapsto (m \mapsto \beta(x, m))$$

is an isomorphism.

Proof. Choose a K-basis $e_1, \dots, e_n$ contained in M, and let $e_1^*, \dots, e_n^*$ be its dual basis for β. Put

$$N = Ae_1 + \dots + Ae_n$$

Since M is finitely generated, clearing denominators gives $0 \neq d \in A$ such that

$$N \subseteq M \subseteq \frac{1}{d}N$$

Reversing inclusions under duality gives

$$dN^* \subseteq M^* \subseteq N^*$$

The module $N^* = Ae_1^* + \dots + Ae_n^*$ is finite free. Since A is Noetherian, its submodule M^* is finitely generated; because it contains dN^*, it spans V. Hence M^* is a lattice.

The displayed map is injective because M spans V and β is nondegenerate. For surjectivity, tensor $f \in M^\vee$ with K to obtain a functional $f_K : V \to K$. Perfection of β gives a unique $x \in V$ such that $f_K(m) = \beta(x, m)$. Since $f(M) \subseteq A$, this x lies in M^*. □

Proposition 3.2.2 (Localization and Biduality). Let $S \subseteq A$ be multiplicatively closed. Then

$$\left(S^{-1}M\right)^* = S^{-1}(M^*)$$

If A is Dedekind and β is symmetric, then every A-lattice satisfies

$$(M^*)^* = M$$

Proof. One inclusion in the localization formula is immediate. For the other, choose finitely many generators m_i of M. If $x \in \left(S^{-1}M\right)^*$, clear the finitely many denominators occurring in $\beta(x, m_i)$ to find $s \in S$ with $sx \in M^*$.

For biduality, equality can be checked after localization at every nonzero prime $\mathfrak{p}$. The ring $A_\mathfrak{p}$ is a DVR, so the localized lattice is free. A basis and its dual basis show directly that the double dual equals the original free lattice. The localization formula then gives the global equality. □

Example (Trace-Dual Lattices). Let L/K be a finite separable extension. The trace pairing

$$\beta(x, y) = \mathrm{Tr}_{L/K}(xy)$$

is symmetric and perfect. For an A-lattice $M \subseteq L$, its trace dual is therefore

$$M^* = \left\{x \in L \mid \mathrm{Tr}_{L/K}(xM) \subseteq A\right\}$$

This construction will control the size of the integral closure of A in L.

3.3 Dedekind Extensions

Definition 3.3.1 (Integral Closure and the Dedekind-Extension Setup). Let A be a domain with fraction field K, and let L/K be a finite extension. An element $x \in L$ is *integral over A* if it satisfies a monic polynomial in $A[T]$. The *integral closure* of A in L is

$$B = \{x \in L \mid x \text{ is integral over } A\}$$

In this section we use the following setup:

1. A is a Dedekind domain with fraction field K;
2. L/K is finite and separable;
3. B is the integral closure of A in L.

The extension B/A will be called a *finite separable Dedekind extension.*

The set B is a subring of L. The important point is not integral closedness, which is built into the definition, but finite generation over A.

Proposition 3.3.1 (Clearing Denominators in a Finite Extension). Every $x \in L$ can be written as

$$x = \frac{b}{a}$$

for some $b \in B$ and $0 \neq a \in A$. Consequently,

$$KB = L, \quad \mathrm{Frac}(B) = L$$

Proof. Let

$$f(T) = T^n + c_{n-1}T^{n-1} + \ldots + c_0 \in K[T]$$

be the minimal polynomial of x. Choose $0 \neq a \in A$ such that $ac_i \in A$ for every i. The element ax satisfies the monic polynomial

$$T^n + ac_{n-1}T^{n-1} + a^2c_{n-2}T^{n-2} + \ldots + a^nc_0$$

in $A[T]$. Thus $b = ax$ belongs to B and $x = \frac{b}{a}$. It follows that B spans L over K and that every element of L is a quotient of two elements of B. □

Lemma 3.3.1 (Trace and Norm Preserve Integrality). For every $b \in B$,

$$\mathrm{Tr}_{L/K}(b) \in A, \quad \mathrm{N}_{L/K}(b) \in A$$

Proof. The minimal polynomial $m_{b,K}$ lies in $A[T]$. Indeed, its roots in a splitting field are integral over A, so its coefficients are integral over A; the coefficients also lie in K, and A is integrally closed in K. The minimal-polynomial formulas for trace and norm express them in terms of these coefficients, and therefore put both elements in A. □

Theorem 3.3.1 (The Integral Closure Is a Lattice). In the Dedekind-extension setup, B is an A-lattice in L. In particular, it is a finite A-module of rank $[L : K]$.

Proof. By the denominator-clearing proposition, B spans L. Choose a K-basis $e_1, \ldots, e_n$ of L contained in B, and set

$$M = Ae_1 + \ldots + Ae_n \subseteq B$$

Use the perfect trace pairing on L and define

$$B^* = \left\{x \in L \mid \mathrm{Tr}_{L/K}(xB) \subseteq A\right\}$$

If $b, c \in B$, then $bc \in B$, so the preceding lemma gives $\mathrm{Tr}_{L/K}(bc) \in A$. Hence $B \subseteq B^*$. Since $M \subseteq B$, duality reverses containment and gives

$$M \subseteq B \subseteq B^* \subseteq M^*$$

The dual-lattice theorem says that M^* is a finite A-module. Since A is Noetherian, its submodule B is finitely generated. Together with $KB = L$, this proves that B is an A-lattice. Its rank is $\dim_K(L) = [L : K]$. □

> “The trace pairing traps the integral closure between two lattices; this is the finiteness mechanism behind Dedekind extensions.”

Lemma 3.3.2 (Dimension under an Integral Extension). Let C/A be an integral extension of domains. If $\mathfrak{q}_0 \subset \mathfrak{q}_1$ are prime ideals of C, then

$$\mathfrak{q}_0 \cap A \subset \mathfrak{q}_1 \cap A$$

In particular, $\dim(C) \leq \dim(A)$.

Proof. Suppose the two contractions were equal to $\mathfrak{p}$. The domain $C/\mathfrak{q}_0$ is integral over $A/\mathfrak{p}$. After inverting all nonzero elements of $A/\mathfrak{p}$, it becomes an algebraic domain over the field $\mathrm{Frac}(A/\mathfrak{p})$, hence a field. But $\mathfrak{q}_1/\mathfrak{q}_0$ is disjoint from these denominators and would localize to a nonzero proper prime ideal of that field, a contradiction. Thus strict chains contract to strict chains. □

> **Theorem 3.3.2 (Integral Closures of Dedekind Domains).** Let A be a Dedekind domain with fraction field K, let L/K be a finite separable extension, and let B be the integral closure of A in L. Then B is a Dedekind domain with fraction field L.

Proof. We verify the three defining properties.

1. The ring B is integrally closed in L. If $x \in L$ is integral over B, then transitivity of integrality makes x integral over A, so $x \in B$.
2. The ring B is Noetherian. It is finite over the Noetherian ring A by the lattice theorem, and every finite algebra over a Noetherian ring is Noetherian.
3. We have $\dim(B) \leq \dim(A) = 1$ by the dimension lemma. Moreover, B is not a field. Indeed, $B \cap K = A$ because A is integrally closed in K; if B were a field, it would contain K and force $K = B \cap K = A$. Therefore $\dim(B) = 1$.

Finally, the denominator-clearing proposition gives $\operatorname{Frac}(B) = L$. □

> **Remark (What Separability Is Doing).** Separability makes the trace pairing perfect, which proves that B is finite over A. The conclusion that the integral closure is Dedekind also holds for arbitrary finite extensions, but in the inseparable case its Noetherianity requires the deeper Krull–Akizuki theorem; the integral closure need not be finite as an A -module.

3.4 Splitting Primes

Continue with the Dedekind-extension setup, and let $n = [L : K]$. Since B is Dedekind, the extension of a nonzero prime ideal $\mathfrak{p} \subseteq A$ is nonzero. It is also proper: $B_{\mathfrak{p}}$ is free of positive rank over the DVR $A_{\mathfrak{p}}$, so its reduction modulo $\mathfrak{p}$ is nonzero. It therefore has a unique prime-ideal factorization. Write $\mathfrak{q}|\mathfrak{p}$ for the primes that occur, so

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}}}$$

The primes in this product are the points of $\operatorname{Spec} B$ lying over the point $\mathfrak{p}$ of $\operatorname{Spec} A$.

> **Proposition 3.4.1 (Primes Lying Above a Prime).** For a nonzero prime $\mathfrak{q} \subseteq B$ and a nonzero prime $\mathfrak{p} \subseteq A$, the following are equivalent:
>
> 1. $\mathfrak{q}$ occurs in the factorization of $\mathfrak{p}B$;
> 2. $\mathfrak{q}$ contains $\mathfrak{p}B$;
> 3. $\mathfrak{q} \cap A = \mathfrak{p}$.
>
> In this case we write $\mathfrak{q}|\mathfrak{p}$ and say that $\mathfrak{q}$ *lies above* $\mathfrak{p}$.

Proof. In a Dedekind domain, a prime divides an integral ideal exactly when it contains that ideal, so the first two conditions are equivalent. If $\mathfrak{p}B \subseteq \mathfrak{q}$, then $\mathfrak{p} \subseteq \mathfrak{q} \cap A$; maximality of $\mathfrak{p}$ forces equality. Conversely, $\mathfrak{q} \cap A = \mathfrak{p}$ implies $\mathfrak{p}B \subseteq \mathfrak{q}$. □

Definition 3.4.1 (Ramification Index and Residue Degree). Let $\mathfrak{q}|\mathfrak{p}$. The exponent

$$e_{\mathfrak{q}/\mathfrak{p}} := v_{\mathfrak{q}}(\mathfrak{p}B)$$

is the *ramification index*. The inclusion $A \to B$ induces an extension of residue fields

$$\kappa(\mathfrak{p}) := A/\mathfrak{p} \to \kappa(\mathfrak{q}) := B/\mathfrak{q}$$

Its degree

$$f_{\mathfrak{q}/\mathfrak{p}} := [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]$$

is the *residue degree* or *inertia degree*. We also put

$$g_{\mathfrak{p}} := |\{\mathfrak{q} : \mathfrak{q}|\mathfrak{p}\}|$$

Theorem 3.4.1 (The Fundamental Identity). For every nonzero prime $\mathfrak{p}$ of A,

$$\sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = [L : K]$$

Proof. First compute the dimension of the fiber algebra. Localizing at $\mathfrak{p}$, the module $B_{\mathfrak{p}}$ is finite and torsion-free over the DVR $A_{\mathfrak{p}}$, hence free. Its rank is n because $K \otimes_A B \simeq L$. Therefore

$$\dim_{\kappa(\mathfrak{p})}(B/\mathfrak{p}B) = n$$

On the other hand, the prime powers in the factorization of $\mathfrak{p}B$ are pairwise coprime, so the Chinese remainder theorem gives

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} (B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}})$$

For $e = e_{\mathfrak{q}/\mathfrak{p}}$, the filtration by powers of $\mathfrak{q}$ has e successive quotients, each one-dimensional over $\kappa(\mathfrak{q})$. Hence

$$\dim_{\kappa(\mathfrak{p})}(B/\mathfrak{q}^e) = e f_{\mathfrak{q}/\mathfrak{p}}$$

Taking dimensions in the product decomposition proves the identity. □

Definition 3.4.2 (Ramified, Inert, and Split Primes). Let $\mathfrak{p}$ be a nonzero prime of A.

1. The extension is *unramified at* $\mathfrak{q}|\mathfrak{p}$ if $e_{\mathfrak{q}/\mathfrak{p}} = 1$ and $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. It is unramified above $\mathfrak{p}$ if this holds for every $\mathfrak{q}|\mathfrak{p}$.
2. The prime $\mathfrak{p}$ is *inert* if $\mathfrak{p}B$ is prime and the extension is unramified there. Equivalently, $g_{\mathfrak{p}} = 1$, $e = 1$, and $f = n$.
3. The prime $\mathfrak{p}$ *splits completely* if $g_{\mathfrak{p}} = n$; equivalently, $e = f = 1$ for every prime above it.
4. The prime $\mathfrak{p}$ is *totally ramified* if there is one prime above it with $e = n$; the fundamental identity then forces $f = 1$.

Behavior	*Prime data*	*Fiber picture*
split completely	$g = n$, all $e = f = 1$	many distinct rational points
inert	$g = 1, e = 1, f = n$	one reduced point with a larger residue field
totally ramified	$g = 1, e = n, f = 1$	one point of multiplicity n
mixed	$\sum ef = n$	several points with varying degrees and lengths

Remark (The Geometric Fiber). Put $X := \operatorname{Spec} B$, $Y := \operatorname{Spec} A$, and let $\pi : X \to Y$ be the morphism induced by $A \to B$. The extended ideal $\mathfrak{p}B$, or equivalently the ideal sheaf $\mathfrak{p}\mathcal{O}_X$, is the ideal-theoretic pullback of $\mathfrak{p}$. It cuts out the *scheme-theoretic fiber* above $\mathfrak{p}$:

$$X \times_Y \operatorname{Spec}(\kappa(\mathfrak{p})) \simeq \operatorname{Spec}(B/\mathfrak{p}B) = V_X(\mathfrak{p}B)$$

Thus $\mathfrak{p}B$ means more than just the set of primes above $\mathfrak{p}$: the ideal also remembers the infinitesimal thickness of the fiber. Indeed, the factorization of $\mathfrak{p}B$ and the Chinese remainder theorem give

$$B/\mathfrak{p}B \simeq \prod_{\mathfrak{q}|\mathfrak{p}} B/\mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

The underlying points of this fiber are precisely the $\mathfrak{q}|\mathfrak{p}$. Its component at $\mathfrak{q}$ is a thickened point of length $e_{\mathfrak{q}/\mathfrak{p}}$ over its residue field $\kappa(\mathfrak{q})$. The residue degree $f_{\mathfrak{q}/\mathfrak{p}}$ measures how far that point is from being $\kappa(\mathfrak{p})$-rational. Consequently, its contribution to the degree of the whole fiber is $e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$, and

$$B \otimes_A \kappa(\mathfrak{p}) \simeq B/\mathfrak{p}B$$

has total dimension $[L : K]$ over $\kappa(\mathfrak{p})$.

There is an equivalent divisor picture. A nonzero prime $\mathfrak{p}$ is a closed point, hence a prime divisor, on the arithmetic curve Y. Pulling it back to X gives

$$\pi^*[\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{q}]$$

In other words, the ideal identity $\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$ is exactly the divisor-theoretic statement that the point $\mathfrak{p}$ pulls back to the points above it, counted with ramification multiplicity. Notice that the residue degrees $f_{\mathfrak{q}/\mathfrak{p}}$ do not appear as divisor coefficients; they record the degrees of the resulting points instead.

The fiber is finite étale exactly when every $e = 1$ and every residue-field extension is separable. If $\kappa(\mathfrak{p})$ is perfect, as it is for a finite field, this reduces to the condition $e = 1$ for every prime above $\mathfrak{p}$.

Example (Splitting in $\mathbb{Z}[i]/\mathbb{Z}$). Let $A = \mathbb{Z}$, $B = \mathbb{Z}[i]$, and $L = \mathbb{Q}(i)$. The behavior of a rational prime is determined by the factorization of $T^2 + 1$ modulo p.

1. The prime 2 is totally ramified:

$$(2) = (1 + i)^2$$

2. If $p \equiv 1 \bmod 4$, then -1 is a square modulo p. Choosing $a^2 \equiv -1 \bmod p$ gives two distinct primes

$$(p) = (p, i - a)(p, i + a)$$

so p splits completely.

3. If $p \equiv 3 \bmod 4$, then $T^2 + 1$ is irreducible modulo p, so (p) is inert and its residue degree is 2.

In all three cases the identity $\sum ef = 2 = [\mathbb{Q}(i) : \mathbb{Q}]$ is visible directly.

3.5 Rings of Integers

Definition 3.5.1 (Algebraic Integers and the Ring of Integers). An algebraic number $\alpha \in \overline{\mathbb{Q}}$ is an *algebraic integer* if it is integral over $\mathbb{Z}$, equivalently, if it is a root of a monic polynomial in $\mathbb{Z}[T]$.

A *number field* is a finite extension $K/\mathbb{Q}$. Its *ring of integers* is

$$\mathcal{O}_K := \{\alpha \in K \mid \alpha \text{ is integral over } \mathbb{Z}\}$$

Thus $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K.

Theorem 3.5.1 (The Ring of Integers Is Dedekind). For every number field K, the ring $\mathcal{O}_K$ is a Dedekind domain with fraction field K. Moreover, if $n = [K : \mathbb{Q}]$, then $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n.

Proof. The ring $\mathbb{Z}$ is a Dedekind domain with fraction field $\mathbb{Q}$. Every finite extension of $\mathbb{Q}$ is separable because $\mathrm{char}(\mathbb{Q}) = 0$. The theorem on integral closures of Dedekind domains therefore applies to the integral closure $\mathcal{O}_K$ of $\mathbb{Z}$ in K, proving that it is Dedekind and has fraction field K.

The lattice theorem makes $\mathcal{O}_K$ a finitely generated $\mathbb{Z}$-module that spans K over $\mathbb{Q}$. It is torsion-free because it lies in the field K. Every finitely generated torsion-free module over the PID $\mathbb{Z}$ is free, and its rank is

$$\dim_{\mathbb{Q}}(\mathbb{Q} \otimes_{\mathbb{Z}} \mathcal{O}_K) = \dim_{\mathbb{Q}}(K) = n$$

□

Remark (What the Theorem Buys Us). Every nonzero ideal of $\mathcal{O}_K$ factors uniquely into prime ideals, every localization $(\mathcal{O}_K)_{\mathfrak{p}}$ is a DVR, and the class group measures the failure of principal factorization. Thus the entire ideal-theoretic framework of Chapter 1 applies uniformly to every number field.

Lemma 3.5.1 (Trace and Norm Criterion for Quadratic Fields). Let $K/\mathbb{Q}$ be a quadratic extension. For $\alpha \in K$, let $\overline{\alpha}$ denote its image under the nontrivial $\mathbb{Q}$-automorphism of K. Then

$$\alpha \in \mathcal{O}_K \Leftrightarrow \mathrm{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z} \text{ and } \mathrm{N}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$$

Proof. The two conjugates of α are α and $\overline{\alpha}$, so

$$\begin{aligned}(T-\alpha)(T-\overline{\alpha}) &= T^2 - (\alpha+\overline{\alpha})T + \alpha\overline{\alpha} \\ &= T^2 - \mathrm{Tr}_{K/\mathbb{Q}}(\alpha)T + \mathrm{N}_{K/\mathbb{Q}}(\alpha)\end{aligned}$$

Suppose first that α is integral. Its conjugate $\overline{\alpha}$ is integral as well, because a $\mathbb{Q}$-automorphism preserves every polynomial in $\mathbb{Z}[T]$. Hence the trace and norm are rational algebraic integers, and therefore lie in $\mathbb{Z}$. Here we use the elementary fact that a rational algebraic integer is an integer: applying the rational-root argument to a reduced fraction shows that its denominator is 1.

Conversely, if the trace and norm lie in $\mathbb{Z}$, the displayed factorization is a monic polynomial in $\mathbb{Z}[T]$ having α as a root. Thus α is integral. □

Theorem 3.5.2 (Ring of Integers of a Quadratic Field). Let $d \neq 1$ be a squarefree integer and put $K = \mathbb{Q}(\sqrt{d})$. Then

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}\left[\left(1+\sqrt{d}\right)/2\right] & d \equiv 1 \bmod 4 \\ \mathbb{Z}\left[\sqrt{d}\right] & d \not\equiv 1 \bmod 4 \end{cases}$$

Equivalently, if

$$\omega = \begin{cases} \left(1+\sqrt{d}\right)/2 & d \equiv 1 \bmod 4 \\ \sqrt{d} & d \not\equiv 1 \bmod 4 \end{cases}$$

then $\{1, \omega\}$ is an *integral basis* of K and $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\omega$.

Proof. Let $\alpha = a + b\sqrt{d}$ with $a, b \in \mathbb{Q}$. By the preceding lemma and the formulas from Chapter 2,

$$\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = 2a, \quad \mathrm{N}_{K/\mathbb{Q}}(\alpha) = a^2 - db^2$$

Suppose first that α is integral. Set $m := 2a \in \mathbb{Z}$. The discriminant of its characteristic polynomial is

$$m^2 - 4\mathrm{N}_{K/\mathbb{Q}}(\alpha) = 4db^2 \in \mathbb{Z}$$

Write $b = \frac{u}{v}$ in lowest terms with $v > 0$. The last integrality condition implies $v^2 \mid 4d$. Since d is squarefree, no odd prime can divide v, and 4 cannot divide v. Hence $v \mid 2$, so $n := 2b \in \mathbb{Z}$ and

$$\alpha = \left(m + n\sqrt{d}\right)/2$$

The norm is integral precisely when

$$m^2 - dn^2 \equiv 0 \bmod 4$$

If $d \equiv 1 \bmod 4$, this says $m^2 \equiv n^2 \bmod 4$. Since every square is congruent to 0 or 1 modulo 4, the integers m and n have the same parity. Consequently,

$$\alpha = (m-n)/2 + n\left(1+\sqrt{d}\right)/2 \in \mathbb{Z}\left[\left(1+\sqrt{d}\right)/2\right]$$

If $d \not\equiv 1 \bmod 4$, squarefreeness forces $d \equiv 2$ or $3 \bmod 4$. If n were odd, then dn^2 would be congruent to 2 or 3 modulo 4, neither of which is a square modulo 4. Thus n is even, and the same congruence then forces m to be even. It follows that $\alpha \in \mathbb{Z}\left[\sqrt{d}\right]$. This proves the inclusion of $\mathcal{O}_K$ in the displayed ring in both cases.

For the reverse inclusion, $\sqrt{d}$ is integral because it satisfies $T^2 - d = 0$. When $d \equiv 1 \bmod 4$, the element $\omega = \left(1 + \sqrt{d}\right)/2$ satisfies

$$\omega^2 - \omega + (1 - d)/4 = 0$$

whose coefficients are integers. Thus ω is integral. Since the algebraic integers form a ring, $\mathbb{Z}[\omega] \subseteq \mathcal{O}_K$, completing the proof. □

In particular,

$$\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i], \quad \mathcal{O}_{\mathbb{Q}\left(\sqrt{-5}\right)} = \mathbb{Z}\left[\sqrt{-5}\right]$$

so the two running examples from the previous chapters are genuinely rings of integers and hence Dedekind domains.

4 Ideal Norms

Throughout this chapter, we retain the Dedekind-extension setup from Chapter 3: A is a Dedekind domain with fraction field K, the extension L/K is finite and separable of degree n, and B is the integral closure of A in L. Thus B is a Dedekind domain and an A-lattice of rank n.

The field norm sends elements of L to elements of K. We now construct its ideal-theoretic counterpart

$$\mathrm{N}_{B/A} : \mathrm{FracId}(B) \to \mathrm{FracId}(A)$$

The main subtlety is that B need not be free over A. It is only finite projective in general, so a definition that depends on choosing a global A-basis would miss precisely the cases where the class group is nontrivial.

4.1 Module Indices

The ordinary index $[\mathbb{Z} : 2\mathbb{Z}] = 2$ measures the change of volume between two lattices. Over a Dedekind domain the answer is not naturally a single element of A, because there may be no global bases. The correct replacement is a fractional ideal, obtained by measuring the determinant locally at every prime.

Definition 4.1.1 (Local Module Index). Let V be an r-dimensional K-vector space, let $M, N \subseteq V$ be A-lattices, and let $\mathfrak{p}$ be a nonzero prime of A. The localized modules $M_\mathfrak{p}$ and $N_\mathfrak{p}$ are free $A_\mathfrak{p}$-modules of rank r.

Choose an $A_\mathfrak{p}$-module isomorphism

$$\varphi_\mathfrak{p} : M_\mathfrak{p} \to N_\mathfrak{p}$$

and let $\Phi_\mathfrak{p} : V \to V$ be its unique K-linear extension. The *local module index* is the principal fractional $A_\mathfrak{p}$-ideal

$$\left[M_\mathfrak{p} : N_\mathfrak{p}\right]_{A_\mathfrak{p}} := \det\left(\Phi_\mathfrak{p}\right) A_\mathfrak{p}$$

Remark (Why It Is Well Defined). Any two choices of $\varphi_\mathfrak{p}$ differ by automorphisms of the free modules $M_\mathfrak{p}$ and $N_\mathfrak{p}$. Their determinants are units of $A_\mathfrak{p}$, so they generate the same principal fractional ideal. The generator is not canonical, but the ideal is.

Definition 4.1.2 (Global Module Index). The *module index* of N in M is

$$[M : N]_A := \bigcap_{0 \neq \mathfrak{p} \subseteq A} \left[M_\mathfrak{p} : N_\mathfrak{p}\right]_{A_\mathfrak{p}} \subseteq K$$

where the intersection takes place inside K.

Proposition 4.1.1 (Local-to-Global Existence). The module index $[M : N]_A$ is a nonzero fractional ideal of A, and for every nonzero prime $\mathfrak{p}$,

$$([M : N]_A)_{\mathfrak{p}} = \left[M_{\mathfrak{p}} : N_{\mathfrak{p}}\right]_{A_{\mathfrak{p}}}$$

Proof. By commensurability of lattices, there is a nonzero $c \in A$ such that $cM \subseteq N$ and $cN \subseteq M$. If $\mathfrak{p}$ does not contain c, then c is a unit in $A_{\mathfrak{p}}$ and $M_{\mathfrak{p}} = N_{\mathfrak{p}}$. Hence the local index is $A_{\mathfrak{p}}$ away from the finitely many prime divisors of (c).

At each remaining prime the local index is a unique integer power of $\mathfrak{p}A_{\mathfrak{p}}$. The product of these finitely many powers, allowing negative exponents, is a nonzero fractional ideal J of A with exactly the prescribed localizations. The local description of fractional ideals from Chapter 1 then gives $J = [M : N]_A$ and proves the claimed localization formula. □

Proposition 4.1.2 (Calculus of Module Indices). For A-lattices M, N, P in the same K-vector space,

$$[M : N]_A[N : P]_A = [M : P]_A$$

In particular,

$$[M : M]_A = A, \quad [M : N]_A^{-1} = [N : M]_A$$

If $N \subseteq M$, then $[M : N]_A$ is an integral ideal of A. If M and N are free and bases of M and N are related by a matrix $C \in \mathrm{Mat}_r(K)$, with the N-basis equal to the M-basis multiplied by C, then

$$[M : N]_A = (\det C)$$

Proof. Localize at $\mathfrak{p}$. Isomorphisms $M_{\mathfrak{p}} \to N_{\mathfrak{p}} \to P_{\mathfrak{p}}$ compose, and determinants multiply. This proves the first identity locally and hence globally. The inverse formulas follow immediately. If $N \subseteq M$, the change-of-basis matrix has entries in $A_{\mathfrak{p}}$ at every prime, so every local determinant has nonnegative valuation; therefore the global index is integral. The free case is the same determinant computation before localization. □

Definition 4.1.3 (The Geometric Module Index). Put $X = \operatorname{Spec} A$ and suppose first that $N \subseteq M$. The corresponding finite projective modules give rank-r vector bundles

$$\mathscr{E}_N := \widetilde{N} \to \mathscr{E}_M := \widetilde{M}$$

with the same generic fiber V. Their quotient

$$\mathscr{Q} := \mathscr{E}_M / \mathscr{E}_N$$

is a *torsion coherent sheaf* supported at finitely many closed points. Let $X^{(1)}$ denote the set of codimension-one points of X; these are precisely its closed points. For $x \in X^{(1)}$, define

$$\ell_x(\mathscr{Q}) := \mathrm{length}_{\mathscr{O}_{X,x}}(\mathscr{Q}_x)$$

The *geometric module index* is the effective divisor

$$D_{M/N} := \sum_{x \in X^{(1)}} \ell_x(\mathcal{Q})[x]$$

For arbitrary lattices $M, N \subseteq V$, commensurability supplies a common sublattice $P \subseteq M \cap N$. Define the relative divisor

$$D_{M,N} := D_{M/P} - D_{N/P}$$

The next theorem shows that this divisor is independent of P and is an equivalent definition of the module index.

Theorem 4.1.1 (Equivalence of the Determinant and Divisor Definitions). Recall the ideal–divisor isomorphism from Chapter 1,

$$\Phi(I) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(I)[\mathfrak{p}]$$

For any two A-lattices $M, N \subseteq V$,

$$\Phi([M : N]_A) = D_{M,N}$$

Equivalently, the module index may be defined geometrically by

$$[M : N]_A = \Phi^{-1}\left(D_{M,N}\right)$$

In particular, if $N \subseteq M$, then

$$v_{\mathfrak{p}}([M : N]_A) = \mathrm{length}_{A_{\mathfrak{p}}}\left(M_{\mathfrak{p}}/N_{\mathfrak{p}}\right)$$

Proof. First suppose that $N \subseteq M$. Fix a nonzero prime $\mathfrak{p}$ and a uniformizer π of $A_{\mathfrak{p}}$. The elementary-divisor theorem gives a basis $e_1, \ldots, e_r$ of $M_{\mathfrak{p}}$ and nonnegative integers $a_1, \ldots, a_r$ such that

$$N_{\mathfrak{p}} = A_{\mathfrak{p}}\pi^{a_1}e_1 \oplus \ldots \oplus A_{\mathfrak{p}}\pi^{a_r}e_r$$

Relative to this basis, an isomorphism $M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is represented by the diagonal matrix

$$\mathrm{diag}(\pi^{a_1}, \ldots, \pi^{a_r})$$

Its determinant gives

$$v_{\mathfrak{p}}([M : N]_A) = a_1 + \ldots + a_r$$

On the geometric side,

$$M_{\mathfrak{p}}/N_{\mathfrak{p}} \simeq A_{\mathfrak{p}}/(\pi^{a_1}) \oplus \ldots \oplus A_{\mathfrak{p}}/(\pi^{a_r})$$

and the length of this module is also $a_1 + \ldots + a_r$. Thus the coefficient of $[\mathfrak{p}]$ in $\Phi([M : N]_A)$ equals the coefficient of the corresponding closed point in $D_{M/N}$. This proves the result for an inclusion.

For general M, N, choose $P \subseteq M \cap N$. The cocycle identity gives

$$[M : N]_A = [M : P]_A[N : P]_A^{-1}$$

Applying Φ and using the inclusion case yields

$$\Phi([M:N]_A) = D_{M/P} - D_{N/P} = D_{M,N}$$

The left-hand side does not depend on P, so neither does the relative divisor. This also proves that the geometric prescription is equivalent to the determinant definition. □

Corollary 4.1.1 (Determinant Line, Fitting Ideal, and Degeneracy Divisor). The relative determinant line of two lattices satisfies

$$[M:N]_A \mathscr{O}_X \simeq \det(\mathscr{E}_N) \otimes \det(\mathscr{E}_M)^\vee$$

If $N \subseteq M$, then

$$[M:N]_A = \mathrm{Fitt}_0(M/N)$$

and $D_{M/N}$ is the vanishing divisor of the determinant of the inclusion $\mathscr{E}_N \to \mathscr{E}_M$.

Proof. The statements can be checked over each DVR $A_\mathfrak{p}$. In the diagonal form used above, both the relative determinant line and the zeroth Fitting ideal are generated by $\pi^{a_1+\ldots+a_r}$. The determinant of the inclusion vanishes to this same order, proving all three claims. □

Remark (What the Divisor Remembers). Locally, the integers $a_1, \ldots, a_r$ record the changes of scale in the individual vector-bundle directions. The geometric module index retains only their sum, which is simultaneously a determinant valuation, the length of a torsion quotient, and the multiplicity of a closed point in a divisor. For noncomparable lattices, positive coefficients record relative zeros and negative coefficients record relative poles.

“The genuine geometric object behind the module index is the relative divisor measuring where, and by how much, two lattices fail to coincide.”

Remark (Rank One and the Colon Ideal). When $V = K$, its lattices are precisely the nonzero fractional ideals of A. For fractional ideals M, N,

$$[M:N]_A = (N:M) = NM^{-1}$$

where

$$(N:M) := \{x \in K \mid xM \subseteq N\}$$

Notice the reversed order: module index generalizes $[\mathbb{Z} : 2\mathbb{Z}] = 2$, whereas a colon ideal behaves like a ratio.

Theorem 4.1.2 (Module Index from a Quotient). Suppose $N \subseteq M$ and the finite torsion module M/N has a cyclic decomposition

$$M/N \simeq A/I_1 \oplus \ldots \oplus A/I_s$$

for nonzero integral ideals $I_1, \ldots, I_s$ of A. Then

$$[M:N]_A = I_1 \ldots I_s$$

Proof. Localize at a nonzero prime $\mathfrak{p}$ and choose a uniformizer π of the DVR $A_{\mathfrak{p}}$. Choose bases of $M_{\mathfrak{p}}$ and $N_{\mathfrak{p}}$. The matrix expressing the latter basis in terms of the former has Smith normal form

$$\mathrm{diag}\left(\pi^{d_1}, \dots, \pi^{d_r}\right)$$

Its determinant has valuation $d_1 + \dots + d_r$. On the other hand, the localized cyclic decomposition shows that the same module has total length

$$\sum_{j=1}^{s} v_{\mathfrak{p}}(I_j)$$

Uniqueness of the invariant factors over a PID identifies these two integers. Thus the localizations of $[M : N]_A$ and $I_1 \dots I_s$ agree at every prime, proving the equality. □

> **Example (The Classical Lattice Index).** Let $A = \mathbb{Z}$, $M = \mathbb{Z}^2$, and
>
> $$N = \mathbb{Z}(2, 0) + \mathbb{Z}(1, 3)$$
>
> The change-of-basis matrix has determinant 6, so
>
> $$[M : N]_{\mathbb{Z}} = (6)$$
>
> Correspondingly, M/N is a finite abelian group of order 6. The ideal-valued module index is therefore the Dedekind-domain version of lattice covolume.

4.2 The Ideal Norm

Every nonzero fractional B-ideal is an A-lattice in L, so the module index can be applied to the pair B, I.

> **Definition 4.2.1 (Relative Ideal Norm).** For a nonzero fractional ideal I of B, its *ideal norm from B to A* is
>
> $$\mathrm{N}_{B/A}(I) := [B : I]_A$$
>
> We also set $\mathrm{N}_{B/A}((0)) := (0)$.

If $I \subseteq B$ is integral, then $\mathrm{N}_{B/A}(I)$ is an integral ideal of A. For a genuinely fractional ideal the norm may have negative prime exponents, exactly as it should for a homomorphism between fractional ideal groups.

> **Proposition 4.2.1 (Compatibility with the Field Norm).** For every $\alpha \in L^{\times}$,
>
> $$\mathrm{N}_{B/A}(\alpha B) = \mathrm{N}_{L/K}(\alpha) A$$

Proof. At every nonzero prime $\mathfrak{p}$ of A, multiplication by α gives an isomorphism of free $A_{\mathfrak{p}}$-modules

$$B_{\mathfrak{p}} \to \alpha B_{\mathfrak{p}}$$

Its K-linear extension is multiplication by α on L, whose determinant is $\mathrm{N}_{L/K}(\alpha)$. Thus all local module indices are the localizations of the principal ideal $\mathrm{N}_{L/K}(\alpha) A$. □

Theorem 4.2.1 (Multiplicativity of the Ideal Norm). For nonzero fractional ideals I, J of B,

$$\mathrm{N}_{B/A}(IJ) = \mathrm{N}_{B/A}(I)\mathrm{N}_{B/A}(J)$$

Hence $\mathrm{N}_{B/A} : \mathrm{FracId}(B) \to \mathrm{FracId}(A)$ is a group homomorphism.

Proof. It is enough to compare localizations at every nonzero prime $\mathfrak{p}$ of A. The semilocal Dedekind domain $B_\mathfrak{p}$ is a PID, so there are $\alpha, \beta \in L^\times$ with

$$I_\mathfrak{p} = \alpha B_\mathfrak{p}, \quad J_\mathfrak{p} = \beta B_\mathfrak{p}$$

The principal-ideal formula and multiplicativity of the field norm give

$$\begin{aligned} \mathrm{N}_{B_\mathfrak{p}/A_\mathfrak{p}}\left(I_\mathfrak{p} J_\mathfrak{p}\right) &= \mathrm{N}_{L/K}(\alpha\beta)A_\mathfrak{p} \\ &= \mathrm{N}_{L/K}(\alpha)\mathrm{N}_{L/K}(\beta)A_\mathfrak{p} \end{aligned}$$

This is the product of the two localized ideal norms. Equality at all $\mathfrak{p}$ proves the result. □

Corollary 4.2.1 (Norm and Relative Module Index). For nonzero fractional B-ideals I and J,

$$[I : J]_A = \mathrm{N}_{B/A}\left(I^{-1}J\right) = \mathrm{N}_{B/A}((J : I))$$

Proof. The colon-ideal identity $(J : I) = I^{-1}J$ holds because every nonzero fractional ideal of B is invertible. Using the cocycle rule for module indices and multiplicativity of the norm gives

$$\begin{aligned} [I : J]_A = [I : B]_A[B : J]_A &= [B : I]_A^{-1}[B : J]_A \\ &= \mathrm{N}_{B/A}\left(I^{-1}J\right) \end{aligned}$$

□

Proposition 4.2.2 (The Norm Is Generated by Element Norms). For every fractional ideal I of B,

$$\mathrm{N}_{B/A}(I) = \left(\mathrm{N}_{L/K}(\alpha) : \alpha \in I\right)$$

where the right-hand side denotes the fractional A-ideal generated by all field norms of elements of I.

Proof. Localize at $\mathfrak{p}$ and write $I_\mathfrak{p} = \gamma B_\mathfrak{p}$. Every element of $I_\mathfrak{p}$ is γb with $b \in B_\mathfrak{p}$, so its field norm lies in $\mathrm{N}_{L/K}(\gamma)A_\mathfrak{p}$. Conversely, write $\gamma = \frac{\alpha}{s}$ with $\alpha \in I$ and $s \in A$, $s \notin \mathfrak{p}$. Since $\mathrm{N}_{L/K}(s) = s^n$ is a unit of $A_\mathfrak{p}$, the norm of α generates the same localized ideal as the norm of γ. The two sides therefore agree after localization at every prime. □

4.2.1 Norms of Prime Ideals

Let $\mathfrak{q}$ be a nonzero prime of B lying above $\mathfrak{p}$, and recall the residue degree

$$f_{\mathfrak{q}/\mathfrak{p}} = [B/\mathfrak{q} : A/\mathfrak{p}]$$

Theorem 4.2.2 (Norm of a Prime Ideal). If $\mathfrak{q}|\mathfrak{p}$, then

$$\mathrm{N}_{B/A}(\mathfrak{q}) = \mathfrak{p}^{f_{\mathfrak{q}/\mathfrak{p}}}$$

Proof. As a vector space over $A/\mathfrak{p}$, the residue field $B/\mathfrak{q}$ has dimension $f = f_{\mathfrak{q}/\mathfrak{p}}$. Hence, as an A-module,

$$B/\mathfrak{q} \simeq (A/\mathfrak{p})^f$$

Applying the quotient formula for the module index gives

$$[B : \mathfrak{q}]_A = \mathfrak{p}^f$$

and the left-hand side is the definition of $\mathrm{N}_{B/A}(\mathfrak{q})$. □

Corollary 4.2.2 (The Valuation Formula). For every nonzero fractional B-ideal I and every nonzero prime $\mathfrak{p}$ of A,

$$v_{\mathfrak{p}}\big(\mathrm{N}_{B/A}(I)\big) = \sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)$$

Equivalently,

$$\mathrm{N}_{B/A}(I) = \prod_{\mathfrak{p}} \mathfrak{p}^{\sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)}$$

Proof. Factor I into prime ideals of B, apply multiplicativity, and use the formula for the norm of each prime factor. Only finitely many valuations are nonzero. □

Proposition 4.2.3 (Ideal Norm as Pushforward of Divisors). Let $\pi : \operatorname{Spec} B \to \operatorname{Spec} A$. Define the pushforward on prime divisors by

$$\pi_*[\mathfrak{q}] := f_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{p}], \quad \mathfrak{p} = \mathfrak{q} \cap A$$

and extend it $\mathbb{Z}$-linearly. If Φ_A and Φ_B denote the ideal–divisor correspondences for A and B, then

$$\Phi_A\big(\mathrm{N}_{B/A}(I)\big) = \pi_*\Phi_B(I)$$

for every nonzero fractional B-ideal I. Thus the ideal norm is exactly divisor pushforward written multiplicatively.

Proof. Both sides are homomorphisms from $\mathrm{FracId}(B)$ to the divisor group of $\operatorname{Spec} A$, so it is enough to check a prime ideal $\mathfrak{q}|\mathfrak{p}$. By the prime-norm theorem,

$$\Phi_A\big(\mathrm{N}_{B/A}(\mathfrak{q})\big) = \Phi_A\big(\mathfrak{p}^{f_{\mathfrak{q}/\mathfrak{p}}}\big)$$

$$= f_{\mathfrak{q}/\mathfrak{p}}[\mathfrak{p}] = \pi_*\Phi_B(\mathfrak{q})$$

Multiplicativity proves the result for every fractional ideal. Explicitly, the coefficient of $[\mathfrak{p}]$ on either side is

$$\sum_{\mathfrak{q}|\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} v_{\mathfrak{q}}(I)$$

which recovers the valuation formula above. □

Corollary 4.2.3 (Norm after Extending an Ideal). For every nonzero fractional ideal J of A,

$$\mathrm{N}_{B/A}(JB) = J^n$$

Proof. Under the ideal–divisor correspondence, extension of ideals is divisor pullback:

$$\Phi_B(JB) = \pi^* \Phi_A(J)$$

Applying the pushforward proposition gives

$$\Phi_A\Big(\mathrm{N}_{B/A}(JB)\Big) = \pi_* \pi^* \Phi_A(J)$$

For a prime divisor $[\mathfrak{p}]$, the fundamental identity from Chapter 3 gives

$$\pi_* \pi^* [\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{p}] = n[\mathfrak{p}]$$

Therefore the right-hand side is $n\Phi_A(J) = \Phi_A(J^n)$. Since Φ_A is injective, the desired ideal identity follows. □

Remark (Pullback versus Pushforward). The two divisor operations are determined on closed points by

$$\pi^*[\mathfrak{p}] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{q}]$$

$$\pi_*[\mathfrak{q}] = f_{\mathfrak{q}/\mathfrak{p}} [\mathfrak{p}]$$

Thus ramification indices e occur in pullback, whereas residue degrees f occur in pushforward. Their composite is multiplication by the generic degree:

$$\pi_* \pi^* D = nD$$

Pushforward also preserves principal divisors in the precise form

$$\pi_* \mathrm{div}_L(\alpha) = \mathrm{div}_K\Big(\mathrm{N}_{L/K}(\alpha)\Big)$$

This is the divisor version of the principal-ideal formula. Consequently, pushforward respects linear equivalence, and the ideal norm induces a map on class groups

$$\mathrm{Cl}(B) \to \mathrm{Cl}(A), \quad [I] \mapsto \Big[\mathrm{N}_{B/A}(I)\Big]$$

> “Extending an ideal pulls a prime upward and counts ramification; taking its norm pushes primes downward and counts residue-field degree.”

4.2.2 Absolute Norms in Number Fields

Definition 4.2.2 (Absolute Ideal Norm). Let K be a number field and let $0 \neq I \subseteq \mathcal{O}_K$ be an integral ideal. Its *absolute norm* is the positive integer

$$\mathrm{N}(I) := |\mathcal{O}_K / I|$$

Proposition 4.2.4 (Basic Formulas for the Absolute Norm). For nonzero integral ideals $I, J \subseteq \mathcal{O}_K$,

1. The absolute norm is multiplicative:

$$\mathrm{N}(IJ) = \mathrm{N}(I)\mathrm{N}(J)$$

2. If $0 \neq \alpha \in \mathcal{O}_K$, then

$$\mathrm{N}(\alpha\mathcal{O}_K) = \left|\mathrm{N}_{K/\mathbb{Q}}(\alpha)\right|$$

3. If $\mathfrak{q}$ lies above the rational prime p, then

$$\mathrm{N}(\mathfrak{q}) = p^{f_{\mathfrak{q}/(p)}}$$

Proof. The ideal norm $\mathrm{N}_{\mathcal{O}_K/\mathbb{Z}}(I)$ is an integral ideal of $\mathbb{Z}$. By the quotient formula for module indices, its positive generator is the finite group index $|\mathcal{O}_K / I|$. The three formulas now follow from multiplicativity, compatibility with the field norm, and the prime-ideal formula. □

Example (Norms in the Gaussian Integers). In $\mathbb{Z}[i]$, the principal prime ideal $\mathfrak{q} = (2+i)$ lies above 5. Since $\mathrm{N}_{\mathbb{Q}(i)/\mathbb{Q}}(2+i) = 5$,

$$\mathrm{N}_{\mathbb{Z}[i]/\mathbb{Z}}(\mathfrak{q}) = (5), \quad \mathrm{N}(\mathfrak{q}) = 5$$

On the other hand,

$$(5) = (2+i)(2-i)$$

and therefore

$$\mathrm{N}((5)) = 25$$

This is a useful distinction: a prime above 5 has norm 5, while the extended ideal $5\mathbb{Z}[i]$ has norm $5^{[\mathbb{Q}(i):\mathbb{Q}]} = 25$.

4.3 The Dedekind–Kummer Theorem

The Dedekind–Kummer theorem turns factorization of a polynomial over a residue field into factorization of a prime ideal in an integral closure. It is one of the basic bridges between polynomial arithmetic and ideal arithmetic.

> *Core idea.* Dedekind–Kummer transforms the ideal factorization of $\mathfrak{p}B$ into the factorization of a single polynomial over the residue field $\kappa(\mathfrak{p})$. For number fields over $\mathbb{Q}$, this is precisely the passage from ideal factorization in a number ring to polynomial factorization over $\mathbb{F}_p$.

Theorem 4.3.1 (Primitive Element Theorem). Every finite separable field extension is simple: if L/K is finite separable, then there exists $\alpha \in L$ such that

$$L = K(\alpha)$$

In our setting, B spans L over K. After multiplying α by a suitable nonzero element of A, we may therefore choose an integral primitive element $\theta \in B$ with $L = K(\theta)$. Thus the primitive element theorem supplies a single algebraic coordinate for L/K; the remaining question is whether the powers of this coordinate generate the whole integral closure B.

4.3.1 Power Bases and the Index Divisor

Fix such an integral primitive element $\theta \in B$, and let

$$f(T) \in A[T]$$

be its monic minimal polynomial. If $n = [L : K]$, then

$$A[\theta] = A + A\theta + \dots + A\theta^{n-1}$$

is a free A-lattice in L, but it need not equal B.

Definition 4.3.1 (Monogenic Extensions and the Index Ideal). The extension B/A is *monogenic* if

$$B = A[\theta]$$

for some $\theta \in B$. In this case $\{1, \theta, \dots, \theta^{n-1}\}$ is an integral power basis of B.

For an integral primitive element θ, define its *index ideal* by

$$\mathfrak{i}_\theta := [B : A[\theta]]_A$$

A nonzero prime $\mathfrak{p}$ of A is *good for* θ if $v_\mathfrak{p}(\mathfrak{i}_\theta) = 0$; equivalently, $\mathfrak{p}$ does not divide the index ideal.

Proposition 4.3.1 (The Index Divisor Measures Failure of Monogenicity). For every nonzero prime $\mathfrak{p}$ of A,

$$v_\mathfrak{p}(\mathfrak{i}_\theta) = \operatorname{length}_{A_\mathfrak{p}}\left(B_\mathfrak{p}/A_\mathfrak{p}[\theta]\right)$$

Consequently, $\mathfrak{p}$ is good for θ exactly when

$$B_\mathfrak{p} = A_\mathfrak{p}[\theta]$$

Under the ideal–divisor correspondence, the *index divisor* is

$$D_\theta := \Phi_A(\mathfrak{i}_\theta) = \sum_{\mathfrak{p}} \operatorname{length}_{A_\mathfrak{p}}\left(B_\mathfrak{p}/A_\mathfrak{p}[\theta]\right)[\mathfrak{p}]$$

It is supported at the finitely many primes where the power basis generated by θ fails to be an integral basis locally.

Proof. Apply the geometric module-index formula to the inclusion $A[\theta] \subseteq B$. The coefficient at $\mathfrak{p}$ is the length of the localized quotient. A finite-length module over the local ring $A_\mathfrak{p}$ has length zero exactly when it is zero, which gives the local equality criterion. Finiteness of the support follows because $\mathfrak{i}_\theta$ is a nonzero integral ideal. □

Remark (Why an Index Condition Appears). The polynomial f always describes the order $A[\theta]$. It describes the maximal order B at $\mathfrak{p}$ only when $\mathfrak{p}$ is absent from the index divisor. Dedekind–Kummer is therefore a local monogenicity theorem: one generator need not work everywhere, but it works at every prime outside a finite obstruction divisor.

4.3.2 Statement and Proof

Theorem 4.3.2 (Dedekind–Kummer). Let $\theta \in B$ satisfy $L = K(\theta)$, let $f \in A[T]$ be its monic minimal polynomial, and let $\mathfrak{p}$ be a nonzero prime of A that is good for θ. Factor the reduction of f into distinct monic irreducibles over the residue field $\kappa(\mathfrak{p}) = A/\mathfrak{p}$:

$$\overline{f} = \prod_{i=1}^{r} \overline{g}_i^{e_i}$$

Choose any monic lift $g_i \in A[T]$ of each $\overline{g}_i$, and put

$$\mathfrak{q}_i := \mathfrak{p}B + g_i(\theta)B = (\mathfrak{p}, g_i(\theta))$$

Then the $\mathfrak{q}_i$ are precisely the distinct prime ideals of B above $\mathfrak{p}$, and

$$\mathfrak{p}B = \prod_{i=1}^{r} \mathfrak{q}_i^{e_i}$$

Moreover,

$$f_{\mathfrak{q}_i/\mathfrak{p}} = \deg(\overline{g}_i)$$

In particular, if $B = A[\theta]$, the conclusion holds for every nonzero prime $\mathfrak{p}$ of A.

Proof. Since $\mathfrak{p}$ is good for θ, the index-divisor criterion gives

$$B_{\mathfrak{p}} = A_{\mathfrak{p}}[\theta]$$

Localizing does not change the fiber over $\mathfrak{p}$, because every element of A outside $\mathfrak{p}$ becomes a unit modulo $\mathfrak{p}$. Therefore

$$B/\mathfrak{p}B \simeq B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}$$

$$\simeq \kappa(\mathfrak{p})[T]/\left(\overline{f}\right)$$

The Chinese remainder theorem now gives the scheme-theoretic decomposition

$$B/\mathfrak{p}B \simeq \prod_{i=1}^{r} \kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)$$

The maximal ideals of the right-hand side are indexed by the distinct polynomials $\overline{g}_i$. Their inverse images in B are exactly $\mathfrak{q}_i = (\mathfrak{p}, g_i(\theta))$, so these are all the primes above $\mathfrak{p}$. Furthermore,

$$B/\mathfrak{q}_i \simeq \kappa(\mathfrak{p})[T]/(\overline{g}_i)$$

is a field of degree $\deg(\overline{g}_i)$ over $\kappa(\mathfrak{p})$. This proves the residue-degree formula.

It remains to identify the ramification exponents. Localize the fiber at the point $\mathfrak{q}_i$. All the factors $\overline{g}_j$ with $j \neq i$ become units, and hence

$$B_{\mathfrak{q}_i}/\mathfrak{p}B_{\mathfrak{q}_i} \simeq \left(\kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)\right)_{(\overline{g}_i)}$$

The maximal ideal on the right has nilpotence order e_i. On the left, $B_{\mathfrak{q}_i}$ is a DVR and

$$\mathfrak{p}B_{\mathfrak{q}_i} = \left(\mathfrak{q}_i B_{\mathfrak{q}_i}\right)^{e_{\mathfrak{q}_i/\mathfrak{p}}}$$

so the maximal ideal has nilpotence order $e_{\mathfrak{q}_i/\mathfrak{p}}$. These orders are equal, proving $e_{\mathfrak{q}_i/\mathfrak{p}} = e_i$ and hence the asserted prime-ideal factorization. □

The theorem can be remembered through the following dictionary.

$\overline{f}$ *modulo* $\mathfrak{p}$	$\mathfrak{p}B$ *in* B
irreducible factor $\overline{g}_i$	prime ideal $\mathfrak{q}_i = (\mathfrak{p}, g_i(\theta))$
$\deg(\overline{g}_i)$	residue degree $f_{\mathfrak{q}_i/\mathfrak{p}}$
multiplicity e_i	ramification index $e_{\mathfrak{q}_i/\mathfrak{p}}$

Remark (The Fiber and Divisor Pictures). Dedekind–Kummer can be read directly from the fiber

$$\mathrm{Spec}(B/\mathfrak{p}B) \simeq \sqcup_{i=1}^{r} \mathrm{Spec}\left(\kappa(\mathfrak{p})[T]/\left(\overline{g}_i^{e_i}\right)\right)$$

Each irreducible factor $\overline{g}_i$ gives one closed point of residue degree $\deg(\overline{g}_i)$, while its exponent e_i gives the scheme-theoretic thickness of that point. In divisor language,

$$\pi^*[\mathfrak{p}] = \sum_{i=1}^{r} e_i[\mathfrak{q}_i], \quad \pi_*[\mathfrak{q}_i] = \deg(\overline{g}_i)[\mathfrak{p}]$$

Thus the factorization of $\overline{f}$ simultaneously displays the pullback and pushforward data of the closed point $[\mathfrak{p}]$.

4.3.3 Simple Applications

Corollary 4.3.1 (Splitting in Quadratic Fields). Let $d \neq 1$ be squarefree, let $K = \mathbb{Q}\left(\sqrt{d}\right)$, and define

$$\omega = \begin{cases} \left(1+\sqrt{d}\right)/2 & d \equiv 1 \bmod 4 \\ \sqrt{d} & d \not\equiv 1 \bmod 4 \end{cases}$$

By Chapter 3, $\mathcal{O}_K = \mathbb{Z}[\omega]$. The minimal polynomial of ω is

$$F(T) = \begin{cases} T^2 - T + (1-d)/4 & d \equiv 1 \bmod 4 \\ T^2 - d & d \not\equiv 1 \bmod 4 \end{cases}$$

For every rational prime p, the factorization of F modulo p determines the factorization of (p) in $\mathcal{O}_K$:

1. If $\overline{F}$ is irreducible, then (p) is inert.
2. If $\overline{F}$ is a product of two distinct linear factors, then (p) splits into two distinct primes.
3. If $\overline{F}$ is the square of a linear factor, then (p) is ramified.

Proof. The ring of integers is monogenic with generator ω, so there is no index obstruction. The three cases are exactly the three possible factorizations of a monic quadratic polynomial over $\mathbb{F}_p$, and Dedekind–Kummer translates their degrees and multiplicities into residue degrees and ramification indices. □

Remark (The Discriminant Test in Degree Two). The polynomial discriminant of F is

$$\Delta_K = \begin{cases} d & d \equiv 1 \bmod 4 \\ 4d & d \not\equiv 1 \bmod 4 \end{cases}$$

A quadratic polynomial has a repeated factor modulo p exactly when its discriminant vanishes modulo p. Hence

$$p \text{ ramifies in } K \Leftrightarrow p \mid \Delta_K$$

For odd p not dividing Δ_K, the prime splits or remains inert according as Δ_K is or is not a square modulo p.

Example (Running the Algorithm in $\mathbb{Z}[i]$). Here $\theta = i$, $B = \mathbb{Z}[i]$, and $f(T) = T^2 + 1$.

1. Modulo 2,
$$\overline{f} = (T+1)^2$$
Thus $(2) = (2, i+1)^2 = (1+i)^2$ is totally ramified.
2. Modulo 3, the polynomial $T^2 + 1$ is irreducible. Thus (3) is inert and has residue degree 2.
3. Modulo 5,
$$\overline{f} = (T-2)(T+2)$$
Hence
$$(5) = (5, i-2)(5, i+2)$$
and (5) splits completely.

Example (Three Behaviors in $\mathbb{Q}(\zeta_5)$). Let ζ_5 be a primitive fifth root of unity. Using the standard fact $\mathcal{O}_{\mathbb{Q}(\zeta_5)} = \mathbb{Z}[\zeta_5]$, Dedekind–Kummer applies to

$$f(T) = T^4 + T^3 + T^2 + T + 1$$

1. The reduction of f modulo 2 is irreducible, so (2) is inert with residue degree 4.
2. Modulo 5,
$$\overline{f} = (T-1)^4$$

so

$$(5) = (5, \zeta_5 - 1)^4$$

is totally ramified.

3. Modulo 11,

$$\overline{f} = (T - 4)(T - 9)(T - 5)(T - 3)$$

so (11) splits completely into four distinct primes of residue degree 1.

Example (Why the Index Condition Cannot Be Dropped). Let $K = \mathbb{Q}(\sqrt{5})$ and choose the integral primitive element $\theta = \sqrt{5}$. The maximal order is

$$B = \mathbb{Z}\left[\left(1 + \sqrt{5}\right)/2\right]$$

and the power order $\mathbb{Z}[\theta]$ has index ideal

$$[B : \mathbb{Z}[\theta]]_{\mathbb{Z}} = (2)$$

Indeed, if $\omega = \left(1 + \sqrt{5}\right)/2$, then $\theta = 2\omega - 1$, so the two power bases differ by a matrix of determinant 2.

The polynomial $T^2 - 5$ reduces modulo 2 to $(T - 1)^2$. Applying the theorem without checking the index would incorrectly suggest ramification. But the integral generator ω has minimal polynomial

$$T^2 - T - 1$$

whose reduction $T^2 + T + 1$ is irreducible over $\mathbb{F}_2$. Therefore (2) is actually inert in $\mathbb{Q}(\sqrt{5})$. The failure occurs exactly at the prime supporting the index divisor.

4.4 The Conductor

Dedekind–Kummer compares the power order $A[\theta]$ with the integral closure B. The index ideal measures the size of their difference as A-lattices. The *conductor* records a complementary piece of information: it is the largest ideal of the maximal order B that is already contained in the smaller order.

Definition 4.4.1 (Orders and the Conductor). An *A-order* in L is an A-subalgebra $C \subseteq L$ that is an A-lattice in L. Its integral closure is B. The *conductor of C in B* is the colon ideal

$$\mathfrak{f}_{B/C} := (C : B) := \{x \in L \mid xB \subseteq C\}$$

Proposition 4.4.1 (Equivalent Descriptions of the Conductor). Let $C \subseteq B$ be an A-order. Then

$$\mathfrak{f}_{B/C} = \mathrm{Ann}_C(B/C)$$

Moreover, $\mathfrak{f}_{B/C}$ is a nonzero ideal of both C and B, and it is the largest B-ideal contained in C.

Proof. Since $1 \in B$, every x satisfying $xB \subseteq C$ already lies in C. Such an x annihilates B/C, and the converse is immediate; this proves the displayed equality.

If $x \in \mathfrak{f}_{B/C}$ and $b \in B$, then

$$(bx)B \subseteq xB \subseteq C$$

so bx again belongs to the conductor. Thus it is a B-ideal, and hence also a C-ideal. The lattices B and C are commensurable, so some $0 \neq a \in A$ satisfies $aB \subseteq C$; therefore the conductor is nonzero. Finally, every B-ideal $J \subseteq C$ satisfies $JB \subseteq C$, hence $J \subseteq \mathfrak{f}_{B/C}$. □

> *Core idea.* The index measures how large B/C is, while the conductor measures how much of B can be multiplied back into C. One is a determinant; the other is an annihilator.

Proposition 4.4.2 (The Conductor Locus and the Index Divisor). For every nonzero prime $\mathfrak{p}$ of A, the following are equivalent.

1. $C_{\mathfrak{p}} = B_{\mathfrak{p}}$.
2. $\mathfrak{f}_{B/C} B_{\mathfrak{p}} = B_{\mathfrak{p}}$.
3. There is an $s \in \mathfrak{f}_{B/C} \cap A$ with $s \notin \mathfrak{p}$.
4. $v_{\mathfrak{p}}([B : C]_A) = 0$.

Consequently,

$$\sqrt{[B : C]_A} = \sqrt{\mathfrak{f}_{B/C} \cap A}$$

Thus the index ideal and the conductor have the same bad primes, although their multiplicities need not agree.

Proof. Localization commutes with the colon ideal because B is finite over A:

$$\mathfrak{f}_{B/C} B_{\mathfrak{p}} = \left(C_{\mathfrak{p}} : B_{\mathfrak{p}}\right)$$

The right-hand side contains 1 exactly when $C_{\mathfrak{p}} = B_{\mathfrak{p}}$. The localized conductor is the unit ideal exactly when it contains an element of A outside $\mathfrak{p}$, which gives the third condition. Finally, the module-index formula gives

$$v_{\mathfrak{p}}([B : C]_A) = \mathrm{length}_{A_{\mathfrak{p}}}\left(B_{\mathfrak{p}}/C_{\mathfrak{p}}\right)$$

so its valuation vanishes exactly when the two localized orders coincide. The equivalent conditions show that the two ideals have the same prime divisors, proving the radical identity. Equivalently, one may use

$$[B : C]_A = \mathrm{Fitt}_0^A(B/C) \subseteq \mathrm{Ann}_A(B/C) = \mathfrak{f}_{B/C} \cap A$$

together with the fact that the zeroth Fitting ideal and the annihilator of a finite torsion module have the same support. □

Remark (The Geometry of an Order). Put $S \coloneqq \operatorname{Spec} A$, $X_C \coloneqq \operatorname{Spec} C$, and $X_B \coloneqq \operatorname{Spec} B$. Because an A-order is a finite projective A-module of rank $n = [L : K]$, the map

$$\pi_C : X_C \to S$$

is finite and flat of degree n. Its generic fiber is

$$X_C \times_S \operatorname{Spec} K \simeq \operatorname{Spec} L \simeq X_B \times_S \operatorname{Spec} K$$

Thus an order is a one-dimensional arithmetic model of the field L: all orders have the same generic fiber, but they may have different closed fibers. Since B/C is torsion over A, these models differ at only finitely many closed points.

The finite birational map

$$\nu : X_B \to X_C$$

is the normalization of X_C. The scheme X_B is regular, whereas X_C may have nonnormal curve singularities. Geometrically, normalization repairs those singular points and may separate several branches lying over one closed point of X_C.

Remark (The Conductor Square). Write $\mathfrak{f} = \mathfrak{f}_{B/C}$ and put

$$Z_B \coloneqq \operatorname{Spec}(B/\mathfrak{f}), \quad Z_C \coloneqq \operatorname{Spec}(C/\mathfrak{f})$$

Since $\mathfrak{f}$ is an ideal in both rings, it defines finite closed subschemes $Z_B \subseteq X_B$ and $Z_C \subseteq X_C$. They fit into the *conductor square*

$$\begin{tikzcd}
Z_B = \operatorname{Spec}(B/\mathfrak{f}) \arrow[r] \arrow[d] & X_B = \operatorname{Spec} B \arrow[d, "\nu"'] \\
Z_C = \operatorname{Spec}(C/\mathfrak{f}) \arrow[r] & X_C = \operatorname{Spec} C
\end{tikzcd}$$

Algebraically, the square is encoded by the pullback identity

$$C \simeq B \times_{B/\mathfrak{f}} (C/\mathfrak{f})$$

Geometrically, X_C is reconstructed from its normalization X_B by replacing the finite subscheme Z_B with Z_C. This operation is called *pinching*. Outside Z_C, normalization is already an isomorphism, so the conductor is the scheme-theoretic interface along which the singular model is glued back together.

Remark (The Conductor Divisor). Since B is Dedekind, the conductor has a unique prime-ideal factorization

$$\mathfrak{f} = \prod_{\mathfrak{q}} \mathfrak{q}^{c_{\mathfrak{q}}}$$

and therefore determines the effective *conductor divisor*

$$D_{\mathrm{cond}}(B/C) \coloneqq \sum_{\mathfrak{q}} c_{\mathfrak{q}} [\mathfrak{q}]$$

Locally at $\mathfrak{q}$,

$$\mathfrak{f}B_{\mathfrak{q}} = \left(\mathfrak{q}B_{\mathfrak{q}}\right)^{c_{\mathfrak{q}}}$$

so $c_{\mathfrak{q}}$ measures the thickness of the correction locus along $[\mathfrak{q}]$. Its support maps precisely onto the support of the index divisor $\Phi_A([B : C]_A)$ in Spec A.

Remark (Connection with Dedekind–Kummer). For $C = A[\theta]$, a prime $\mathfrak{p}$ is good for θ exactly when it lies outside the conductor locus. Hence Dedekind–Kummer reads the factorization of $\mathfrak{p}B$ from $\overline{f}$ exactly where the normalization Spec $B \to$ Spec C is locally an isomorphism. The index condition and the conductor condition describe the same exceptional set.

Example (Quadratic Orders).

Let K be a quadratic number field, write $B = \mathcal{O}_K = \mathbb{Z}[\omega]$, and let $m \geq 1$. The subring

$$C_m \coloneqq \mathbb{Z} + mB = \mathbb{Z}[m\omega]$$

is an order of index m. Its conductor is

$$\mathfrak{f}_{B/C_m} = mB$$

Indeed, write $\omega^2 = t\omega - u$ with $t, u \in \mathbb{Z}$. For $x = a + b\omega \in B$, the conditions $x, x\omega \in C_m$ imply $m \mid b$ and $m \mid a + bt$, hence $m \mid a$. Thus $x \in mB$; the reverse inclusion is immediate. Moreover,

$$[B : C_m]_{\mathbb{Z}} = (m), \quad \mathfrak{f}_{B/C_m} \cap \mathbb{Z} = (m), \quad \operatorname{disc}(C_m) = m^2 \operatorname{disc}(B)$$

For $K = \mathbb{Q}\left(\sqrt{5}\right)$ and $C = \mathbb{Z}\left[\sqrt{5}\right]$, one has $m = 2$ and $\mathfrak{f}_{B/C} = 2B$, explaining again why the prime 2 is exactly the exceptional prime for the generator $\sqrt{5}$.

5 Galois Extensions

Galois theory studies a field extension through its symmetries. Separability ensures that all conjugates are distinct, while normality ensures that they remain inside the same field. When both conditions hold, the automorphism group remembers the entire lattice of intermediate fields.

5.1 Definitions and Basic Theory

Throughout this section, L/K is a finite extension and $\overline{K}$ is a fixed algebraic closure of K containing L.

5.1.1 Automorphisms and Normality

Definition 5.1.1 (Galois Group and Fixed Fields). Let $\mathbf{Fld}_K$ be the category whose objects are fields equipped with an embedding of K, and whose morphisms are field homomorphisms fixing K pointwise. Regard L as an object of $\mathbf{Fld}_K$. The *Galois group* of the extension is, by definition, its automorphism group in this category:

$$\mathrm{Gal}(L/K) \coloneqq \mathrm{Aut}_{\mathbf{Fld}_K}(L)$$

Concretely,

$$\mathrm{Gal}(L/K) = \{\sigma \in \mathrm{Aut}(L) \mid \sigma(a) = a \text{ for every } a \in K\}$$

with group law given by composition. This group is defined for every finite extension; the condition that the extension itself be Galois is introduced below. If $H \subseteq \mathrm{Gal}(L/K)$ is a subgroup, its *fixed field* is

$$L^H \coloneqq \{x \in L \mid \sigma(x) = x \text{ for every } \sigma \in H\}$$

Remark (The Categorical Meaning of a Fixed Field). Regard a group H as the one-object groupoid BH. An action of H on L by K-automorphisms is precisely a functor

$$\rho_H : BH \to \mathbf{Fld}_K, \quad * \mapsto L, \quad \sigma \mapsto \sigma$$

The fixed field is the *limit* of this action diagram:

$$L^H \simeq \lim_{\mathrm{B}\,H} \rho_H$$

Concretely, this limit is the simultaneous equalizer

$$L^H = \bigcap_{\sigma \in H} \mathrm{Eq}(\sigma, \mathrm{id}_L)$$

Its universal property says that a morphism $R \to L$ in $\mathbf{Fld}_K$ factors through L^H exactly when it is invariant under every element of H:

$$\mathrm{Hom}_{\mathbf{Fld}_K}\left(R, L^H\right) \simeq \left\{f \in \mathrm{Hom}_{\mathbf{Fld}_K}(R, L) \mid \sigma \circ f = f \text{ for every } \sigma \in H\right\}$$

Remark (The Galois Connection Before the Galois Theorem). Put $G = \mathrm{Gal}(L/K)$. Intermediate fields and subgroups form poset categories, and the assignments

$$E \mapsto \mathrm{Gal}(L/E), \quad H \mapsto L^H$$

are contravariant. For every intermediate field E and subgroup $H \subseteq G$,

$$H \subseteq \mathrm{Gal}(L/E) \Leftrightarrow E \subseteq L^H$$

This adjunction of order-reversing maps is a *Galois connection*. It always gives closure relations

$$E \subseteq L^{\mathrm{Gal}(L/E)}, \quad H \subseteq \mathrm{Gal}\left(L/L^H\right)$$

The fundamental theorem proved below says that, when L/K is finite Galois, both inclusions are equalities. In categorical language, the Galois connection becomes an anti-equivalence of the two poset categories.

Lemma 5.1.1 (Extension of Embeddings). Let $K \subseteq E \subseteq L$, and let $\tau : E \to \overline{K}$ be a K-embedding. Then τ extends to a K-embedding

$$\tilde{\tau} : L \to \overline{K}$$

Proof. Choose a tower

$$E = E_0 \subseteq E_1 \subseteq \ldots \subseteq E_r = L, \quad E_i = E_{i-1}(\alpha_i)$$

Suppose the embedding has been extended to E_{i-1}. Apply it to the coefficients of the minimal polynomial of α_i. The resulting polynomial has a root in the algebraically closed field $\overline{K}$, and choosing such a root extends the embedding to E_i. Induction reaches L. □

Definition 5.1.2 (Normal and Galois Extensions). The extension L/K is *normal* if every irreducible polynomial $f \in K[T]$ having one root in L splits completely over L.

It is *Galois* if it is both normal and separable.

Proposition 5.1.1 (Equivalent Forms of Normality). The following conditions are equivalent.

1. The extension L/K is normal.
2. Every K-embedding $\sigma : L \to \overline{K}$ satisfies $\sigma(L) = L$.
3. The field L is the splitting field over K of a family of polynomials in $K[T]$; since L/K is finite, the family may be replaced by one polynomial.

Proof. Suppose L/K is normal. For $\alpha \in L$, every conjugate of α over K lies in L, so every K-embedding sends L into L. Its image has the same degree over K as L, hence it equals L.

Conversely, let $\alpha \in L$ and let $\beta \in \overline{K}$ be any root of the minimal polynomial of α. The embedding $K(\alpha) \to \overline{K}$ sending α to β extends to L. By the second condition its image is L, so $\beta \in L$. Thus every minimal polynomial with a root in L splits in L.

If L/K is finite and normal, choose generators of L and take the product of their minimal polynomials. Its splitting field is L. Conversely, a K-embedding permutes the roots of any polynomial over K, so it preserves its splitting field. This proves the last equivalence. □

5.1.2 Artin's Theorem and Galois Characterizations

Lemma 5.1.2 (Artin Independence of Embeddings). Distinct field homomorphisms $\sigma_1, \ldots, \sigma_r : L \to \Omega$ are linearly independent over Ω as functions from L to Ω.

Proof. Suppose there is a nontrivial relation of minimal length

$$a_1\sigma_1 + \ldots + a_r\sigma_r = 0$$

and scale it so that $a_1 = 1$. Choose $x \in L$ with $\sigma_1(x) \neq \sigma_r(x)$. Evaluate the relation at xy and subtract $\sigma_1(x)$ times the relation evaluated at y. This gives a shorter nontrivial relation among $\sigma_2, \ldots, \sigma_r$, a contradiction. □

Theorem 5.1.1 (Artin's Fixed-Field Theorem). Let H be a finite subgroup of $\mathrm{Aut}(L)$ and put $F = L^H$. Then

$$[L : F] = |H|, \quad \mathrm{Gal}(L/F) = H$$

Proof. Write $H = \{\sigma_1, \ldots, \sigma_r\}$. Given $r + 1$ elements $x_1, \ldots, x_{r+1} \in L$, the homogeneous system

$$\sum_{j=1}^{r+1} a_j\sigma_i(x_j) = 0, \quad 1 \leq i \leq r$$

has a nonzero solution (a_j) in L^{r+1}. Choose one with minimal support and normalize a nonzero coordinate to 1. For every $\tau \in H$, applying τ to the equation indexed by $\tau^{-1} \circ \sigma_i$ shows that $(\tau(a_j))$ is another solution. Subtracting the two solutions would reduce the support unless $\tau(a_j) = a_j$ for every j. Hence all a_j lie in F, and the equation for the identity automorphism gives an F-linear relation among the x_j. Therefore $[L : F] \leq r$.

The preceding lemma says that the r automorphisms in H are linearly independent over L. Once $[L : F]$ is finite, the L-vector space of F-linear maps $L \to L$ has dimension $[L : F]$, so $r \leq [L : F]$. Equality follows. Finally, $H \subseteq \mathrm{Gal}(L/F)$ and every finite extension has at most its degree many automorphisms, so $\mathrm{Gal}(L/F) = H$. □

Theorem 5.1.2 (Characterizations of a Finite Galois Extension). Put $G = \mathrm{Gal}(L/K)$ and $n = [L : K]$. The following conditions are equivalent.

1. The extension L/K is Galois.
2. The field L is the splitting field over K of a separable polynomial.
3. The extension has the maximal possible number of automorphisms:

$$|G| = n$$

4. The base field is exactly the fixed field:

$$K = L^G$$

Proof. By Chapter 2, separability gives exactly n embeddings $L \to \overline{K}$. Normality says that all of them have image L, so they are precisely the elements of G and $|G| = n$. Conversely, if $|G| = n$, then these automorphisms exhaust all possible embeddings. Hence there are n distinct embeddings and all have image L, which gives separability and normality.

The equivalence with being the splitting field of a separable polynomial follows from the normality criterion. Finally, Artin's theorem gives

$$\left[L : L^G\right] = |G|$$

Since $K \subseteq L^G$, this equality shows that $|G| = n$ exactly when $K = L^G$. □

> *Core idea.* A finite extension is Galois precisely when every K-embedding of L into an algebraic closure is already a symmetry of L itself.

5.1.3 Conjugates and the Galois Action

Proposition 5.1.2 (Orbits Are Conjugates). Suppose L/K is Galois with group G, and let $\alpha \in L$. Define its stabilizer

$$G_\alpha \coloneqq \{\sigma \in G \mid \sigma(\alpha) = \alpha\}$$

Then the roots of the minimal polynomial $m_{\alpha,K}$ are exactly the distinct elements in the orbit $G \cdot \alpha$. In particular,

$$\deg(m_{\alpha,K}) = |G \cdot \alpha| = [G : G_\alpha]$$

and, choosing one representative from every coset,

$$m_{\alpha,K}(T) = \prod_{\sigma \in G/G_\alpha} (T - \sigma(\alpha))$$

Moreover,

$$K(\alpha) = L^{G_\alpha}$$

Proof. Every $\sigma(\alpha)$ is a root of $m_{\alpha,K}$. Conversely, an embedding $K(\alpha) \to \overline{K}$ sending α to any conjugate extends to an embedding of L. Since L/K is normal, the extension belongs to G. Thus conjugates and orbit elements coincide. The orbit–stabilizer theorem gives the degree formula. Finally, $K(\alpha) \subseteq L^{G_\alpha}$, and Artin's theorem together with the degree formula shows that both fields have the same degree over K. □

Corollary 5.1.1 (Trace and Norm in a Galois Extension). If L/K is Galois with group G, then for every $\alpha \in L$,

$$\mathrm{Tr}_{L/K}(\alpha) = \sum_{\sigma \in G} \sigma(\alpha), \quad \mathrm{N}_{L/K}(\alpha) = \prod_{\sigma \in G} \sigma(\alpha)$$

Thus trace and norm are respectively the additive and multiplicative symmetrizations of an element under the Galois action.

5.1.4 The Fundamental Correspondence

Theorem 5.1.3 (Fundamental Theorem of Galois Theory). Let L/K be finite Galois with group G. The assignments

$$E \mapsto \mathrm{Gal}(L/E), \quad H \mapsto L^H$$

give mutually inverse, inclusion-reversing bijections between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \subseteq G$. If $E = L^H$, then

$$[L : E] = |H|, \quad [E : K] = [G : H]$$

Furthermore, E/K is Galois if and only if H is normal in G. In that case restriction induces an isomorphism

$$\mathrm{Gal}(E/K) \simeq G/H$$

Proof. For a subgroup H, Artin's theorem gives $\mathrm{Gal}\left(L/L^H\right) = H$. For an intermediate field E, the extension L/E is separable, and every E-embedding of L is a K-embedding and hence preserves L. Thus L/E is Galois, and the fixed field of $\mathrm{Gal}(L/E)$ is E. This proves that the assignments are inverse. Their order reversal and the degree formulas follow immediately.

If H is normal, then

$$\sigma\left(L^H\right) = L^{\sigma H \sigma^{-1}} = L^H$$

for every $\sigma \in G$. Hence restriction gives a homomorphism $G \to \mathrm{Gal}(E/K)$ with kernel H; the degree formula shows that it is surjective. Conversely, if E/K is normal, every element of G preserves E, so H is the kernel of the restriction map and is therefore normal. □

The correspondence is summarized by the following dictionary.

Subgroup side	*Field side*
$H \subseteq G$	$E = L^H$
$H_1 \subseteq H_2$	$L^{H_2} \subseteq L^{H_1}$
$\lvert H\rvert$	$[L : E]$
$[G : H]$	$[E : K]$
H normal in G	E/K Galois
G/H	$\mathrm{Gal}(E/K)$

5.1.5 The Geometric Characterization

Theorem 5.1.4 (Tensor-Splitting Characterization). Let $G = \mathrm{Gal}(L/K)$. The extension L/K is Galois if and only if the canonical L-algebra map

$$\Psi : L \otimes_K L \to \prod_{\sigma \in G} L, \quad x \otimes y \mapsto (x\sigma(y))_{\sigma \in G}$$

is an isomorphism.

Proof. Suppose L/K is Galois and choose a primitive element θ. Its minimal polynomial has the distinct roots $\sigma(\theta)$ for $\sigma \in G$. Therefore

$$L \otimes_K L \simeq L[T]/\left(m_{\theta,K}\right) \simeq \prod_{\sigma \in G} L[T]/(T - \sigma(\theta)) \simeq \prod_{\sigma \in G} L$$

by the Chinese remainder theorem, and this is precisely Ψ. Conversely, if Ψ is an isomorphism, comparison of dimensions over L gives $[L : K] = |G|$, so the characterization theorem shows that L/K is Galois. □

Remark (The Galois Torsor Picture). Passing to spectra reverses the tensor-splitting isomorphism:

$$\operatorname{Spec} L \times_{\operatorname{Spec} K} \operatorname{Spec} L \simeq \sqcup_{\sigma \in G} \operatorname{Spec} L$$

The component indexed by σ is the graph of that automorphism. Thus $\operatorname{Spec} L \to \operatorname{Spec} K$ is a torsor under the finite constant group G in the étale topology: after pulling the cover back along itself, it becomes a disjoint union of copies of the base. Algebraically, the Galois group is the full deck-transformation group of this finite étale cover.

5.1.6 Basic Examples

Example (Quadratic Extensions). Let $d \in \mathbb{Q}^\times$ be nonsquare and put $L = \mathbb{Q}\left(\sqrt{d}\right)$. The polynomial $T^2 - d$ is separable and splits in L, so $L/\mathbb{Q}$ is Galois with

$$\operatorname{Gal}(L/\mathbb{Q}) = \{\mathrm{id}, c\}, \quad c\left(\sqrt{d}\right) = -\sqrt{d}$$

Thus its Galois group is cyclic of order 2.

Example (A Separable Extension That Is Not Normal). Let $\alpha = \sqrt[3]{2}$ and $L = \mathbb{Q}(\alpha)$. The extension has degree 3 and is separable, but the other roots $\zeta_3\alpha$ and $\zeta_3^2\alpha$ of $T^3 - 2$ do not lie in the real field L. Hence $L/\mathbb{Q}$ is not normal and

$$\operatorname{Gal}(L/\mathbb{Q}) = \{\mathrm{id}\}$$

Its normal closure is $M = \mathbb{Q}(\alpha, \zeta_3)$, and $\operatorname{Gal}(M/\mathbb{Q}) \simeq S_3$.

Example (Finite Fields). For every prime power q and $n \geq 1$, the extension $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois. Its Galois group is cyclic of order n, generated by the Frobenius automorphism

$$\operatorname{Frob}_q : x \mapsto x^q$$

The fixed elements of Frobenius are precisely the roots of $T^q - T$, namely $\mathbb{F}_q$.

Example (Cyclotomic Extensions). Let $m \geq 2$ and let ζ_m be a primitive mth root of unity. The cyclotomic field $\mathbb{Q}(\zeta_m)$ is the splitting field of the separable polynomial $T^m - 1$, so it is Galois over $\mathbb{Q}$. Every automorphism is determined by

$$\sigma_a(\zeta_m) = \zeta_m^a, \quad a \in (\mathbb{Z}/m\mathbb{Z})^\times$$

and this gives a canonical isomorphism

$$\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^\times$$

5.2 Splitting Primes in Galois Extensions

Remark (A–K–L–B–G Convention). From now on, whenever we study primes in a Galois extension, we use the following *standing convention* unless explicitly stated otherwise:

1. A is a Dedekind domain;
2. $K := \mathrm{Frac}(A)$ is its fraction field;
3. L/K is a finite Galois extension;
4. B is the integral closure of A in L;
5. $G := \mathrm{Gal}(L/K)$ is the Galois group.

We further abbreviate

$$n := [L : K] = |G|, \quad \pi : \mathrm{Spec}\, B \to \mathrm{Spec}\, A$$

Thus the letters A, K, L, B, G will always occur in this order: base ring, base field, extension field, integral closure, and symmetry group. Later sections may simply say “retain the A–K–L–B–G convention.”

Under this convention, for a nonzero prime $\mathfrak{p}$ of A, Chapter 3 gives

$$\mathfrak{p}B = \prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}^{e_{\mathfrak{q}/\mathfrak{p}}}$$

with residue degrees $f_{\mathfrak{q}/\mathfrak{p}} = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]$. The Galois hypothesis forces all primes above $\mathfrak{p}$ to look alike.

5.2.1 The Galois Action on Ideals and Primes

Proposition 5.2.1 (The Galois Action on Ideals). For $\sigma \in G$ and a fractional ideal I of B, define

$$\sigma(I) := \{\sigma(x) \mid x \in I\}$$

Then $\sigma(I)$ is a fractional ideal of B, and

$$\sigma(IJ) = \sigma(I)\sigma(J), \quad (\sigma\tau)(I) = \sigma(\tau(I))$$

Thus the fractional ideal group $\mathrm{FracId}(B)$ is a multiplicative G-module. The action preserves prime ideals and restricts to an action on $\mathrm{Spec}\, B$ over $\mathrm{Spec}\, A$.

Proof. If $b \in B$, then b satisfies a monic polynomial $f \in A[T]$. Since every $\sigma \in G$ fixes A,

$$f(\sigma(b)) = \sigma(f(b)) = 0$$

so $\sigma(b)$ is integral over A and lies in B. Applying the same argument to σ^{-1} gives $\sigma(B) = B$. It follows that the image of a finitely generated fractional B-ideal is again such an ideal.

Additivity and multiplicativity of σ give $\sigma(IJ) = \sigma(I)\sigma(J)$, and the composition law is immediate. A ring automorphism sends prime ideals to prime ideals. Finally, if $\mathfrak{q} \cap A = \mathfrak{p}$, then

$$\sigma(\mathfrak{q}) \cap A = \mathfrak{p}$$

because σ fixes every element of A. Hence the action preserves each fiber over Spec A. □

> **Remark (Divisors, Ideal Classes, and Geometry).** The action on fractional ideals becomes an action on divisors by
>
> $$\sigma\left(\sum_{\mathfrak{q}} n_{\mathfrak{q}}[\mathfrak{q}]\right) = \sum_{\mathfrak{q}} n_{\mathfrak{q}}[\sigma(\mathfrak{q})]$$
>
> Principal ideals are preserved because $\sigma(xB) = \sigma(x)B$, so G also acts on the class group $\mathrm{Cl}(B)$, or equivalently on $\mathrm{Pic}(\mathrm{Spec}\, B)$.
>
> Geometrically, each σ is an automorphism over Spec A of the finite map below—the algebro-geometric analogue of a deck transformation:
>
> $$\pi : \mathrm{Spec}\, B \to \mathrm{Spec}\, A$$
>
> The next theorem says that G acts transitively on every closed fiber.

> **Theorem 5.2.1 (Transitivity on Primes Above $\mathfrak{p}$).** For every nonzero prime $\mathfrak{p}$ of A, the action of G on
>
> $$\{\mathfrak{q} \subseteq B \mid \mathfrak{q} \cap A = \mathfrak{p}\}$$
>
> is transitive. Equivalently, the G-orbits of the nonzero primes of B are precisely the fibers of the contraction map
>
> $$\mathrm{Spec}\, B \to \mathrm{Spec}\, A$$

Proof. Suppose two primes $\mathfrak{q}_1, \mathfrak{q}_2$ above $\mathfrak{p}$ lie in distinct G-orbits. The primes in the orbit of $\mathfrak{q}_1$ are distinct maximal ideals and none equals $\mathfrak{q}_2$. By the Chinese remainder theorem, choose $b \in B$ such that

$$b \equiv 0 \bmod \mathfrak{q}_2$$

and

$$b \equiv 1 \bmod \sigma^{-1}(\mathfrak{q}_1) \quad \text{for every} \quad \sigma \in G$$

The Galois norm is

$$\mathrm{N}_{L/K}(b) = \prod_{\sigma \in G} \sigma(b)$$

It belongs to A: it lies in K and is integral over A. Since the product contains the factor $b \in \mathfrak{q}_2$,

$$\mathrm{N}_{L/K}(b) \in \mathfrak{q}_2 \cap A = \mathfrak{p}$$

On the other hand, applying σ to the chosen congruences gives $\sigma(b) \equiv 1 \bmod \mathfrak{q}_1$ for every σ, and hence

$$\mathrm{N}_{L/K}(b) \equiv 1 \bmod \mathfrak{q}_1$$

This contradicts $\mathfrak{p} \subseteq \mathfrak{q}_1$, proving transitivity. □

Remark (The Quotient Picture). Since $L^G = K$ and A is integrally closed,

$$B^G = B \cap K = A$$

Thus $\operatorname{Spec} A = \operatorname{Spec}\left(B^G\right)$ is the affine quotient of $\operatorname{Spec} B$ by G. The transitivity theorem is its pointwise shadow: each closed point below is one orbit of closed points above. The scheme-theoretic fiber contains more information than the orbit alone, because it also remembers ramification thickness and residue-field degrees.

5.2.2 Uniform Ramification and Residue Degrees

Proposition 5.2.2 (Conjugate Primes Have the Same Local Data). Let $\mathfrak{q}|\mathfrak{p}$ and $\sigma \in G$. The map σ induces a $\kappa(\mathfrak{p})$-isomorphism

$$\overline{\sigma}_{\mathfrak{q}} : \kappa(\mathfrak{q}) \to \kappa(\sigma(\mathfrak{q})), \quad x + \mathfrak{q} \mapsto \sigma(x) + \sigma(\mathfrak{q})$$

Moreover,

$$e_{\sigma(\mathfrak{q})/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{p}}, \quad f_{\sigma(\mathfrak{q})/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{p}}$$

In particular, separability of the residue extension is also independent of the chosen prime above $\mathfrak{p}$.

Proof. The residue-field map is well defined because $\sigma(\mathfrak{q})$ is the image of $\mathfrak{q}$, and it fixes $\kappa(\mathfrak{p})$ because σ fixes A. It is an isomorphism with inverse induced by σ^{-1}, proving equality of the residue degrees and invariance of separability.

Since $\sigma(\mathfrak{p}B) = \mathfrak{p}B$, applying σ to the unique prime-ideal factorization of $\mathfrak{p}B$ simply permutes its prime factors without changing their exponents. This proves equality of the ramification indices. □

Definition 5.2.1 (The Galois Splitting Type). By transitivity, the ramification index and residue degree are constant over $\mathfrak{p}$. We may therefore write

$$e_{\mathfrak{p}} := e_{\mathfrak{q}/\mathfrak{p}}, \quad f_{\mathfrak{p}} := f_{\mathfrak{q}/\mathfrak{p}}, \quad g_{\mathfrak{p}} := |\{\mathfrak{q}|\mathfrak{p}\}|$$

The triple

$$\left(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\right)$$

is the *splitting type* of $\mathfrak{p}$ in L.

Corollary 5.2.1 (The Galois Fundamental Identity). For every nonzero prime $\mathfrak{p}$ of A,

$$e_\mathfrak{p} f_\mathfrak{p} g_\mathfrak{p} = [L : K]$$

and the ideal factorization takes the uniform form

$$\mathfrak{p}B = \left(\prod_{\mathfrak{q}|\mathfrak{p}} \mathfrak{q}\right)^{e_\mathfrak{p}}$$

Proof. Substitute the common values $e_\mathfrak{p}$ and $f_\mathfrak{p}$ into the fundamental identity from Chapter 3:

$$[L : K] = \sum_{\mathfrak{q}|\mathfrak{p}} e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}}$$

There are $g_\mathfrak{p}$ equal summands. The uniform factorization follows in the same way. □

Remark (The Fiber and Divisor Pictures). The Chinese remainder theorem gives

$$\operatorname{Spec}(B/\mathfrak{p}B) \simeq \sqcup_{\mathfrak{q}|\mathfrak{p}} \operatorname{Spec}\left(B/\mathfrak{q}^{e_\mathfrak{p}}\right)$$

The group G permutes these components transitively. Every component has the same thickness $e_\mathfrak{p}$ and the same residue degree $f_\mathfrak{p}$. In divisor language,

$$\pi^*[\mathfrak{p}] = e_\mathfrak{p} \sum_{\mathfrak{q}|\mathfrak{p}} [\mathfrak{q}], \quad \pi_*[\mathfrak{q}] = f_\mathfrak{p}[\mathfrak{p}]$$

Thus the identity $e_\mathfrak{p} f_\mathfrak{p} g_\mathfrak{p} = n$ says that the total degree of the fiber is obtained by multiplying its common thickness, the degree of each point, and the number of points.

5.2.3 Extreme Splitting Behaviors

The terminology of Chapter 3 becomes particularly simple in a Galois extension. For the word *inert*, we retain the condition that the residue extension be separable.

Behavior	*Splitting type*	*Fiber picture*
split completely	$(1, 1, n)$	n distinct $\kappa(\mathfrak{p})$-rational points
inert	$(1, n, 1)$	one reduced point of residue degree n
totally ramified	$(n, 1, 1)$	one rational point of thickness n
general	(e, f, g), $efg = n$	g conjugate points, each of degree f and thickness e

Remark (Perfect Residue Fields). If $\kappa(\mathfrak{p})$ is perfect, every finite residue extension is separable. Then $\mathfrak{p}$ is unramified exactly when $e_\mathfrak{p} = 1$, and all three named behaviors in the table are characterized solely by the displayed triples. This applies in particular to rings of integers of number fields, whose residue fields are finite.

Corollary 5.2.2 (Prime-Degree Trichotomy). Suppose $[L : K] = \ell$ is prime. The identity

$$e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}} = \ell$$

leaves exactly three possible splitting types:

$$(\ell, 1, 1), \quad (1, \ell, 1), \quad (1, 1, \ell)$$

If $\kappa(\mathfrak{p})$ is perfect, these are respectively the cases in which $\mathfrak{p}$ is totally ramified, inert, and completely split.

Example (Splitting in the Gaussian Extension). Let $L = \mathbb{Q}(i)$, $K = \mathbb{Q}$, $A = \mathbb{Z}$, and $B = \mathbb{Z}[i]$. This is a quadratic Galois extension, so every rational prime has one of the three splitting types above. Dedekind–Kummer applied to $T^2 + 1$ gives

1. 2 is totally ramified, with type $(2, 1, 1)$;
2. if $p \equiv 1 \bmod 4$, then p splits completely, with type $(1, 1, 2)$;
3. if $p \equiv 3 \bmod 4$, then p is inert, with type $(1, 2, 1)$.

Complex conjugation interchanges the two primes above every split prime, making the transitive Galois action visible.

Remark (Why the Galois Hypothesis Matters). Without normality, the embeddings of L need not preserve L, so there may be too few automorphisms to move one prime above $\mathfrak{p}$ to another. The primes above a fixed base prime can then have different ramification indices and residue degrees. The uniform triple $\left(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\right)$ is a genuinely Galois phenomenon.

5.3 Decomposition and Inertia Groups

Retain the A–K–L–B–G convention, fix a nonzero prime $\mathfrak{p}$ of A, and choose one prime $\mathfrak{q}|\mathfrak{p}$ of B. The full Galois group moves $\mathfrak{q}$ through the entire fiber. To understand the arithmetic near this particular point, we first keep only the automorphisms that preserve it, and then ask how much of their action remains visible on its residue field.

> “The decomposition group remembers the symmetries that preserve the chosen branch $\mathfrak{q}$; the inertia group remembers the symmetries of that branch that become invisible on its reduced closed point.”

5.3.1 The Decomposition Group: Fixing a Branch

Definition 5.3.1 (Decomposition Group). The *decomposition group* of $\mathfrak{q}$ in L/K is the stabilizer of $\mathfrak{q}$ under the Galois action:

$$D_{\mathfrak{q}} = D_{\mathfrak{q}}(L/K) := \mathrm{Stab}_G(\mathfrak{q}) = \{\sigma \in G \mid \sigma(\mathfrak{q}) = \mathfrak{q}\}$$

The choice of $\mathfrak{q}$ is essential. There is generally no preferred point above $\mathfrak{p}$, but all choices give conjugate subgroups.

Proposition 5.3.1 (Conjugacy and the Order of the Decomposition Group). If $\tau \in G$, then

$$D_{\tau(\mathfrak{q})} = \tau D_{\mathfrak{q}} \tau^{-1}$$

Moreover,

$$\left[G : D_{\mathfrak{q}}\right] = g_{\mathfrak{p}}, \quad \left|D_{\mathfrak{q}}\right| = e_{\mathfrak{p}} f_{\mathfrak{p}}$$

Proof. An element $\sigma \in G$ stabilizes $\tau(\mathfrak{q})$ exactly when

$$\tau^{-1}\sigma\tau(\mathfrak{q}) = \mathfrak{q}$$

This is equivalent to $\tau^{-1}\sigma\tau \in D_{\mathfrak{q}}$, proving the conjugacy formula.

The orbit map induces a map

$$G/D_{\mathfrak{q}} \to \{\mathfrak{r} \subseteq B \mid \mathfrak{r} \cap A = \mathfrak{p}\}, \quad \sigma D_{\mathfrak{q}} \mapsto \sigma(\mathfrak{q})$$

It is well defined because two elements of the same left coset have the same effect on $\mathfrak{q}$. Conversely, if $\sigma(\mathfrak{q}) = \tau(\mathfrak{q})$, then $\tau^{-1}\sigma \in D_{\mathfrak{q}}$, so the two elements determine the same coset. The map is therefore injective, and it is surjective by the transitivity theorem. Hence

$$\left[G : D_{\mathfrak{q}}\right] = |\{\mathfrak{r}|\mathfrak{p}\}| = g_{\mathfrak{p}}$$

Finally, the Galois fundamental identity gives

$$\left|D_{\mathfrak{q}}\right| = |G|/g_{\mathfrak{p}} = e_{\mathfrak{p}} f_{\mathfrak{p}}$$

□

Proposition 5.3.2 (Global, Local, and Geometric Characterizations). Restriction to the local ring induces an isomorphism

$$D_{\mathfrak{q}} \simeq \mathrm{Aut}_{\mathbf{Alg}_{A_{\mathfrak{p}}}}\left(B_{\mathfrak{q}}\right)$$

Thus $D_{\mathfrak{q}}$ can equivalently be viewed as

1. the stabilizer of $\mathfrak{q}$ in G;
2. the automorphism group of the local extension $A_{\mathfrak{p}} \subseteq B_{\mathfrak{q}}$;
3. the automorphism group of the pointed cover $(\mathrm{Spec}\, B, \mathfrak{q}) \to (\mathrm{Spec}\, A, \mathfrak{p})$.

Proof. If $\sigma \in D_{\mathfrak{q}}$ and $s \in B \setminus \mathfrak{q}$, then $\sigma(s) \notin \mathfrak{q}$. Hence

$$\frac{b}{s} \mapsto \frac{\sigma(b)}{\sigma(s)}$$

defines an $A_{\mathfrak{p}}$-algebra automorphism of $B_{\mathfrak{q}}$.

Conversely, an $A_{\mathfrak{p}}$-algebra automorphism of $B_{\mathfrak{q}}$ extends uniquely to an automorphism of its fraction field L. It fixes the fraction field K of $A_{\mathfrak{p}}$, so it belongs to G. It also preserves the unique maximal ideal $\mathfrak{q}B_{\mathfrak{q}}$. Since

$$\mathfrak{q}B_{\mathfrak{q}} \cap B = \mathfrak{q}$$

the extended automorphism stabilizes $\mathfrak{q}$ and therefore lies in $D_{\mathfrak{q}}$. The two constructions are inverse to each other. □

> **Remark (Why the Word Decomposition?).** The factorization of $\mathfrak{p}B$ has $g_{\mathfrak{p}}$ conjugate branches. Choosing $\mathfrak{q}$ breaks the global symmetry from G to its stabilizer $D_{\mathfrak{q}}$. The coset space $G/D_{\mathfrak{q}}$ parametrizes the different branches, while $D_{\mathfrak{q}}$ contains the arithmetic that remains after one branch has been selected. In this sense $D_{\mathfrak{q}}$ is the local symmetry group of the decomposition of $\mathfrak{p}$.

5.3.2 The Residue Action and the Inertia Group

Every $\sigma \in D_{\mathfrak{q}}$ preserves $\mathfrak{q}$ and therefore descends to an automorphism of the residue field. This gives a homomorphism

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

defined by

$$\rho_{\mathfrak{q}}(\sigma)(x + \mathfrak{q}) = \sigma(x) + \mathfrak{q}$$

It is well defined because $\sigma(\mathfrak{q}) = \mathfrak{q}$, and it fixes $\kappa(\mathfrak{p})$ because σ fixes A pointwise.

> **Definition 5.3.2 (Inertia Group).** The *inertia group* of $\mathfrak{q}$ is the kernel of the residue action:
>
> $$I_{\mathfrak{q}} = I_{\mathfrak{q}}(L/K) := \ker\left(\rho_{\mathfrak{q}}\right)$$
>
> Equivalently,
>
> $$I_{\mathfrak{q}} = \left\{\sigma \in D_{\mathfrak{q}} \mid \sigma(x) \equiv x \bmod \mathfrak{q} \ \text{ for every } \ x \in B\right\}$$

Thus $D_{\mathfrak{q}}$ preserves the point $\mathfrak{q}$, whereas $I_{\mathfrak{q}}$ preserves both the point and every residue class at that point.

> **Theorem 5.3.1 (Lifting Residue-Field Automorphisms).** The residue extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is normal, and the homomorphism
>
> $$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$
>
> is surjective.

Proof. Put $k := \kappa(\mathfrak{p})$ and $l := \kappa(\mathfrak{q})$. We first construct enough polynomials over k whose roots already lie in l.

Let $\bar{c} \in l$. By the Chinese remainder theorem, choose a lift $c \in B$ such that

$$c + \mathfrak{q} = \bar{c}$$

and

$$c \equiv 0 \bmod \sigma^{-1}(\mathfrak{q}) \ \text{ for every } \ \sigma \in G \setminus D_{\mathfrak{q}}$$

Repeated ideals among the $\sigma^{-1}(\mathfrak{q})$ impose the same condition, and every distinct ideal occurring here is maximal and different from $\mathfrak{q}$, so the Chinese remainder theorem applies. Define

$$P_c(T) := \prod_{\sigma \in G} (T - \sigma(c))$$

The coefficients of P_c are fixed by G, hence lie in K. They are also integral over A, because they are polynomials in the integral elements $\sigma(c)$. Since A is integrally closed, $P_c \in A[T]$.

Reduce P_c modulo $\mathfrak{q}$. If $\sigma \in G \setminus D_{\mathfrak{q}}$, our congruence gives $\sigma(c) \in \mathfrak{q}$. If $\sigma \in D_{\mathfrak{q}}$, its image is $\rho_{\mathfrak{q}}(\sigma)(\bar{c})$. Therefore, for $m := \left|G \setminus D_{\mathfrak{q}}\right|$,

$$\overline{P_c}(T) = T^m \prod_{\sigma \in D_{\mathfrak{q}}} \left(T - \rho_{\mathfrak{q}}(\sigma)(\bar{c})\right)$$

This polynomial lies in $k[T]$, has $\bar{c}$ as a root, and splits in $l[T]$. The minimal polynomial of $\bar{c}$ over k divides $\overline{P_c}$, so it also splits in $l[T]$. Since $\bar{c}$ was arbitrary, l/k is normal.

It remains to prove surjectivity. Let F be the largest subfield of l separable over k. The preceding argument shows that every minimal polynomial over k of an element of F splits in l. Its roots are separable over k, hence lie in F; therefore F/k is Galois.

If $F = k$, then l/k is purely inseparable and has no nonidentity k-automorphisms, so surjectivity is immediate. Otherwise choose a nonzero primitive element $\bar{c}$ with $F = k(\bar{c})$, and make the same CRT choice above. Let $h \in k[T]$ be its minimal polynomial. After removing the zero factors, all roots of $\overline{P_c}(T)/T^m$ are k-conjugates of $\bar{c}$. This quotient belongs to $k[T]$, so its only irreducible factor is h; hence

$$\overline{P_c}(T)/T^m = h(T)^r$$

for some $r \geq 1$. Every root of h must therefore occur as $\rho_{\mathfrak{q}}(\sigma)(\bar{c})$ for some $\sigma \in D_{\mathfrak{q}}$. Since $\bar{c}$ generates F, restriction gives a surjection

$$D_{\mathfrak{q}} \to \mathrm{Gal}(F/k)$$

Finally, restriction from $\mathrm{Gal}(l/k)$ to $\mathrm{Gal}(F/k)$ is injective: an automorphism fixing F fixes the purely inseparable extension l/F pointwise. Its image contains the restrictions of $D_{\mathfrak{q}}$, which already fill $\mathrm{Gal}(F/k)$, so it is an isomorphism. Thus $\rho_{\mathfrak{q}}$ is surjective. □

Corollary 5.3.1 (The Fundamental Exact Sequence). There is a short exact sequence

$$1 \to I_{\mathfrak{q}} \to D_{\mathfrak{q}} \to \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) \to 1$$

Consequently,

$$D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

If

$$f_{\mathfrak{p}}^{\mathrm{i}} := [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_{\mathrm{i}}$$

denotes the inseparable degree of the residue extension, then

$$\left|I_{\mathfrak{q}}\right| = e_{\mathfrak{p}} f_{\mathfrak{p}}^{\mathrm{i}}$$

If the residue extension is separable, then in addition

$$|\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))| = f_{\mathfrak{p}}, \quad \left|I_{\mathfrak{q}}\right| = e_{\mathfrak{p}}$$

Proof. Exactness at $I_\mathfrak{q}$ and $D_\mathfrak{q}$ follows from the definition of the inertia group, and exactness on the right is the surjectivity theorem. The first isomorphism theorem also gives the displayed quotient isomorphism. Since the residue extension is normal, the separable-degree formula from Chapter 2 gives

$$|\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))| = [\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_s$$

The order formula for $D_\mathfrak{q}$ now yields

$$\left|I_\mathfrak{q}\right| = \left(e_\mathfrak{p} f_\mathfrak{p}\right)/[\kappa(\mathfrak{q}) : \kappa(\mathfrak{p})]_s = e_\mathfrak{p} f_\mathfrak{p}^{\mathrm{i}}$$

When the residue extension is separable, its inseparable degree is one and its separable degree is $f_\mathfrak{p}$, proving the final assertions. □

Corollary 5.3.2 (Conjugacy of Inertia Groups). For every $\tau \in G$,

$$I_{\tau(\mathfrak{q})} = \tau I_\mathfrak{q} \tau^{-1}$$

Proof. The corresponding statement for decomposition groups was proved above. Conjugation by τ identifies the residue action at $\mathfrak{q}$ with the residue action at $\tau(\mathfrak{q})$: after using the isomorphism

$$\kappa(\mathfrak{q}) \simeq \kappa(\tau(\mathfrak{q}))$$

induced by τ, the action of $\tau\sigma\tau^{-1}$ is the conjugate of the action of σ. It is the identity on the residue field exactly when σ is, so conjugation carries one kernel onto the other. □

Remark (The Geometry of Inertia). Localizing the fiber at $\mathfrak{q}$ gives the zero-dimensional local scheme

$$\mathrm{Spec}\left(B_\mathfrak{q}/\mathfrak{p}B_\mathfrak{q}\right) = \mathrm{Spec}\left(B_\mathfrak{q}/\mathfrak{q}^{e_\mathfrak{p}}B_\mathfrak{q}\right)$$

Its reduced closed subscheme is $\mathrm{Spec}\,\kappa(\mathfrak{q})$. The decomposition group acts on the entire thickened local fiber, and its quotient $D_\mathfrak{q}/I_\mathfrak{q}$ is precisely the part of this action visible on the reduced point. The inertia group fixes the reduced point together with its residue functions but may still act nontrivially in the transverse, nilpotent direction.

When $\kappa(\mathfrak{p})$ is perfect, $\left|I_\mathfrak{q}\right| = e_\mathfrak{p}$. Thus inertia measures exactly the ramification thickness: the fiber is reduced at $\mathfrak{q}$ if and only if $I_\mathfrak{q}$ is trivial. Over an imperfect residue field, inertia also contains the inseparable factor $f_\mathfrak{p}^{\mathrm{i}}$, which geometrically records the extra multiplicity that appears after passing to an algebraic closure.

5.3.3 Decomposition and Inertia Fields

Definition 5.3.3 (The Two Fixed Fields). Assume that $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. The fixed fields

$$L^{D_\mathfrak{q}} \quad \text{and} \quad L^{I_\mathfrak{q}}$$

are called the *decomposition field* and *inertia field* attached to $\mathfrak{q}$, respectively. Since $I_\mathfrak{q} \subseteq D_\mathfrak{q}$, they form a tower

$$K \subseteq L^{D_{\mathfrak{q}}} \subseteq L^{I_{\mathfrak{q}}} \subseteq L$$

Theorem 5.3.2 (The Field Tower Separates the Three Local Invariants). In the preceding tower,

$$\left[L : L^{I_{\mathfrak{q}}}\right] = e_{\mathfrak{p}}$$

$$\left[L^{I_{\mathfrak{q}}} : L^{D_{\mathfrak{q}}}\right] = f_{\mathfrak{p}}$$

$$\left[L^{D_{\mathfrak{q}}} : K\right] = g_{\mathfrak{p}}$$

Moreover, $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois with group

$$D_{\mathfrak{q}}/I_{\mathfrak{q}} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

Proof. The fundamental theorem of Galois theory gives

$$\left[L : L^{I_{\mathfrak{q}}}\right] = \left|I_{\mathfrak{q}}\right| = e_{\mathfrak{p}}$$

and

$$\left[L^{D_{\mathfrak{q}}} : K\right] = \left[G : D_{\mathfrak{q}}\right] = g_{\mathfrak{p}}$$

The inertia group is normal in $D_{\mathfrak{q}}$ because it is the kernel of $\rho_{\mathfrak{q}}$. Applying Galois theory to the Galois extension $L/L^{D_{\mathfrak{q}}}$, whose Galois group is $D_{\mathfrak{q}}$, shows that $L^{I_{\mathfrak{q}}}/L^{D_{\mathfrak{q}}}$ is Galois with group $D_{\mathfrak{q}}/I_{\mathfrak{q}}$. The fundamental exact sequence identifies this quotient with the residue-field Galois group, whose order is $f_{\mathfrak{p}}$. This proves the middle equality and all the assertions. □

Remark (Reading the Tower from Top to Bottom). Descending from L to the inertia field removes the symmetries responsible for ramification, contributing the factor $e_{\mathfrak{p}}$. Descending from the inertia field to the decomposition field removes the visible residue-field symmetries, contributing $f_{\mathfrak{p}}$. The remaining degree down to K counts the conjugate choices of a prime above $\mathfrak{p}$, contributing $g_{\mathfrak{p}}$. The identity

$$[L : K] = e_{\mathfrak{p}} f_{\mathfrak{p}} g_{\mathfrak{p}}$$

is therefore the degree formula for this field tower.

5.3.4 Behavior in Intermediate Fields

Let $K \subseteq E \subseteq L$, put $B_E := B \cap E$, and let $\mathfrak{q}_E := \mathfrak{q} \cap B_E$. Then B_E is the integral closure of A in E, and $\mathfrak{q}|\mathfrak{q}_E|\mathfrak{p}$.

Proposition 5.3.3 (Compatibility with an Intermediate Field). Put $H := \mathrm{Gal}(L/E)$. Then

$$D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap H$$

and

$$I_{\mathfrak{q}}(L/E) = I_{\mathfrak{q}}(L/K) \cap H$$

Proof. An element of G belongs to $D_{\mathfrak{q}}(L/E)$ exactly when it fixes E pointwise and stabilizes $\mathfrak{q}$. The first condition says that it lies in H, and the second that it lies in $D_{\mathfrak{q}}(L/K)$, proving the first equality.

On this intersection the residue action is simply the restriction of the residue action for L/K. Its kernel consists of the elements acting trivially on $\kappa(\mathfrak{q})$, which proves the second equality. □

Corollary 5.3.3 (Maximality of the Inertia and Decomposition Fields). Continue to assume that $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is separable. For every intermediate field E,

$$e_{\mathfrak{q}_E/\mathfrak{p}} = 1 \Leftrightarrow E \subseteq L^{I_{\mathfrak{q}}}$$

and

$$e_{\mathfrak{q}_E/\mathfrak{p}} = f_{\mathfrak{q}_E/\mathfrak{p}} = 1 \Leftrightarrow E \subseteq L^{D_{\mathfrak{q}}}$$

Thus the inertia field is the largest intermediate field in which the selected branch is unramified, while the decomposition field is the largest one in which that branch is both unramified and residue-trivial.

Proof. Let $H = \mathrm{Gal}(L/E)$, and abbreviate $I := I_{\mathfrak{q}}(L/K)$ and $D := D_{\mathfrak{q}}(L/K)$. By the compatibility proposition and the order formula applied to L/E,

$$e_{\mathfrak{q}/\mathfrak{q}_E} = |I \cap H|$$

The multiplicativity of ramification indices gives

$$e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_E} e_{\mathfrak{q}_E/\mathfrak{p}}$$

Since $e_{\mathfrak{q}/\mathfrak{p}} = |I|$, the equality $e_{\mathfrak{q}_E/\mathfrak{p}} = 1$ holds exactly when $|I \cap H| = |I|$, equivalently $I \subseteq H$. By the Galois correspondence, this is equivalent to $E \subseteq L^I$.

Likewise, the order formula for the decomposition group and multiplicativity of both local invariants give

$$e_{\mathfrak{q}/\mathfrak{q}_E} f_{\mathfrak{q}/\mathfrak{q}_E} = |D \cap H|$$

and

$$e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_E} f_{\mathfrak{q}/\mathfrak{q}_E} e_{\mathfrak{q}_E/\mathfrak{p}} f_{\mathfrak{q}_E/\mathfrak{p}}$$

The product on the left is $|D|$. Hence $e_{\mathfrak{q}_E/\mathfrak{p}} f_{\mathfrak{q}_E/\mathfrak{p}} = 1$ exactly when $D \subseteq H$, equivalently $E \subseteq L^D$. Since both invariants are positive integers, their product is one exactly when each is one. □

Example (The Three Gaussian Behaviors Revisited). Let $L = \mathbb{Q}(i)$ and $G = \{1, \mathrm{conj}\}$ as above.

1. If $p \equiv 1 \bmod 4$, then p splits completely. For either prime $\mathfrak{q}|p$, the nontrivial element exchanges the two branches, so $D_{\mathfrak{q}} = I_{\mathfrak{q}} = \{1\}$.
2. If $p \equiv 3 \bmod 4$, then p is inert. There is only one branch, so $D_{\mathfrak{q}} = G$, but the fiber is unramified and $I_{\mathfrak{q}} = \{1\}$. Thus $D_{\mathfrak{q}}$ acts faithfully on $\kappa(\mathfrak{q}) \simeq \mathbb{F}_{p^2}$ over $\mathbb{F}_p$.

3. The prime 2 is totally ramified. Again there is only one branch, so $D_{\mathfrak{q}} = G$, but now the residue field does not grow. The residue action is trivial and $I_{\mathfrak{q}} = G$; all the nontrivial local symmetry lies in the ramification thickness.

These cases isolate the three numbers in (e, f, g): splitting is detected by the index of $D_{\mathfrak{q}}$, residue-field growth by $D_{\mathfrak{q}}/I_{\mathfrak{q}}$, and ramification by $I_{\mathfrak{q}}$.

5.4 Frobenius Elements and Artin Symbols

Continue with the A–K–L–B–G convention and assume that the residue fields under consideration are finite. For a nonzero prime $\mathfrak{p}$ of A, write

$$\mathrm{N}(\mathfrak{p}) := |\kappa(\mathfrak{p})|$$

This agrees with the absolute ideal norm in the number-field case. If $\mathfrak{q}|\mathfrak{p}$, then $\kappa(\mathfrak{q})$ is also finite. The finite-field Frobenius gives a distinguished automorphism of this residue extension. At an unramified prime, the fundamental exact sequence lets us lift that automorphism uniquely to G.

> “A Frobenius element is the global Galois symmetry whose reduction at one chosen prime is the canonical Frobenius symmetry of the finite residue field. Its conjugacy class records the prime without remembering the choice of a point above it.”

5.4.1 Frobenius on the Residue Field

Proposition 5.4.1 (Finite-Field Frobenius). Let $\mathfrak{q}|\mathfrak{p}$ and put $f = f_{\mathfrak{p}}$. The residue extension is Galois and cyclic:

$$\mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p})) = \langle \varphi_{\mathfrak{q}/\mathfrak{p}} \rangle$$

where

$$\varphi_{\mathfrak{q}/\mathfrak{p}} : x \mapsto x^{\mathrm{N}(\mathfrak{p})}$$

The order of $\varphi_{\mathfrak{q}/\mathfrak{p}}$ is f.

Proof. Put $Q = \mathrm{N}(\mathfrak{p})$. Then

$$\kappa(\mathfrak{p}) \simeq \mathbb{F}_Q, \quad \kappa(\mathfrak{q}) \simeq \mathbb{F}_{Q^f}$$

The map $x \mapsto x^Q$ fixes $\mathbb{F}_Q$ and its rth power is $x \mapsto x^{Q^r}$. Its fixed field inside $\mathbb{F}_{Q^f}$ is $\mathbb{F}_{Q^{\gcd(r,f)}}$, so its smallest positive power equal to the identity is the fth. Thus it generates a cyclic subgroup of order f. Since the extension has degree f, this subgroup is the entire Galois group. □

Remark (Arithmetic versus Geometric Frobenius). We use the *arithmetic Frobenius*

$$x \mapsto x^{\mathrm{N}(\mathfrak{p})}$$

throughout. Many texts in algebraic geometry instead call its inverse the *geometric Frobenius.* With that convention every Frobenius element and Artin symbol below is replaced

> by its inverse. Neither convention changes the resulting conjugacy classes up to applying inversion, but formulas must not mix the two choices.

5.4.2 Frobenius Elements and Frobenius Classes

Suppose now that $\mathfrak{p}$ is unramified. Finite residue fields are perfect, so $I_{\mathfrak{q}} = \{1\}$ and the fundamental exact sequence becomes an isomorphism

$$\rho_{\mathfrak{q}} : D_{\mathfrak{q}} \simeq \mathrm{Gal}(\kappa(\mathfrak{q})/\kappa(\mathfrak{p}))$$

> **Definition 5.4.1 (Frobenius Element).** The *Frobenius element* at $\mathfrak{q}$ is the unique inverse image of the residue-field Frobenius:
>
> $$\mathrm{Frob}_{\mathfrak{q}} := \rho_{\mathfrak{q}}^{-1}\left(\varphi_{\mathfrak{q}/\mathfrak{p}}\right) \in D_{\mathfrak{q}} \subseteq G$$

The restriction to unramified primes is essential. At a ramified prime the map $\rho_{\mathfrak{q}}$ has kernel $I_{\mathfrak{q}}$, so a residue-field Frobenius has an entire $I_{\mathfrak{q}}$-coset of lifts rather than a distinguished lift.

> **Theorem 5.4.1 (Congruence Characterization and Order).** The element $\mathrm{Frob}_{\mathfrak{q}}$ is the unique $\sigma \in G$ satisfying
>
> $$\sigma(x) \equiv x^{\mathrm{N}(\mathfrak{p})} \bmod \mathfrak{q} \quad \text{for every} \quad x \in B$$
>
> Moreover,
>
> $$D_{\mathfrak{q}} = \left\langle \mathrm{Frob}_{\mathfrak{q}} \right\rangle, \quad \mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{q}}\right) = f_{\mathfrak{p}}$$

Proof. By definition, $\mathrm{Frob}_{\mathfrak{q}}$ reduces to $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$, so it satisfies the congruence.

Conversely, suppose $\sigma \in G$ satisfies it. For every $x \in \mathfrak{q}$, the right-hand side is zero modulo $\mathfrak{q}$, hence $\sigma(x) \in \mathfrak{q}$. Thus $\sigma(\mathfrak{q}) \subseteq \mathfrak{q}$. Both are maximal ideals of B, so equality holds and $\sigma \in D_{\mathfrak{q}}$. Its image under $\rho_{\mathfrak{q}}$ is the residue-field Frobenius. Since $\rho_{\mathfrak{q}}$ is an isomorphism, $\sigma = \mathrm{Frob}_{\mathfrak{q}}$.

The same isomorphism identifies $D_{\mathfrak{q}}$ with the cyclic residue-field Galois group, whose generator is $\varphi_{\mathfrak{q}/\mathfrak{p}}$ and whose order is $f_{\mathfrak{p}}$. Its inverse image therefore generates $D_{\mathfrak{q}}$ and has the stated order. □

> **Theorem 5.4.2 (Frobenius Elements Form a Conjugacy Class).** If $\mathfrak{p}$ is unramified, then for every $\tau \in G$,
>
> $$\mathrm{Frob}_{\tau(\mathfrak{q})} = \tau \mathrm{Frob}_{\mathfrak{q}} \tau^{-1}$$
>
> Consequently,
>
> $$\{\mathrm{Frob}_{\mathfrak{r}} : \mathfrak{r} | \mathfrak{p}\}$$
>
> is one conjugacy class in G.

Proof. Let $x \in B$ and put $y = \tau^{-1}(x)$. The congruence characterization at $\mathfrak{q}$ gives

$$\mathrm{Frob}_{\mathfrak{q}}(y) - y^{\mathrm{N}(\mathfrak{p})} \in \mathfrak{q}$$

Applying τ and using that a field automorphism commutes with integer powers yields

$$\tau \mathrm{Frob}_{\mathfrak{q}} \tau^{-1}(x) - x^{\mathrm{N}(\mathfrak{p})} \in \tau(\mathfrak{q})$$

The uniqueness part of the preceding theorem therefore gives the conjugacy formula.

By transitivity, every prime $\mathfrak{r}|\mathfrak{p}$ has the form $\tau(\mathfrak{q})$, so every Frobenius element above $\mathfrak{p}$ is a conjugate of $\mathrm{Frob}_{\mathfrak{q}}$. Conversely, the formula shows that every conjugate is the Frobenius element at a prime above $\mathfrak{p}$. The two sets are equal. □

Definition 5.4.2 (Frobenius Class). For an unramified prime $\mathfrak{p}$, its *Frobenius class* is

$$\mathrm{Frob}_{\mathfrak{p}} := \left\{\mathrm{Frob}_{\mathfrak{q}} : \mathfrak{q}|\mathfrak{p}\right\} \subseteq G$$

It is canonical: it depends on $\mathfrak{p}$ but not on a choice of $\mathfrak{q}|\mathfrak{p}$. If G is abelian, every conjugacy class is a singleton, so all the elements $\mathrm{Frob}_{\mathfrak{q}}$ above $\mathfrak{p}$ are equal; in this case we also denote their common value by $\mathrm{Frob}_{\mathfrak{p}}$.

Remark (The Geometric Meaning of Frobenius). Because $\mathfrak{p}$ is unramified and its residue field is finite, the fiber

$$X_{\mathfrak{p}} := \mathrm{Spec}(B/\mathfrak{p}B)$$

is finite étale over $\mathrm{Spec}\, \kappa(\mathfrak{p})$. Each component is a closed point $x_{\mathfrak{q}} = \mathrm{Spec}\, \kappa(\mathfrak{q})$. The decomposition group is the symmetry group preserving this component, and $\mathrm{Frob}_{\mathfrak{q}}$ is the unique symmetry whose action on its residue field is $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$.

Thus a Frobenius element is a form of *arithmetic monodromy*: it records how the canonical finite-field symmetry is lifted through the cover $\mathrm{Spec}\, B \to \mathrm{Spec}\, A$. Changing the chosen point conjugates the lift, which is why the intrinsic datum attached to $\mathfrak{p}$ is a conjugacy class rather than an individual group element.

5.4.3 Artin Symbols and Splitting

Definition 5.4.3 (Artin Symbol at a Prime Above $\mathfrak{p}$). For an unramified prime $\mathfrak{q}|\mathfrak{p}$, the *Artin symbol* is another notation for its Frobenius element:

$$\left(\frac{L/K}{\mathfrak{q}}\right) := \mathrm{Frob}_{\mathfrak{q}}$$

Our Artin symbols therefore use arithmetic Frobenius. Under the geometric Frobenius convention, the right-hand side would be inverted.

Corollary 5.4.1 (The Frobenius Class Determines the Splitting Type). For every unramified $\mathfrak{p}$ and every $\mathfrak{q}|\mathfrak{p}$,

$$f_{\mathfrak{p}} = \mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{q}}\right), \quad g_{\mathfrak{p}} = [L:K]/\,\mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{q}}\right)$$

In particular,

1. $\mathfrak{p}$ splits completely if and only if $\left(\frac{L/K}{\mathfrak{q}}\right) = 1$;
2. $\mathfrak{p}$ is inert if and only if $\mathrm{Frob}_{\mathfrak{q}}$ generates G.

Proof. The order formula was proved in the congruence-characterization theorem. Since $\mathfrak{p}$ is unramified, $e_{\mathfrak{p}} = 1$, so

$$f_{\mathfrak{p}} g_{\mathfrak{p}} = [L : K]$$

gives the formula for $g_{\mathfrak{p}}$.

Complete splitting is equivalent to $f_{\mathfrak{p}} = 1$, hence to the Frobenius element being the identity. The prime is inert exactly when $g_{\mathfrak{p}} = 1$, equivalently $D_{\mathfrak{q}} = G$. But $D_{\mathfrak{q}} = \left\langle \mathrm{Frob}_{\mathfrak{q}} \right\rangle$, which proves the second criterion. □

Remark (Why One Conjugacy Class Is Enough). Conjugate elements have the same order. Therefore the Frobenius class alone recovers $f_{\mathfrak{p}}$ and, for an unramified prime, the entire splitting type

$$\left(e_{\mathfrak{p}}, f_{\mathfrak{p}}, g_{\mathfrak{p}}\right) = \left(1, \mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{p}}\right), [L : K]/\ \mathrm{ord}\left(\mathrm{Frob}_{\mathfrak{p}}\right)\right)$$

Here the order of a conjugacy class means the common order of any of its elements.

This is the first indication that Frobenius classes encode the arithmetic of prime decomposition very efficiently.

5.4.4 Compatibility in Towers

Proposition 5.4.2 (Functoriality of Artin Symbols). Let $K \subseteq E \subseteq L$, put $B_E = B \cap E$, and let $\mathfrak{q}_E = \mathfrak{q} \cap B_E$. Suppose $\mathfrak{q} | \mathfrak{p}$ is unramified. Then

$$\left(\frac{L/E}{\mathfrak{q}}\right) = \left(\frac{L/K}{\mathfrak{q}}\right)^{f_{\mathfrak{q}_E/\mathfrak{p}}}$$

If E/K is Galois, restriction also gives

$$\left(\frac{E/K}{\mathfrak{q}_E}\right) = \left(\frac{L/K}{\mathfrak{q}}\right)|_E$$

Proof. Put $\sigma = \left(\frac{L/K}{\mathfrak{q}}\right)$ and $d = f_{\mathfrak{q}_E/\mathfrak{p}}$. Since the total ramification index is one, both subextensions are unramified. We have

$$D_{\mathfrak{q}}(L/E) = D_{\mathfrak{q}}(L/K) \cap \mathrm{Gal}(L/E)$$

The group on the right is a subgroup of the cyclic group $D_{\mathfrak{q}}(L/K) = \langle \sigma \rangle$. Its index in that cyclic group is

$$f_{\mathfrak{q}/\mathfrak{p}} / f_{\mathfrak{q}/\mathfrak{q}_E} = f_{\mathfrak{q}_E/\mathfrak{p}} = d$$

by multiplicativity of residue degrees. The unique subgroup of index d in $\langle \sigma \rangle$ is $\left\langle \sigma^d \right\rangle$. Moreover,

$$|\kappa(\mathfrak{q}_E)| = \mathrm{N}(\mathfrak{p})^d$$

so σ^d acts on $\kappa(\mathfrak{q})$ by $x \mapsto x^{|\kappa(\mathfrak{q}_E)|}$. The congruence characterization relative to L/E now shows

$$\left(\frac{L/E}{\mathfrak{q}}\right) = \sigma^d$$

For the second assertion, suppose E/K is Galois. The restriction $\sigma|_E$ stabilizes $\mathfrak{q}_E$ and induces $x \mapsto x^{\mathrm{N}(\mathfrak{p})}$ on its residue field. By uniqueness, it is the Frobenius element of $\mathfrak{q}_E$ in E/K. □

5.4.5 The Artin Map in an Abelian Extension

Suppose now that G is abelian. The Frobenius elements above an unramified prime $\mathfrak{p}$ are all equal, so it is meaningful to put

$$\left(\frac{L/K}{\mathfrak{p}}\right) := \mathrm{Frob}_{\mathfrak{p}}$$

without choosing $\mathfrak{q}|\mathfrak{p}$.

Definition 5.4.4 (Artin Map). Let S be the set of primes of A ramified in L, and let

$$\mathrm{I}_A^S := \left\{I \in \mathrm{FracId}(A) \mid v_{\mathfrak{p}}(I) = 0 \ \text{ for every } \ \mathfrak{p} \in S\right\}$$

be the group of fractional ideals prime to S. The *Artin map* is

$$\mathrm{Art}_{L/K} : \mathrm{I}_A^S \to G$$

determined by

$$\mathrm{Art}_{L/K}\left(\prod_{i=1}^{r} \mathfrak{p}_i^{a_i}\right) := \prod_{i=1}^{r} \left(\frac{L/K}{\mathfrak{p}_i}\right)^{a_i}$$

where the $\mathfrak{p}_i$ are unramified and $a_i \in \mathbb{Z}$.

Proposition 5.4.3 (Well-Definedness and Multiplicativity). The Artin map is a well-defined group homomorphism. On every unramified prime,

$$\mathrm{Art}_{L/K}(\mathfrak{p}) = \mathrm{Frob}_{\mathfrak{p}}$$

Proof. Unique factorization of fractional ideals in the Dedekind domain A writes every element of I_A^S uniquely as a finite product $\prod \mathfrak{p}^{a_{\mathfrak{p}}}$ over primes outside S. The Frobenius class at each such prime is a singleton because G is abelian, so every factor on the right is unambiguous. The product is independent of its ordering because G is abelian. Finally, multiplication of ideals adds their prime exponents, while multiplication in G adds the corresponding exponents of each Artin symbol. This proves both well-definedness and multiplicativity. □

Remark (Why Ramified Primes Are Removed). At a ramified prime the exact sequence has nontrivial kernel $I_{\mathfrak{q}}$. The residue Frobenius therefore determines only a coset of possible lifts in $D_{\mathfrak{q}}$, not a canonical group element. Removing ramified primes is exactly what makes the Artin map canonical.

Remark (Arithmetic Significance). The Artin symbol converts the factorization behavior of primes into group elements. The identity detects complete splitting, the order gives the residue degree, and in a nonabelian extension the conjugacy class is the natural invariant. Chebotarev's density theorem describes how these classes are distributed among primes. In the abelian setting, Artin reciprocity later shows that the Artin map factors through an appropriate ray class group and recovers the Galois group from ideal classes and congruence data.

5.4.6 Basic Examples

Example (Gaussian Integers). Let $L = \mathbb{Q}(i)$ over $K = \mathbb{Q}$. The only ramified rational prime is 2. For every odd prime p,

$$\left(\frac{\mathbb{Q}(i)/\mathbb{Q}}{(p)}\right) = \begin{cases} \mathrm{id} & p \equiv 1 \bmod 4 \\ \mathrm{conj} & p \equiv 3 \bmod 4 \end{cases}$$

Indeed, the first case is complete splitting and the second is inertness. If G is identified with $\{\pm 1\}$ by sending complex conjugation to -1, this Artin symbol is the Legendre symbol $\left(\frac{-1}{p}\right)$.

Example (Cyclotomic Fields). Let $L = \mathbb{Q}(\zeta_m)$ and identify

$$\mathrm{Gal}(L/\mathbb{Q}) \simeq (\mathbb{Z}/m\mathbb{Z})^\times, \quad a \mapsto (\zeta_m \mapsto \zeta_m^a)$$

For every rational prime p not dividing m, the prime is unramified and

$$\left(\frac{\mathbb{Q}(\zeta_m)/\mathbb{Q}}{(p)}\right)(\zeta_m) = \zeta_m^p$$

Thus the Artin symbol of p corresponds exactly to the residue class of p in $(\mathbb{Z}/m\mathbb{Z})^\times$. The order of p modulo m is therefore the residue degree of p in the cyclotomic extension.